

数字系统设计

Digital System Design

---门电路与接口

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任课教师

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- Zhejiang University, Hangzhou, 310027
- Tel: 86-571-87953170 (O)
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Office Hours: 玉泉信电楼 308室（可以钉钉、微信或邮件联系）.

Prerequisites预修课程

- 电子电路基础
- 电子线路
- C语言
- How to learn this Course?
 - Not only listening, thinking and waiting
 - But Exercise, Simulation, Practice!

课程简介

- 课程代码: **671C0050**
- 参考书
 - 阎石,《数字电子技术基础》,第6版,高等教育出版社,2016.
 - 王金明著,《数字系统设计与Verilog HDL》,电子工业出版社,第8版
 - 《数字系统设计-补充讲义》,浙江大学自编讲义
 - Stanford 大学 108A课程notes.
 - R.H.Katz, G.Borriello, *Contemporary Logic Design*, second edition,电子工业出版社,2005.
 - M.M.Mano, 数字设计(第四版),电子工业出版社,2010.
 - » <http://www.prenhall.com/mano>

Other Course Info

- **Website:**

- <http://mypage.zju.edu.cn/wdwd/教学工作/>
- **学在浙大: 数字系统设计**
- Check frequently

- **答疑**

- 玉泉信电楼308室/周一周四下午2:30-5:30
- 上课课间、课后均可
- Email, 钉钉/微信群/数字系统设计, 短信均可

Grading (考核)

Final grades will be computed approximately as follows:

- 平时（含课程作业、期中考试、小测验、Project、出勤等） **40%**

- 期中考试 **10%**
- Homework Sets **15% (12~20)%**
- Project **5% (5~7)%**
- Class Room Check+小测验等 **10% (7~10)%**
- 作业每周一上交, **截止期为课后一周内**有效。

- **Project**

- Project-1必做 (1 members)
- Project-2可选(成绩加分1%~5%, 但不超过平时成绩范围)

- Finial Exam期末**闭卷**考试 - **60%**

授课时间和地点:

- 2025年春夏学期,
 - 周一上午, 第1~2节(8:00-9:35)
 - 星期四上午, 第6~7节(13:25-13:00)
- 地点: 紫金港西2-415(多)
- <http://mypage.zju.edu.cn/wdwd/教学工作/>
- 学在浙大/数字系统设计
- 钉钉/微信/数字系统设计-答疑

课程结构

- 数字理论知识(必备)
 - 数字系统和编码、逻辑代数、门电路
- 数字电路分析与设计
 - 组合逻辑电路
 - 触发器、半导体存贮器、可编程器件
 - 时序逻辑电路
- 脉冲电路与接口
- 控制器与数字系统
 - 状态机
 - 控制器
 - 微码控制器
 - 测试和验证
- 微处理器简介
 - 指令集
 - **4位CPU**介绍

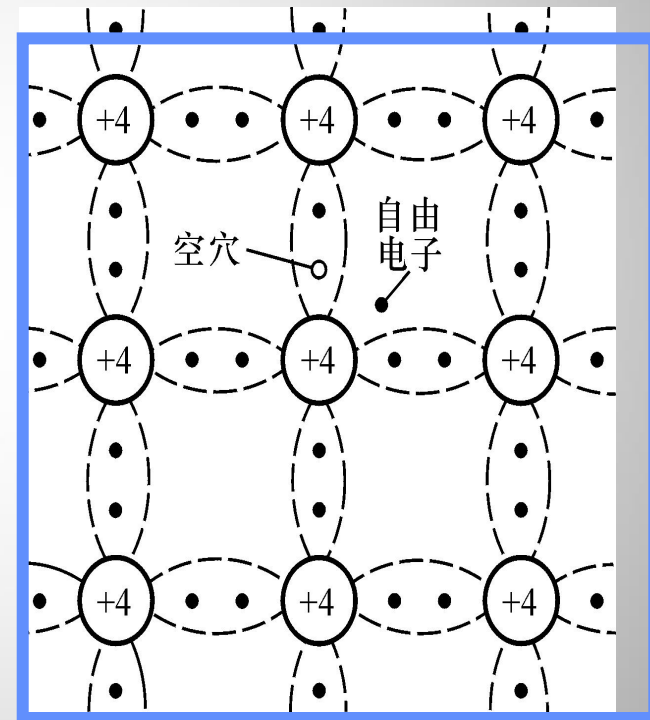
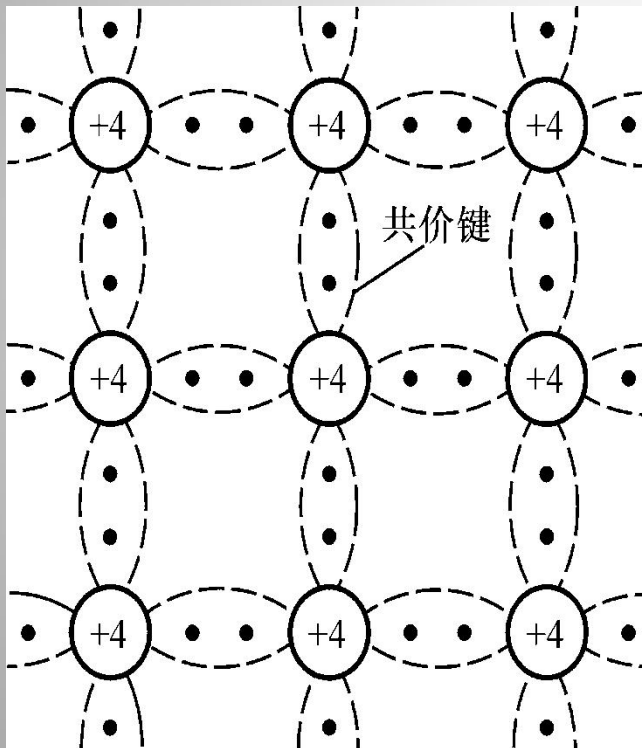
-----门逻辑和接口电路

一、半导体基础知识(补)

半导体基础知识（1）

- 本征半导体：纯净的具有晶体结构的半导体。
- 常用：硅Si，锗Ge

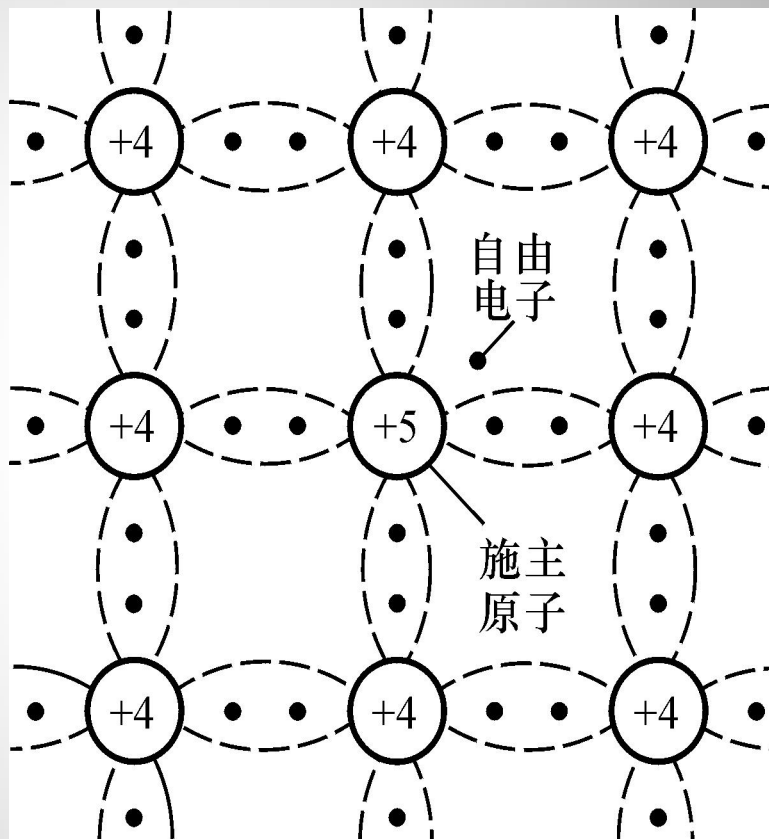
两种载流子



半导体基础知识（2）

- 杂质半导体
- **N型半导体**
(掺磷P或砷As)

多子：自由电子
少子：空穴

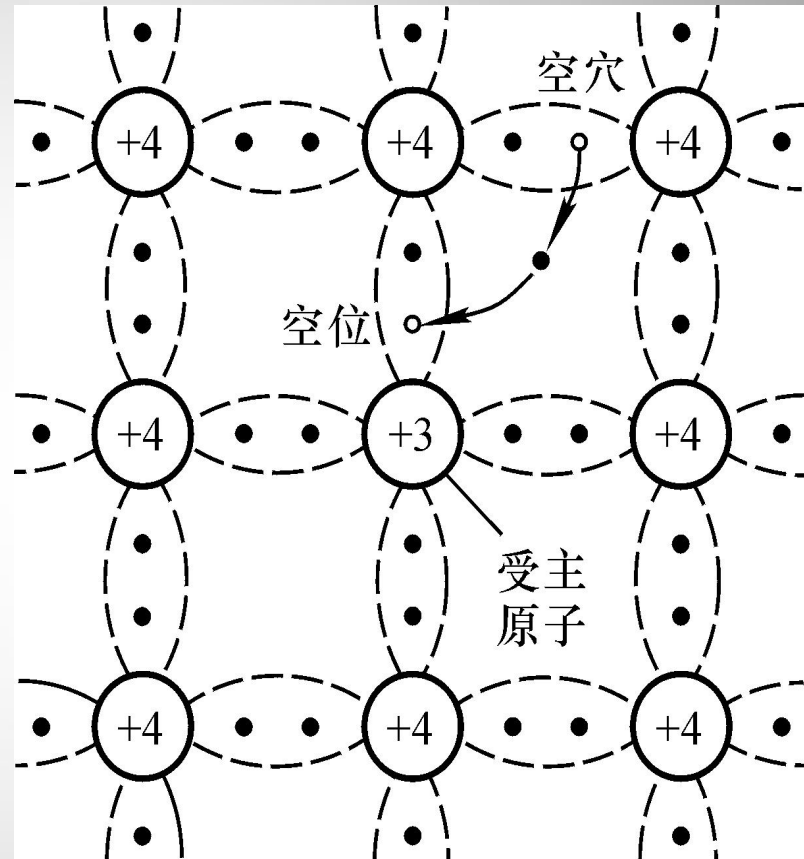


半导体基础知识 (3)

- 杂质半导体
- **P型半导体**(掺硼B)

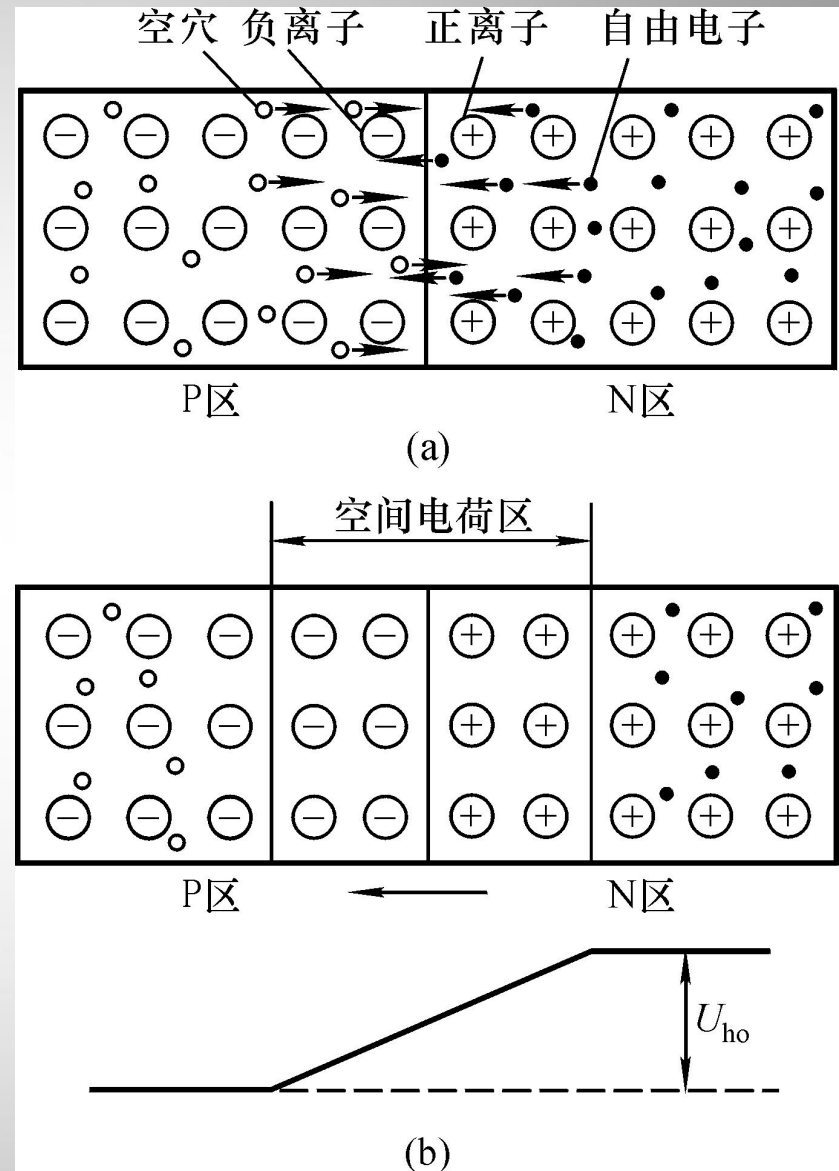
多子：空穴

少子：自由电子



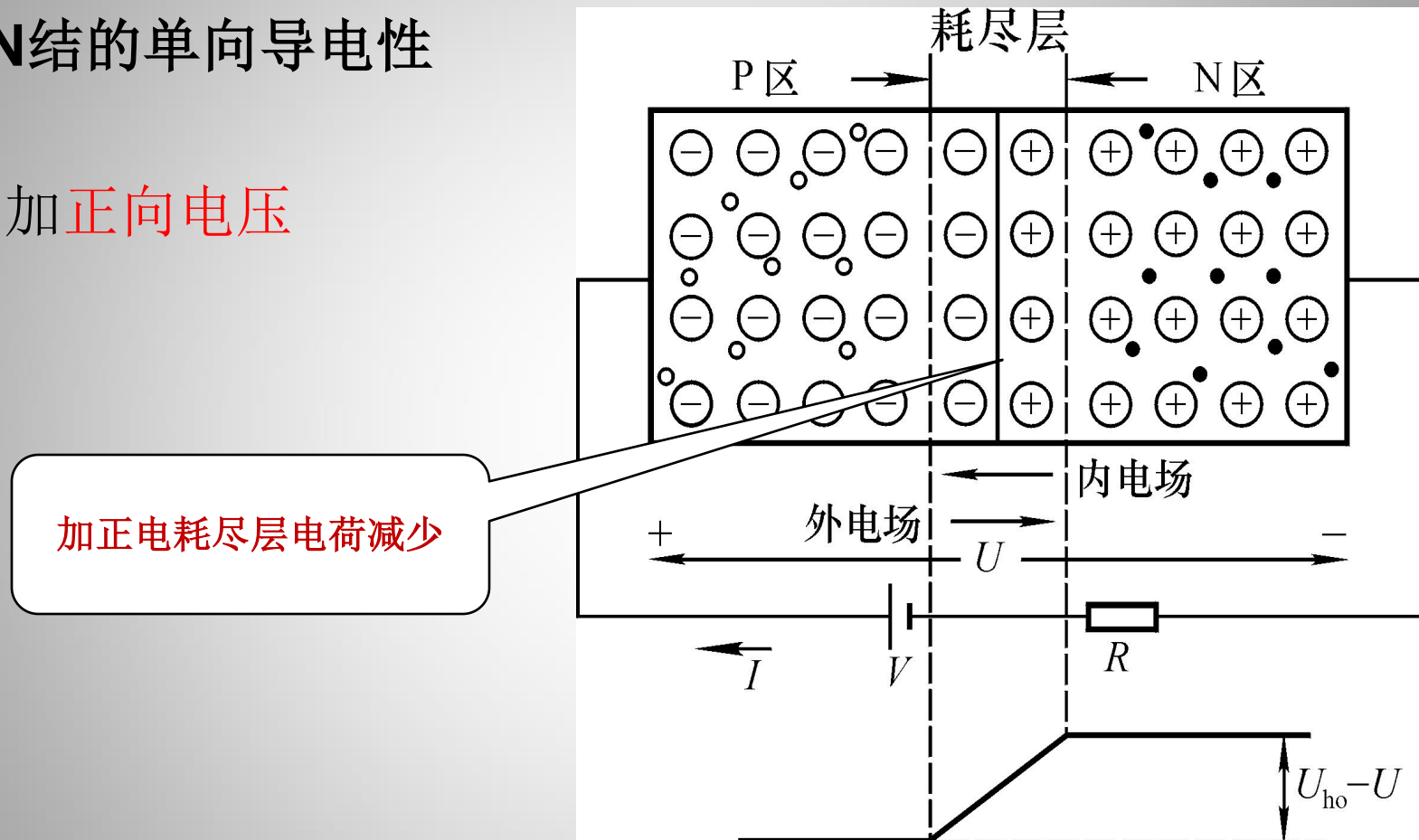
半导体基础知识 (4)

- PN结的形成
- 空间电荷区 (耗尽层)
- 扩散和漂移



半导体基础知识（5）

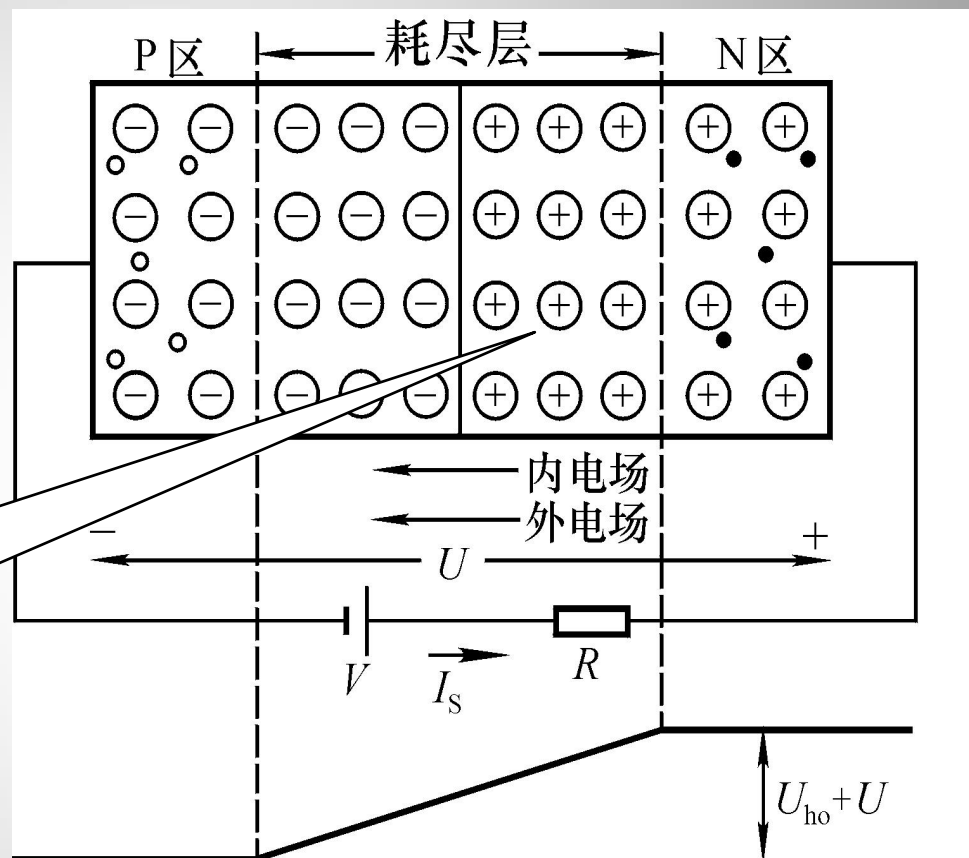
- PN结的单向导电性
- 外加正向电压



半导体基础知识 (6)

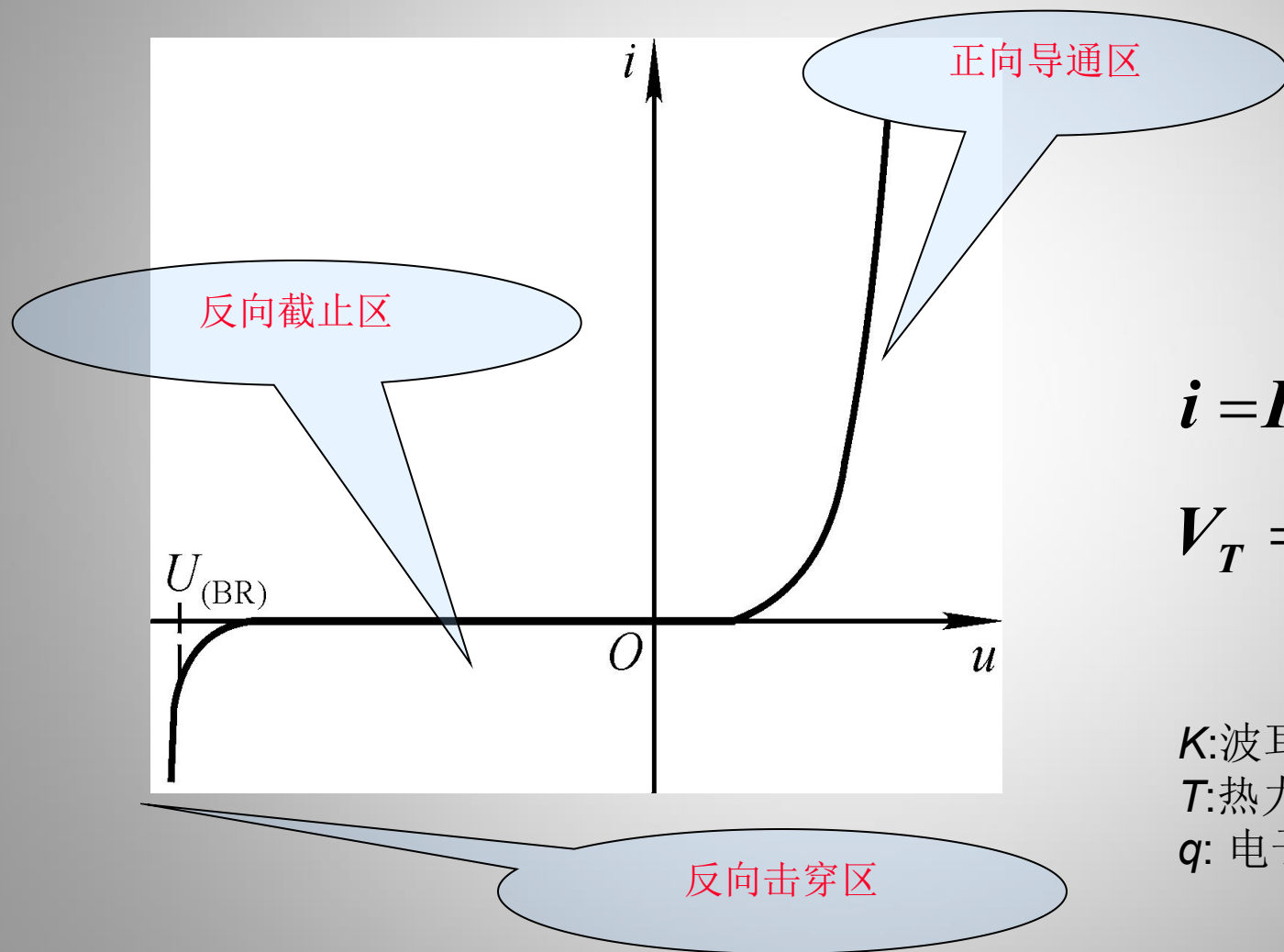
- PN结的单向导电性
- 外加反向电压

加负电耗尽层电荷增加，
产生充放电现象



半导体基础知识 (7)

- PN结的伏安特性



$$i = I_s (e^{V/V_T} - 1)$$

$$V_T = nkT/q$$

K : 玻耳兹曼常数

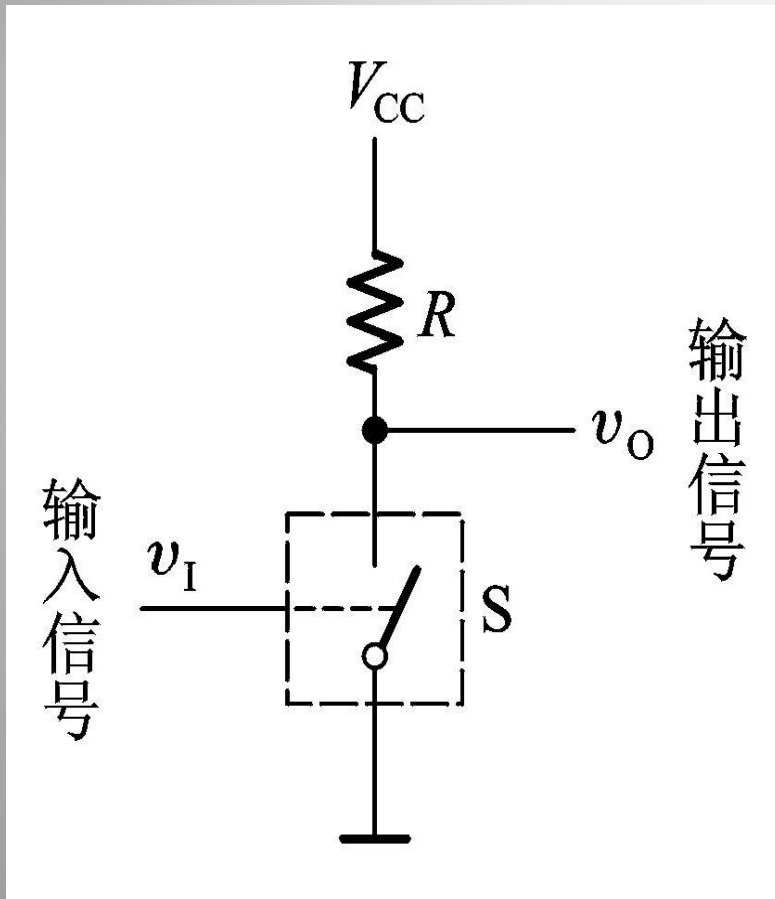
T : 热力学温度

q : 电子电荷

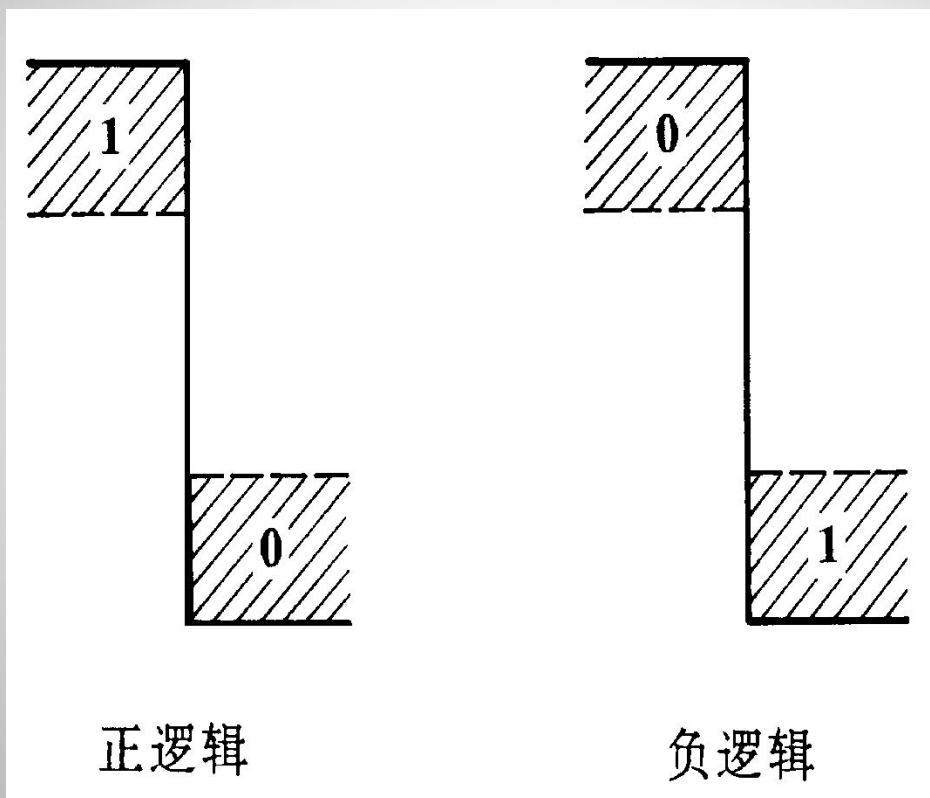
二、逻辑电路的输入输出特性

获得高、低电平的基本原理

高/低电平都允许有一定的变化范围



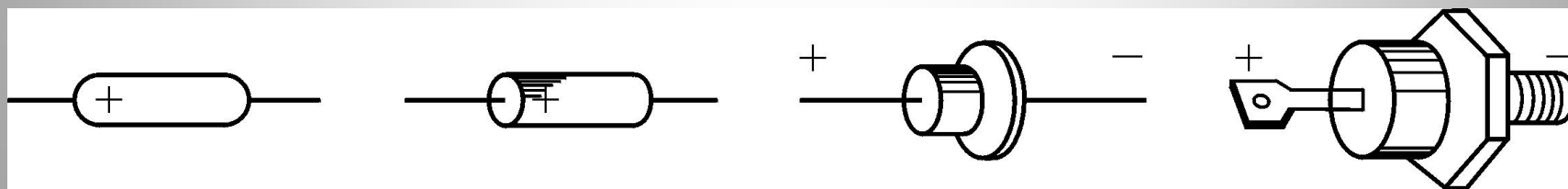
正逻辑：高电平表示1，低电平表示0
负逻辑：高电平表示0，低电平表示1



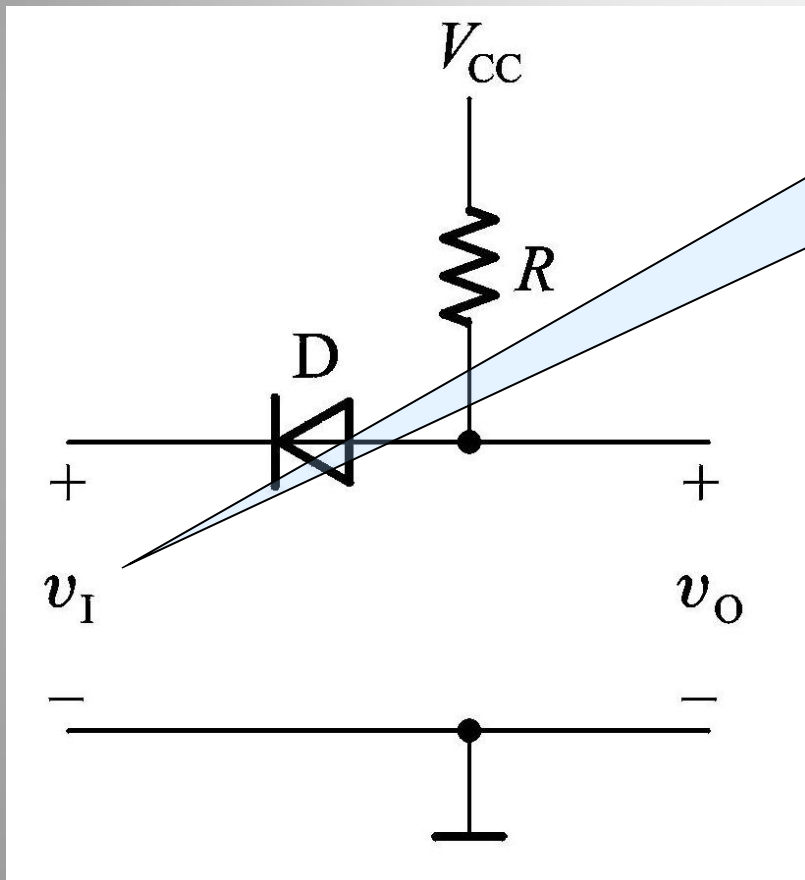
半导体二极管的结构和外特性 (Diode)

□ 二极管的结构:

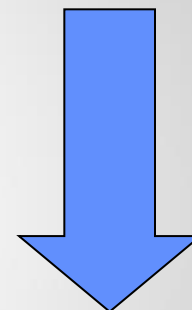
PN结 + 引线 + 封装构成



3. 2. 1 二极管的开关特性:

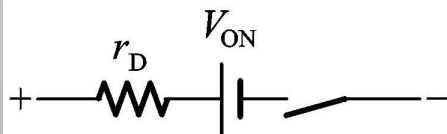
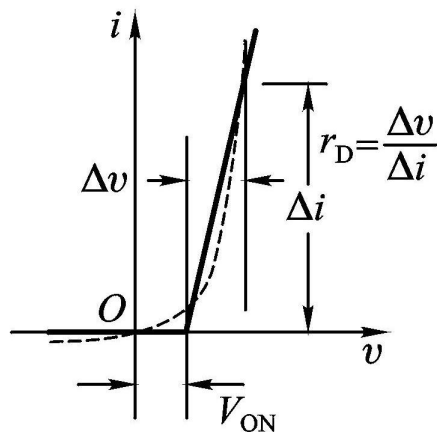
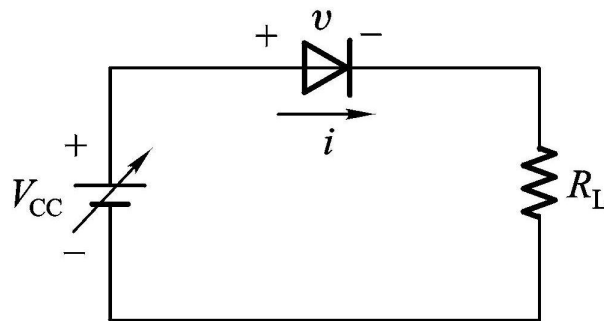


高电平: $V_{IH}=V_{CC}$
低电平: $V_{IL}=0$

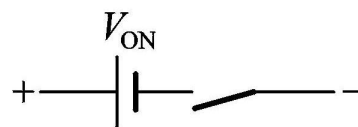
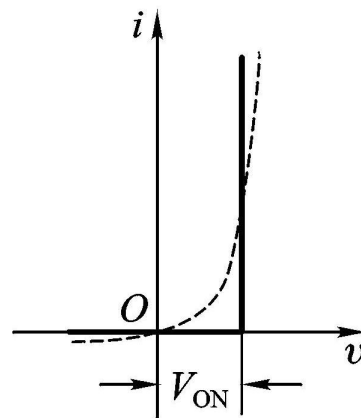


- $V_I=V_{IH}$
D截止, $V_O=V_{OH}=V_{CC}$
- $V_I=V_{IL}$
D导通, $V_O=V_{OL}=0.7V$

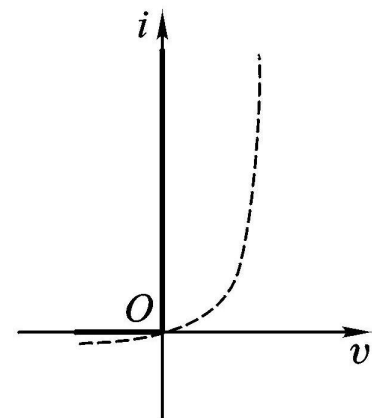
二极管的开关等效电路:



(a)

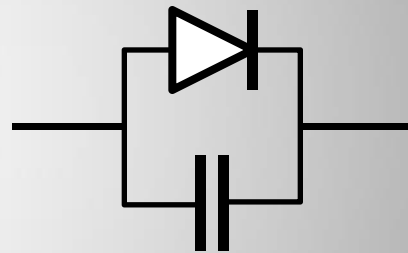
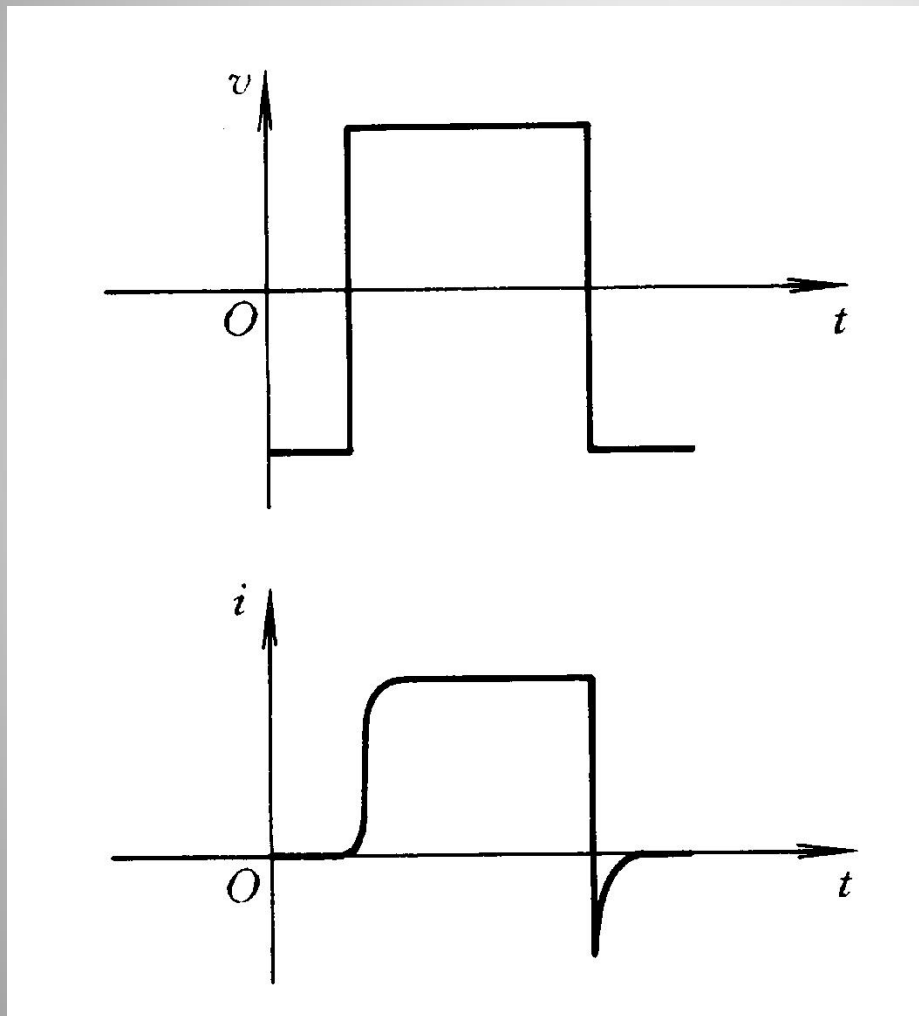


(b)



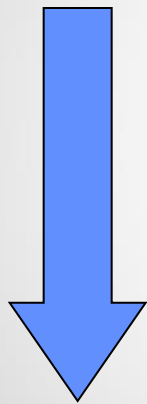
(c)

二极管的动态电流波形:



二极管构成的门电路的缺点

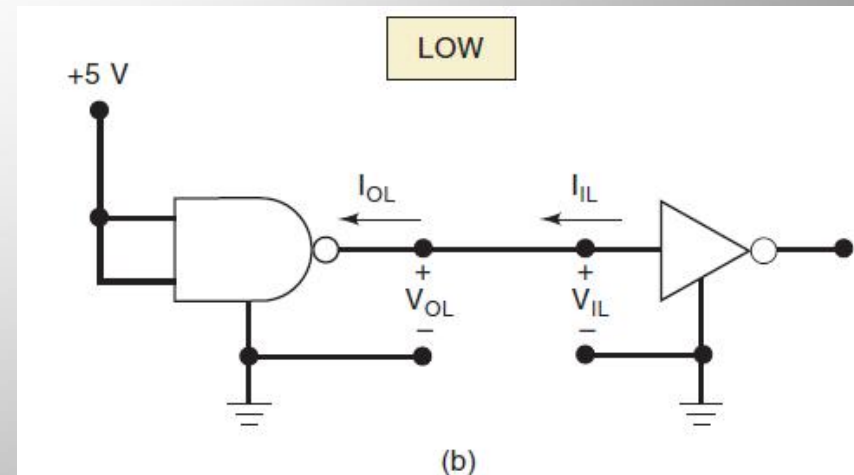
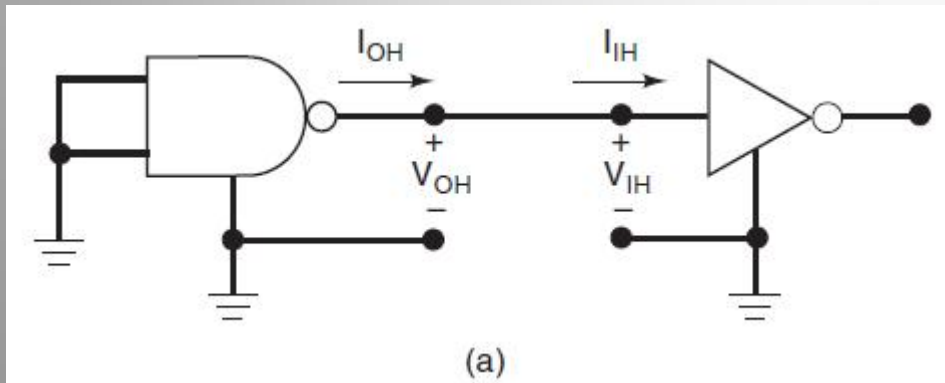
- 电平有偏移
- 带负载能力差



- 只用于IC内部电路

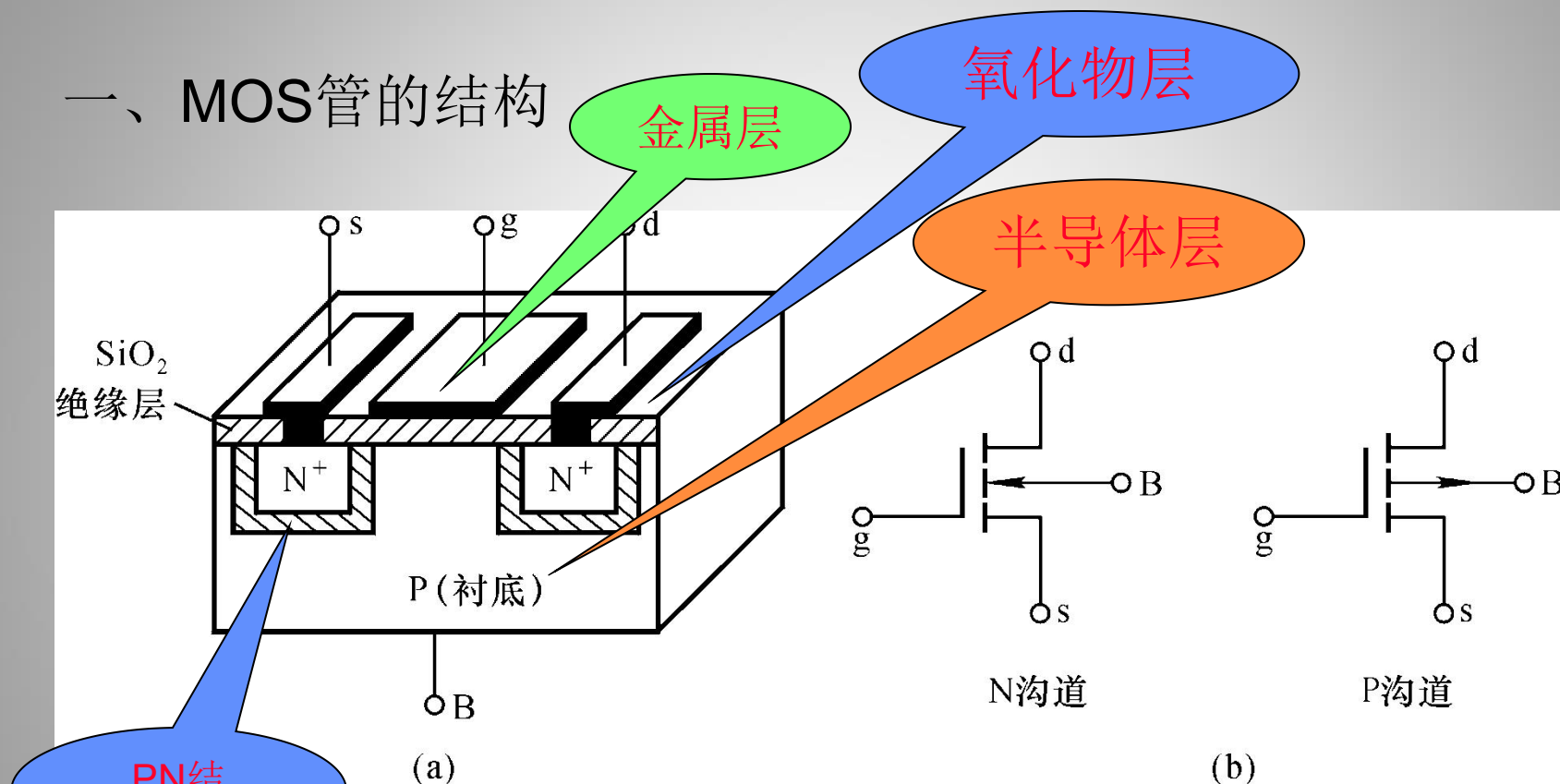
Current and Voltage Parameters

- $V_{IH}(min)$ —High-Level Input Voltage.
- $V_{IL}(max)$ —Low-Level Input Voltage.
- $V_{OH}(min)$ —High-Level Output Voltage.
- $V_{OL}(max)$ —Low-Level Output Voltage.
- I_{IH} —High-Level Input Current.
- I_{IL} —Low-Level Input Current.
- I_{OH} —High-Level Output Current.
- I_{OL} —Low-Level Output Current.

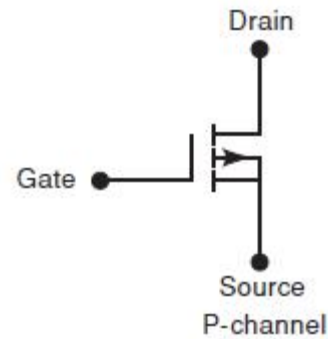
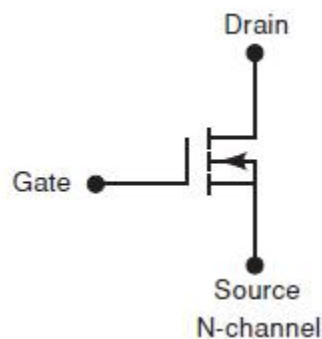


MOS管的开关特性

一、MOS管的结构



S (Source): 源极
G (Gate): 栅极
D (Drain): 漏极
B (Substrate): 衬底



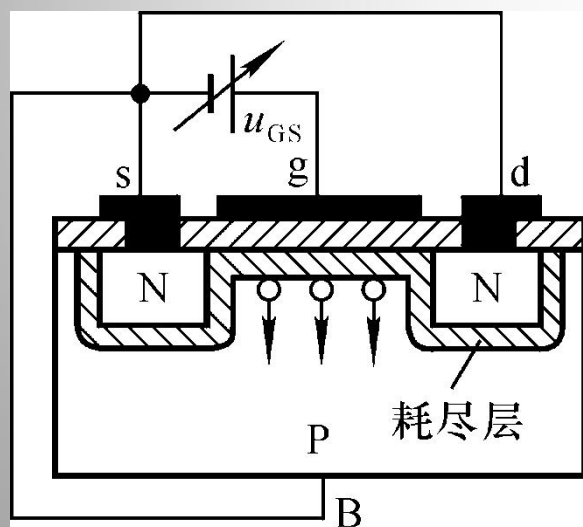
以N沟道增强型为例：

当加 $+V_{DS}$ 时，

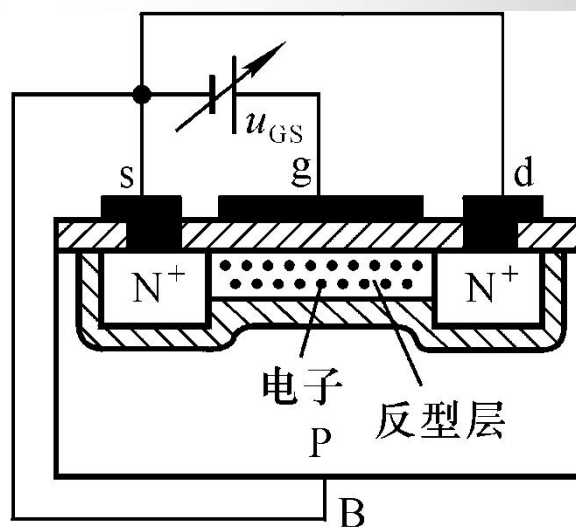
$V_{GS}=0$ 时，D-S间是两个背向PN结串联， $i_D=0$

加上 $+V_{GS}$ ，且足够大至 $V_{GS} > V_{GS(th)}$ ，D-S间形成导电沟道（N型层）

开启电压



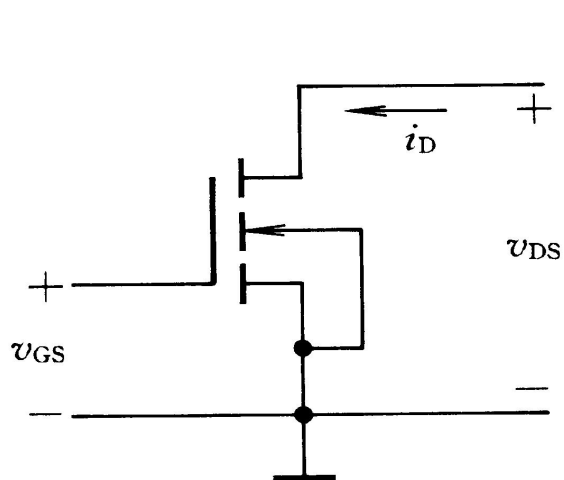
(a)



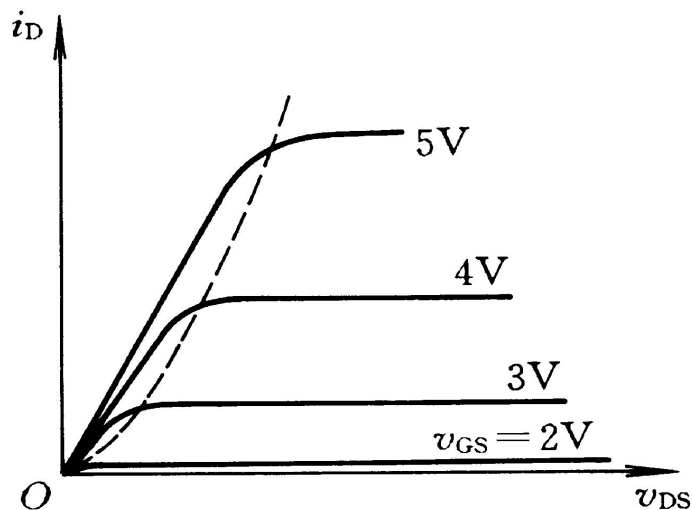
(b)

二、输入特性和输出特性

- ① 输入特性：直流电流为0，看进去有一个输入电容 C_i ，对动态有影响。
- ② 输出特性：
 $i_D = f(V_{DS})$ 对应不同的 V_{GS} 下得一族曲线。



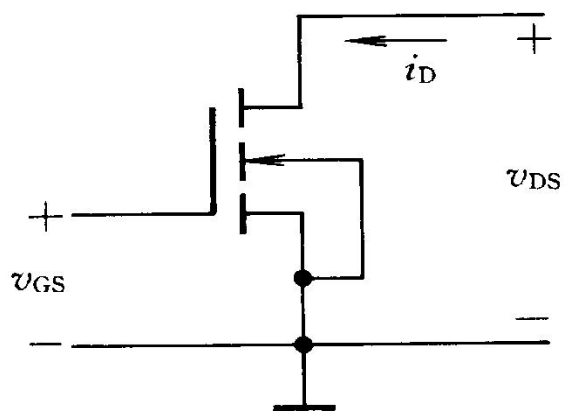
(a)



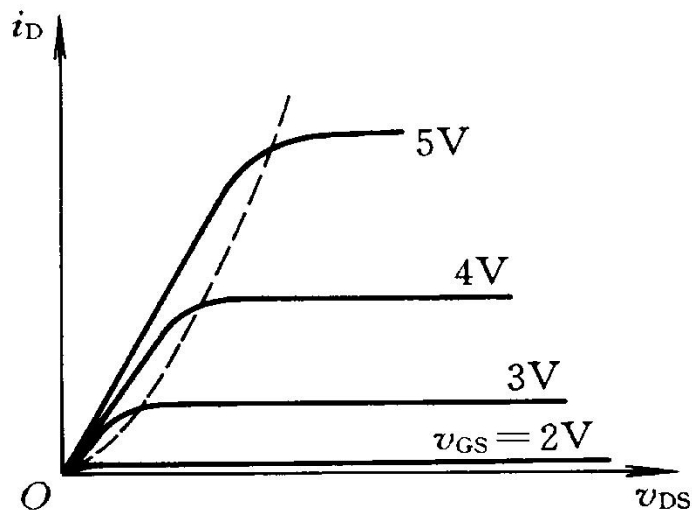
(b)

漏极特性曲线（分三个区域）

- ① 截止区
- ② 恒流区
- ③ 可变电阻区



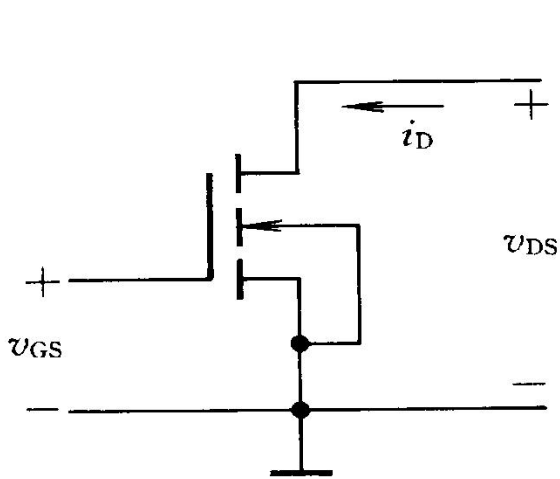
(a)



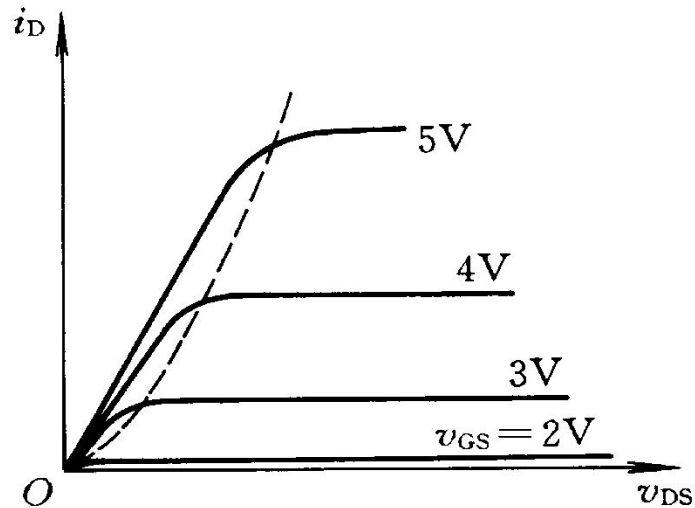
(b)

漏极特性曲线（分三个区域）

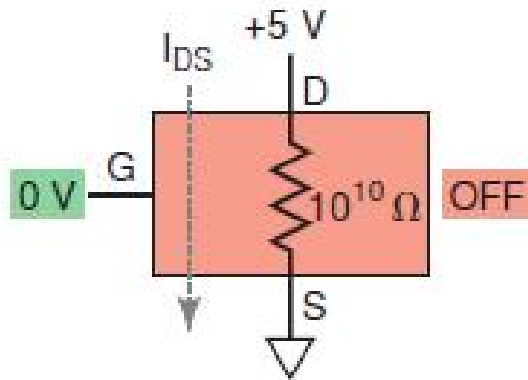
1) 截止区: $V_{GS} < V_{GS(th)}$, $i_D = 0$, $R_{OFF} > 10^9 \Omega$



(a)



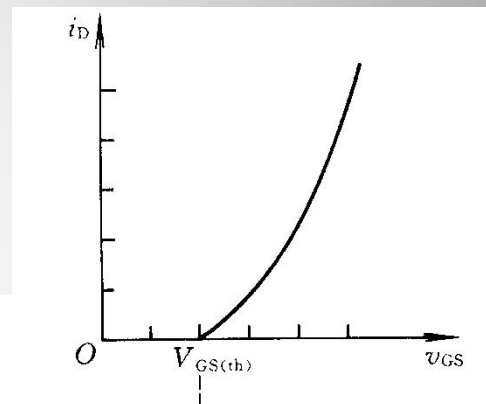
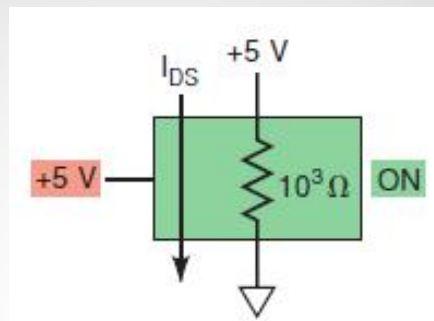
(b)



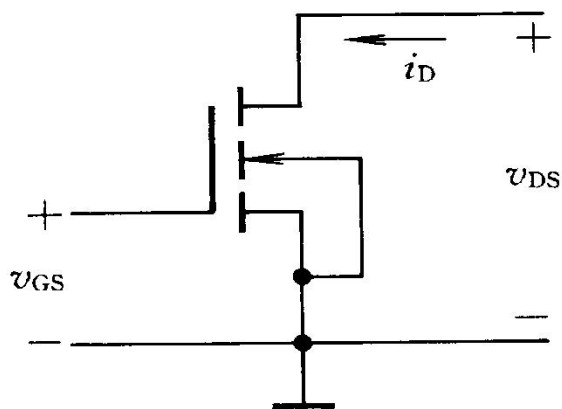
漏极特性曲线（分三个区域）

2) 恒流区: i_D 基本上由 V_{GS} 决定, 与 V_{DS} 关系不大

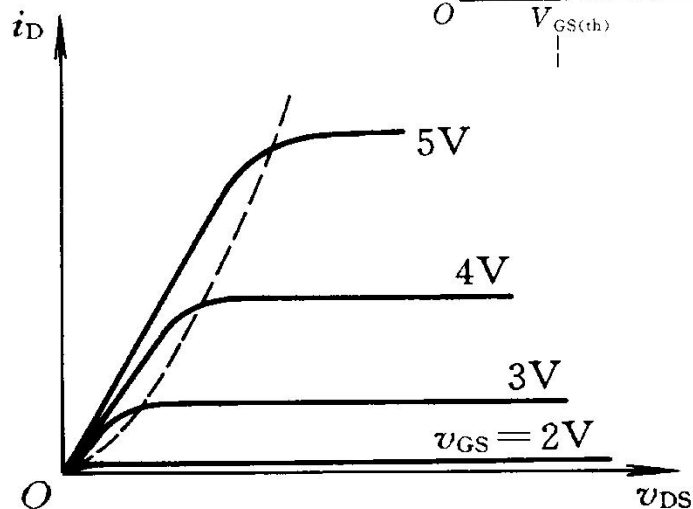
$$i_D = I_{DS} \left(\frac{V_{GS}}{V_{GS(th)}} - 1 \right)^2$$



当 $V_{GS} \gg V_{GS(th)}$ 下, $i_D \propto V_{GS}^2$



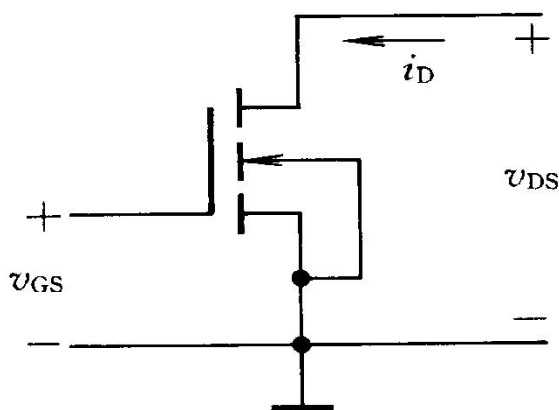
(a)



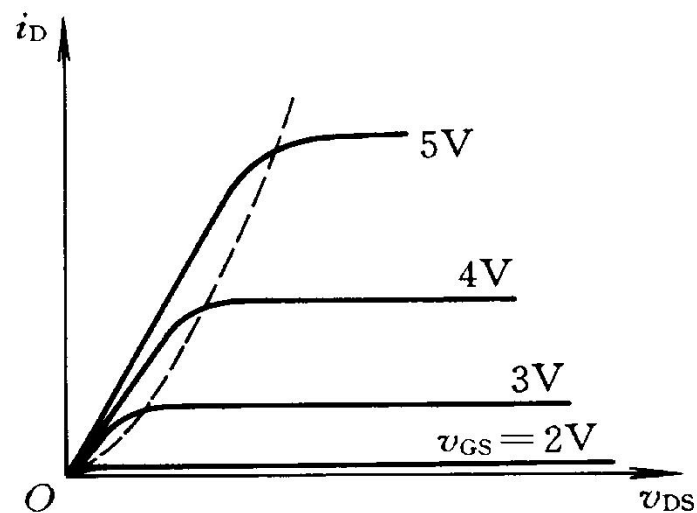
(b)

漏极特性曲线（分三个区域）

3) 可变电阻区：当 V_{DS} 较低（近似为0）， V_{GS} 一定时， $V_{DS}/i_D \approx \text{常数}$ （电阻） 这个电阻受 V_{GS} 控制、可变。



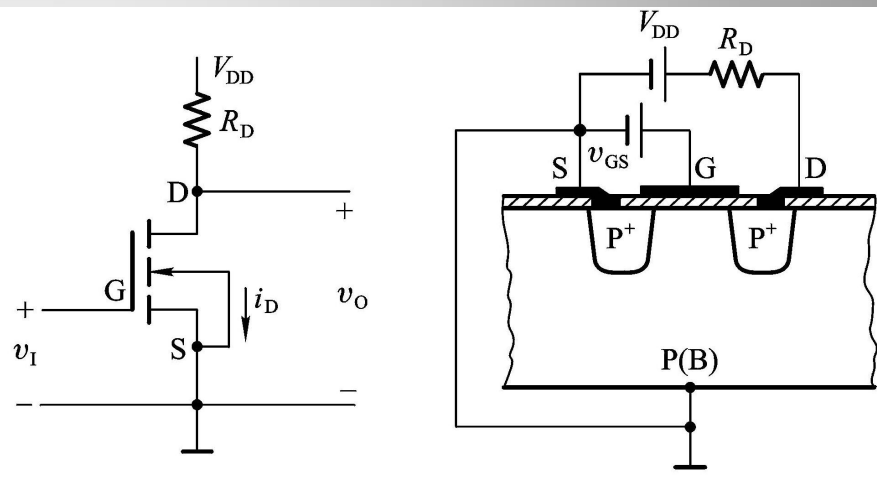
(a)



(b)

三、MOS管的基本开关电路

因为 $R_{OFF} > 10^9 \Omega$, $R_{ON} < 1K\Omega$
只要 $R_{ON} \ll R_D \ll R_{OFF}$, 则:

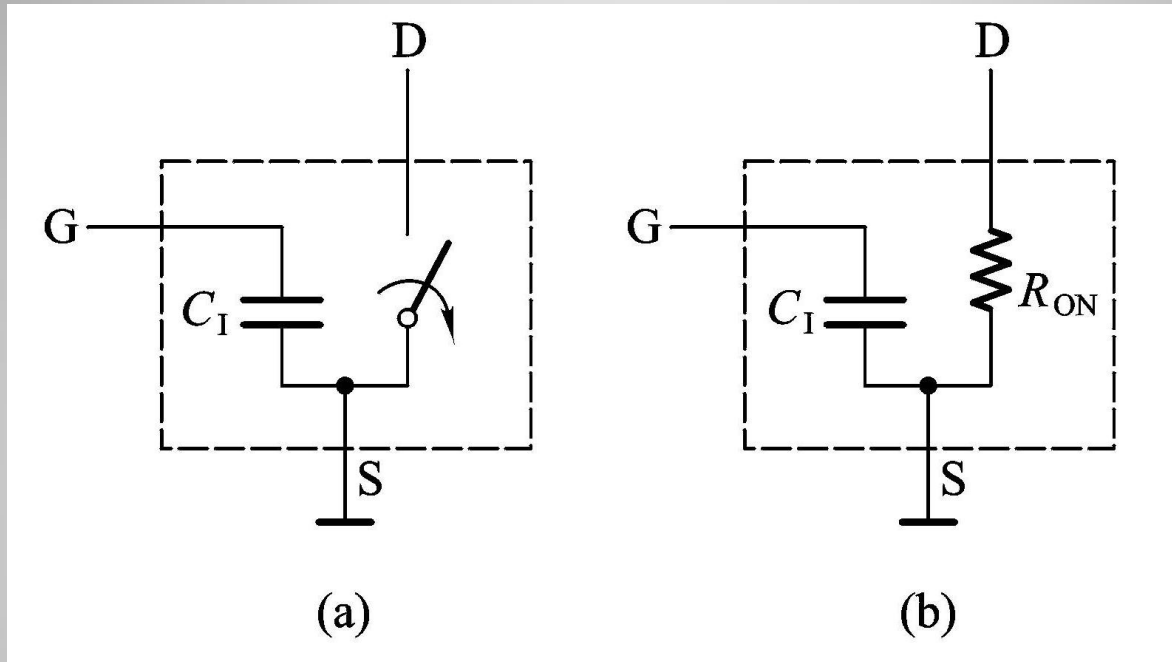


当 $V_I = V_{IL} < V_{GS(th)}$ $\longrightarrow$ T 截止 $\longrightarrow V_O = V_{OH} \approx V_{DD}$

当 $V_I = V_{IH} > V_{GS(th)}$ $\longrightarrow$ T 导通 $\longrightarrow V_O = V_{OL} \approx 0$

所以 MOS 管 $D-S$ 间相当于一个受 V_I 控制的开关。

四、等效电路

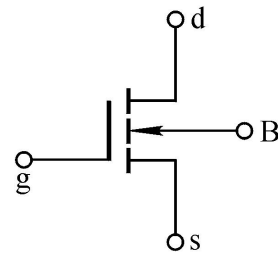
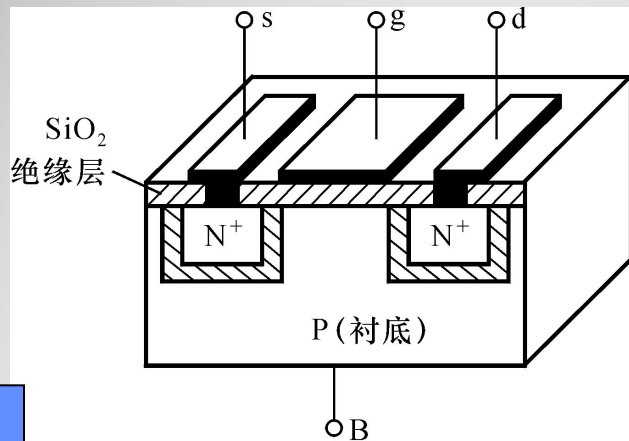


OFF，截止状态

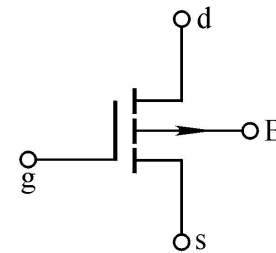
ON，导通状态

五、MOS管的四种类型

- 增强型



N沟道

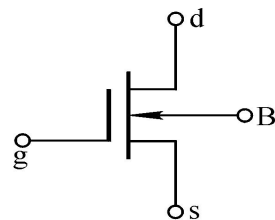
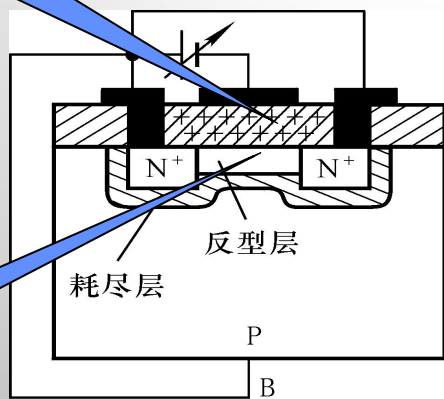


P沟道

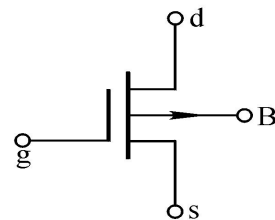
(a)

(b)

- 耗尽型



耗尽型 N 沟道管



耗尽型 P 沟道管

(a)

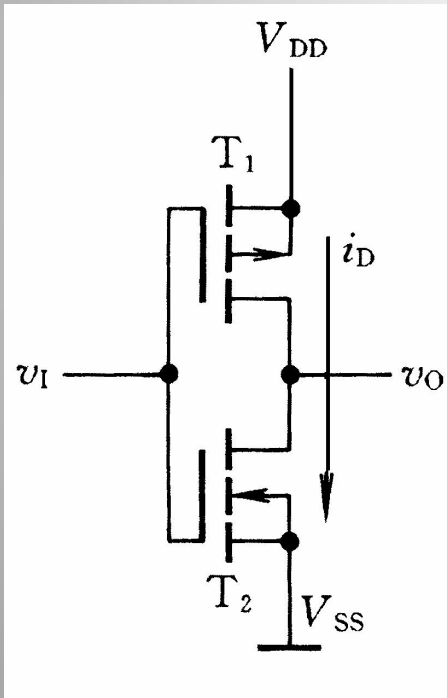
(b)

大量正离子

导电沟道

CMOS反相器的电路结构和工作原理

一、电路结构



$$V_{GS(th)N} = |V_{GS(th)P}|$$

二、电压、电流传输特性

* **AB段**: $V_I < V_{GS(th)N}$

T_1 导通, T_2 截止 $\Rightarrow V_O = V_{OH} = V_{DD}$

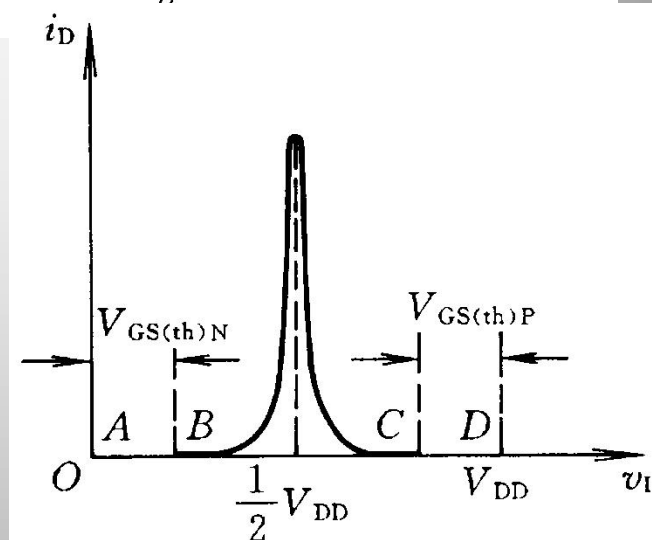
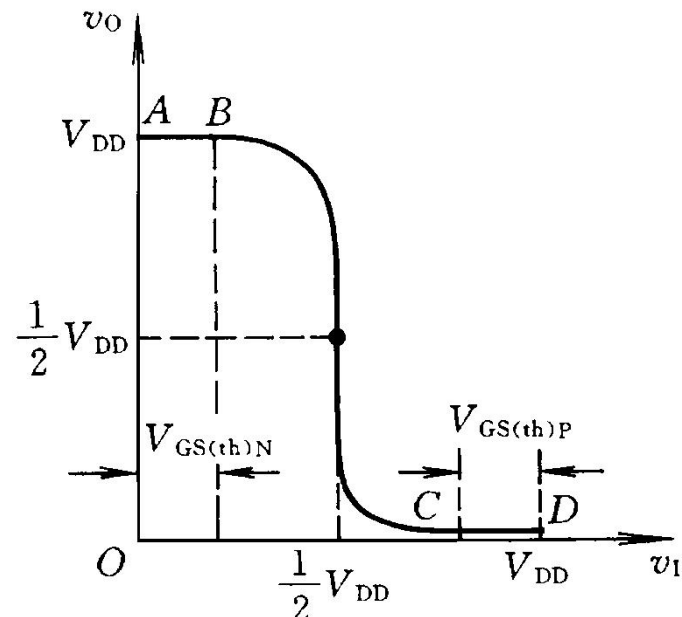
* **CD段**: $V_I > V_{DD} - |V_{GS(th)P}|$

T_2 导通, T_1 截止 $\Rightarrow V_O = V_{OL} = 0$

* **BC段**: $V_{GS(th)N} < V_I < V_{DD} - |V_{GS(th)P}|$

T_1, T_2 同时导通

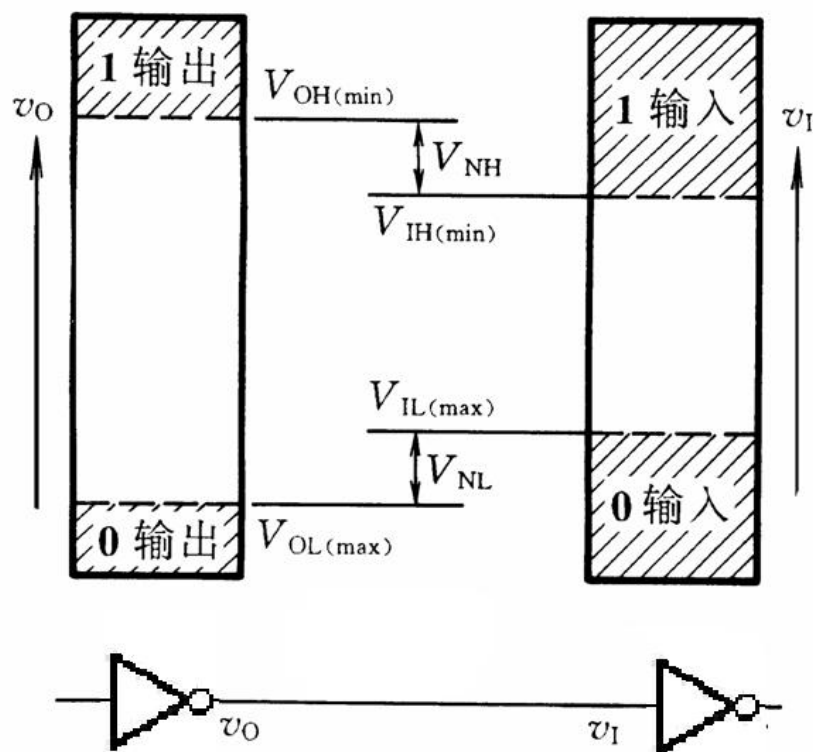
若 T_1, T_2 参数完全对称, $V_I = \frac{1}{2}V_{DD}$ 时, $V_O = \frac{1}{2}V_{DD}$



三、输入噪声容限

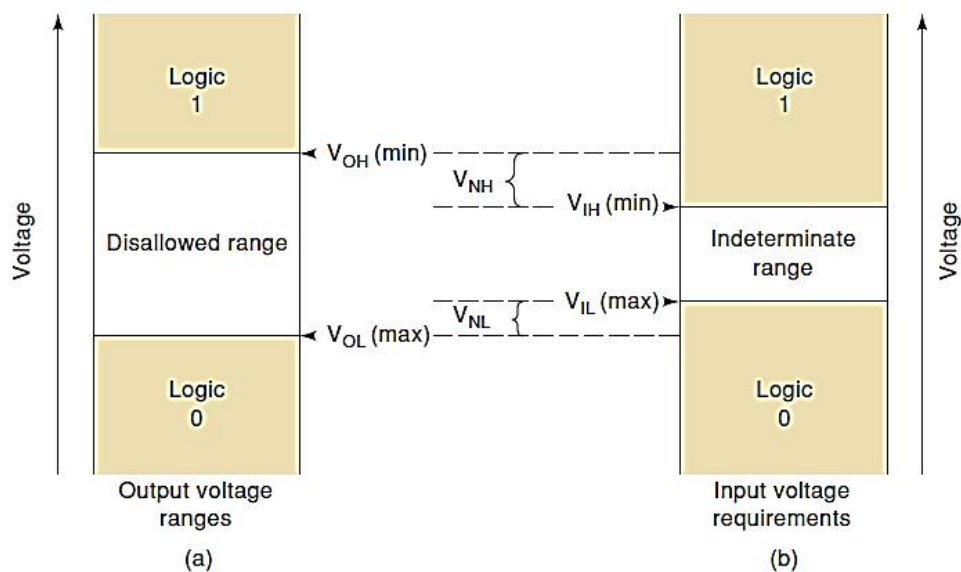
在 V_I 偏离 V_{IH} 和 V_{IL} 的一定范围内， V_O 基本不变；

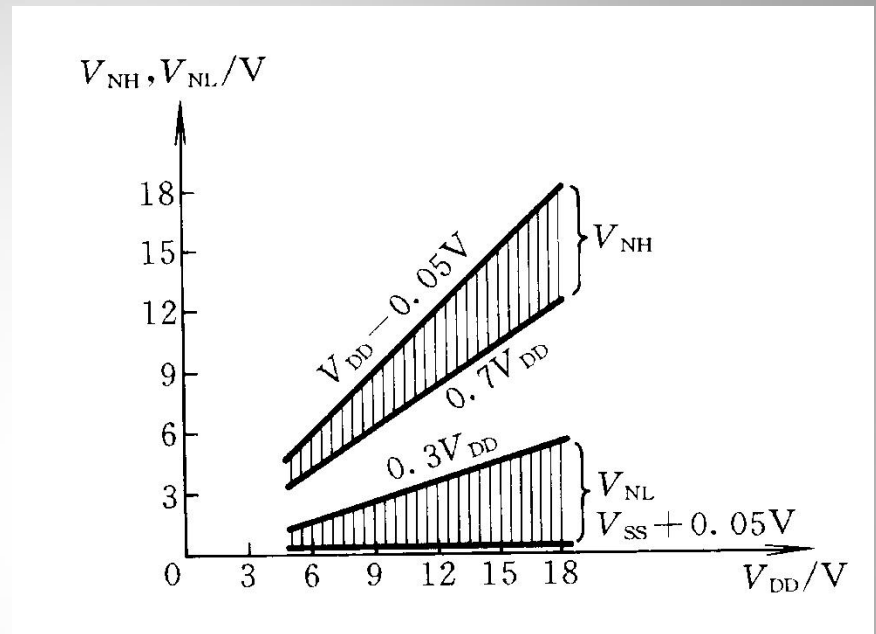
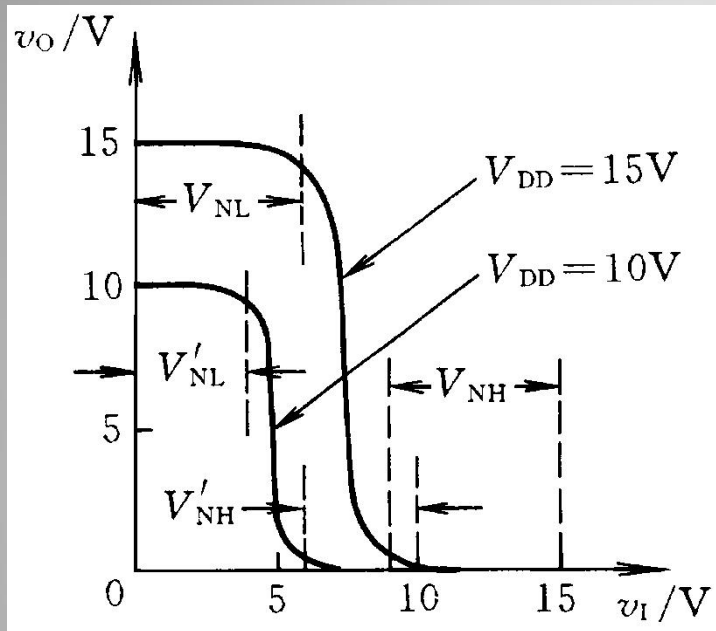
在输出变化允许范围内，允许输入的变化范围称为输入噪声容限



$$V_{NH} = V_{OH(\min)} - V_{IH(\min)}$$

$$V_{NL} = V_{IL(\max)} - V_{OL(\max)}$$

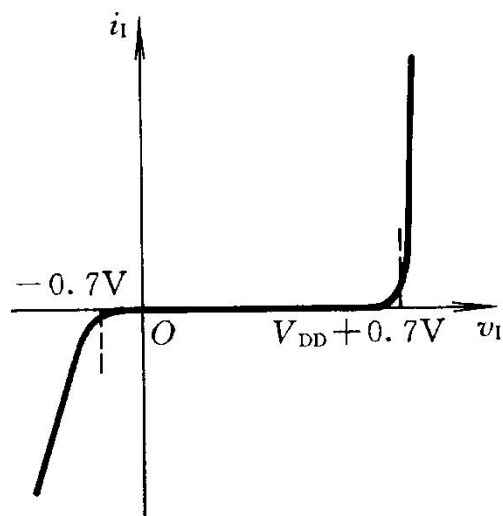
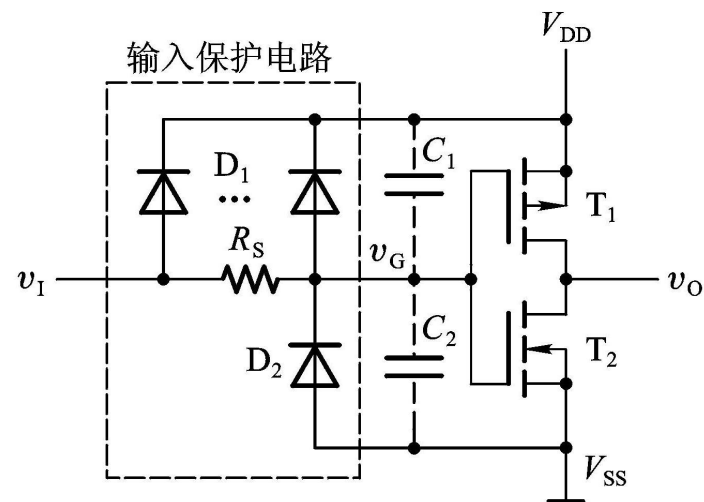
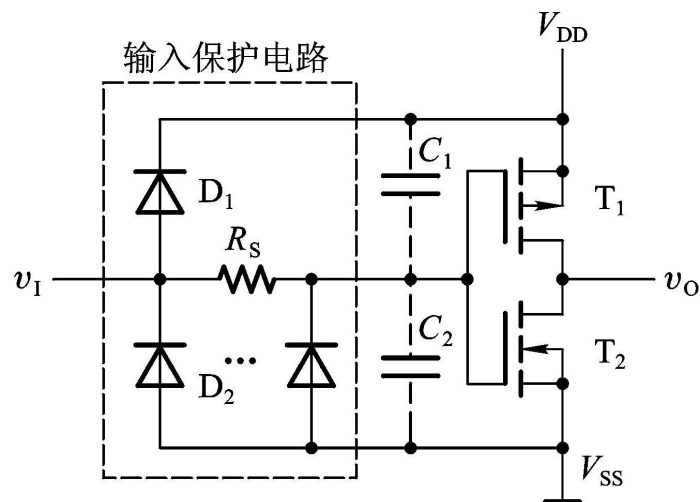




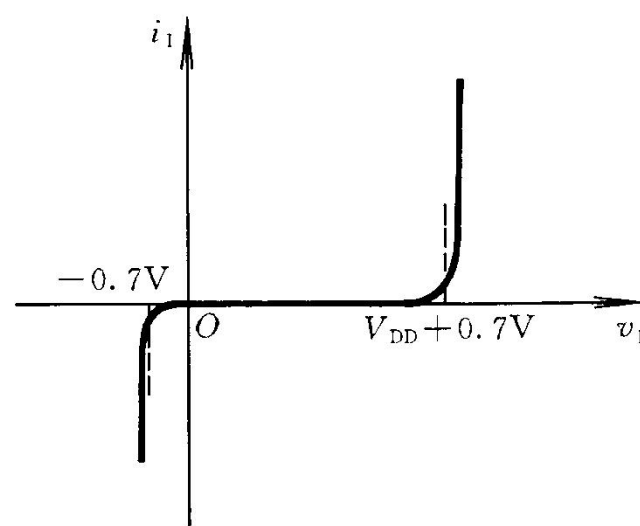
- **结论:** 可以通过提高 V_{DD} 来提高噪声容限

CMOS 反相器的静态输入和输出特性

一、输入特性

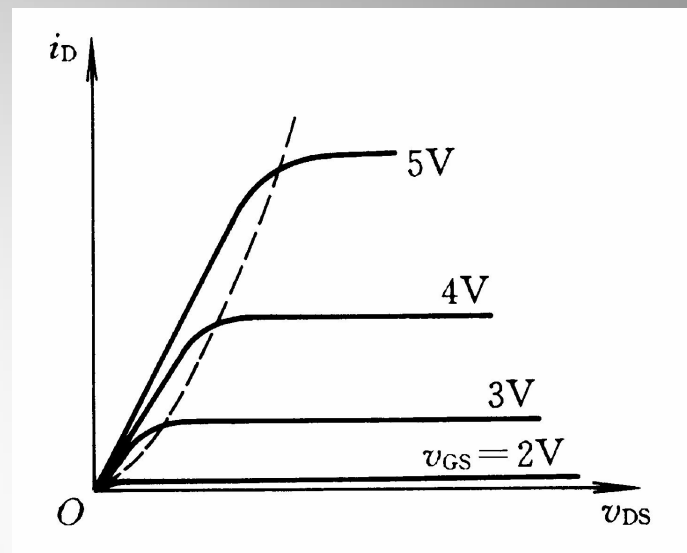
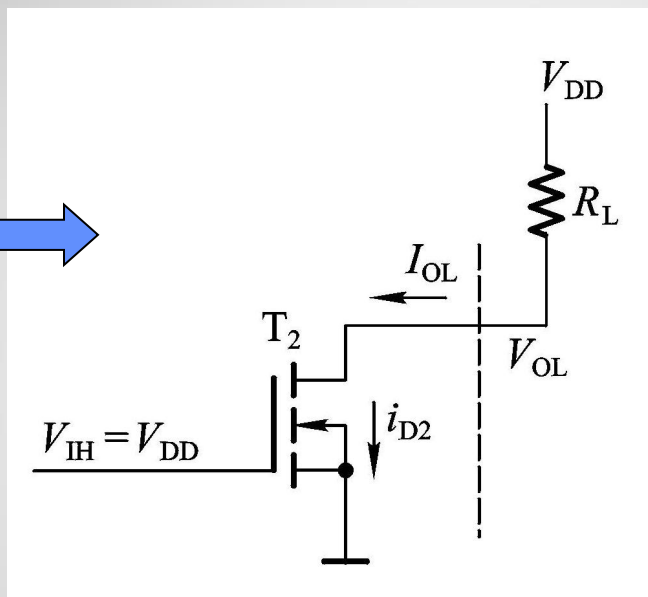
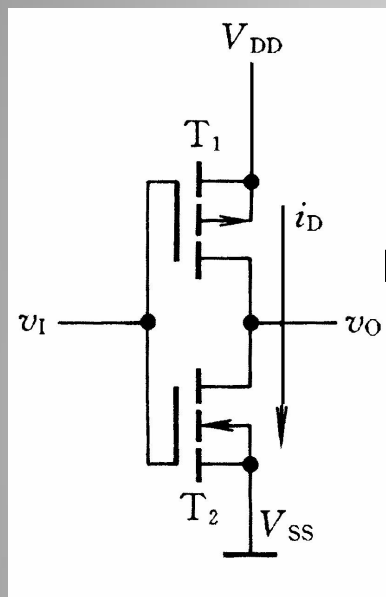


(a)



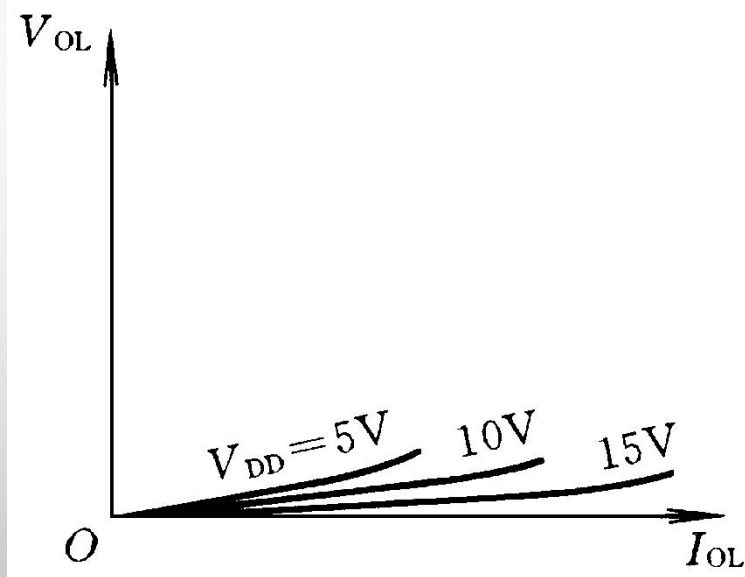
(b)

二、输出特性

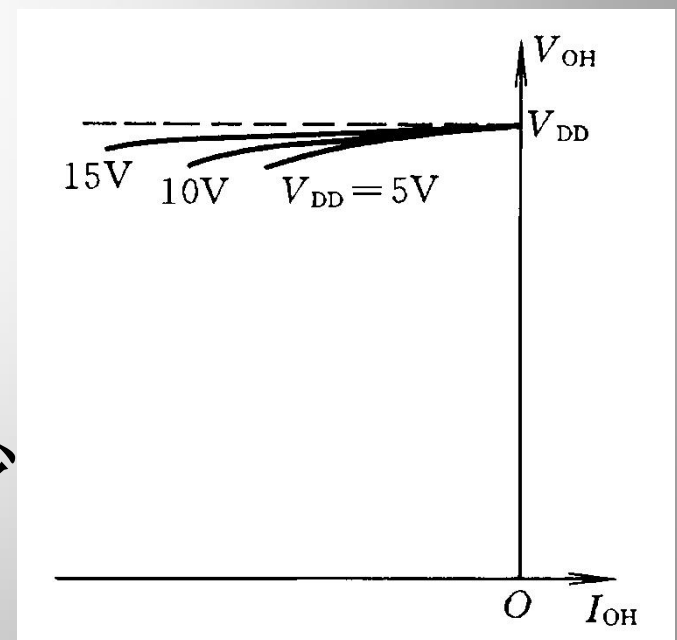
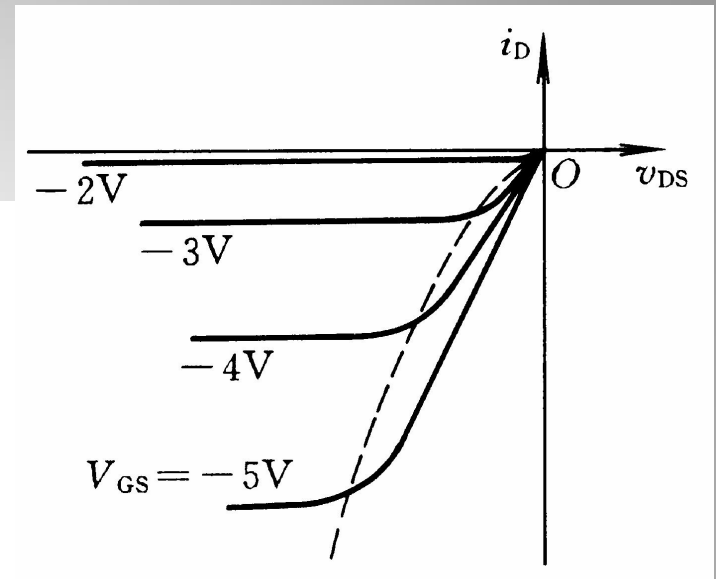
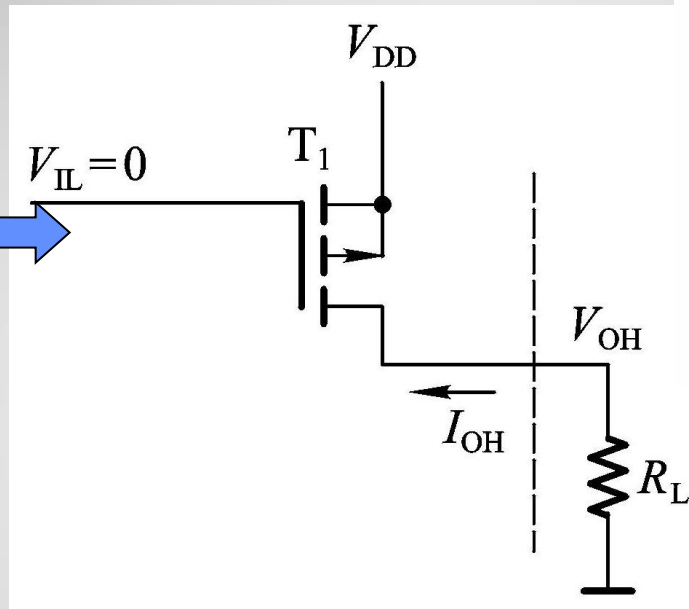
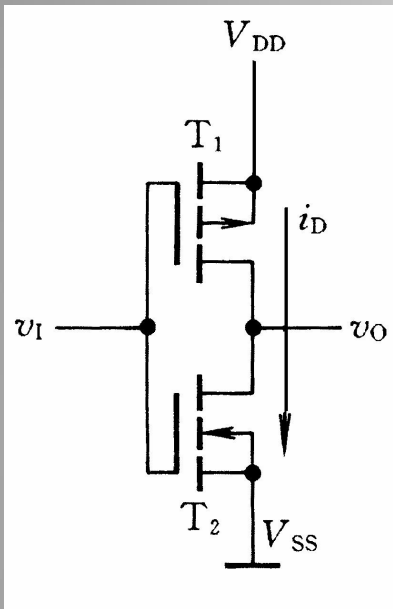


1. 低电平输出特性 $V_{OL} = f(I_{OL})$

同样的 I_{OL} 下, $V_{GS} \uparrow \Rightarrow V_{OL} \downarrow$



三、输出特性



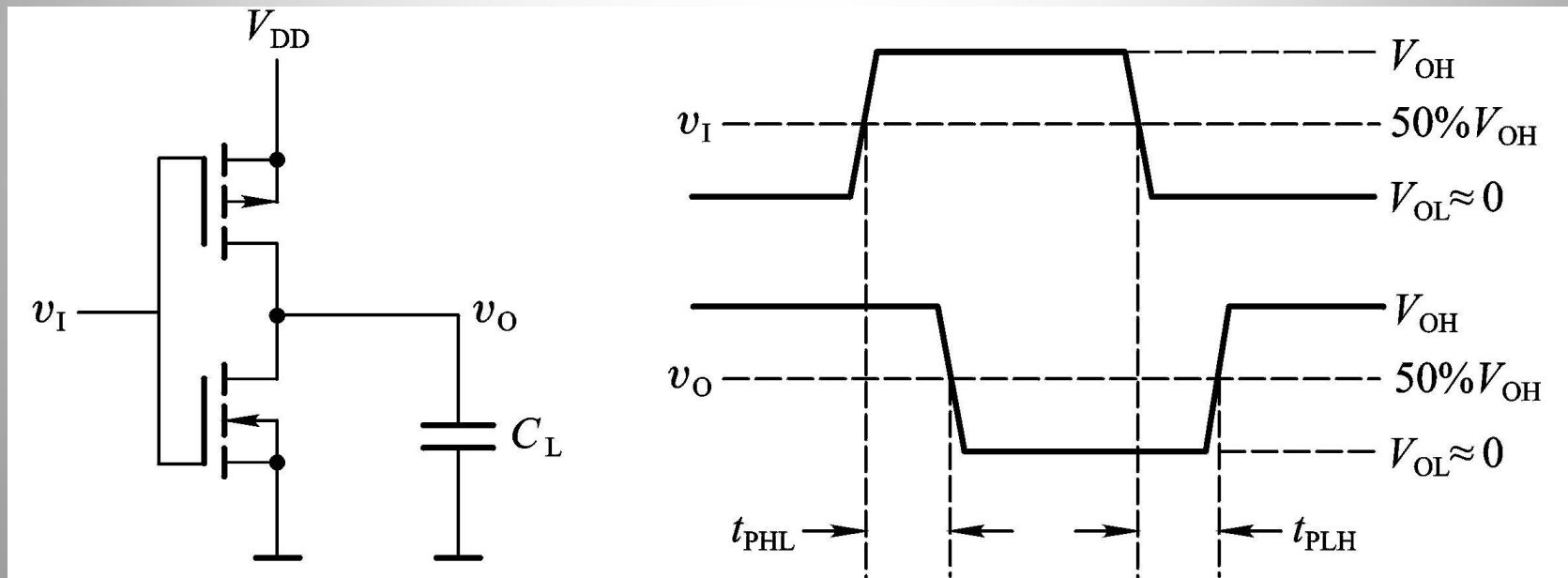
1. 高电平输出特性 $V_{OH} = f(I_{OH})$

同样的 I_{OH} 下, $|V_{GS}| \uparrow \Rightarrow V_{OH} \downarrow$ 越少

3.3.4 CMOS反相器的动态特性

一、传输延迟时间

- 1.原因： C_I 和 C_L 充放电，因为 R_{ON} 较大所以 C_L 充放电影响也较大；
2. t_{PHL} , t_{PLH} 受 C_L 、 V_{DD} 影响；
3. $t_{PHL} \approx t_{PLH}$ 74HC系列为0ns, 74AHC系列为5ns。



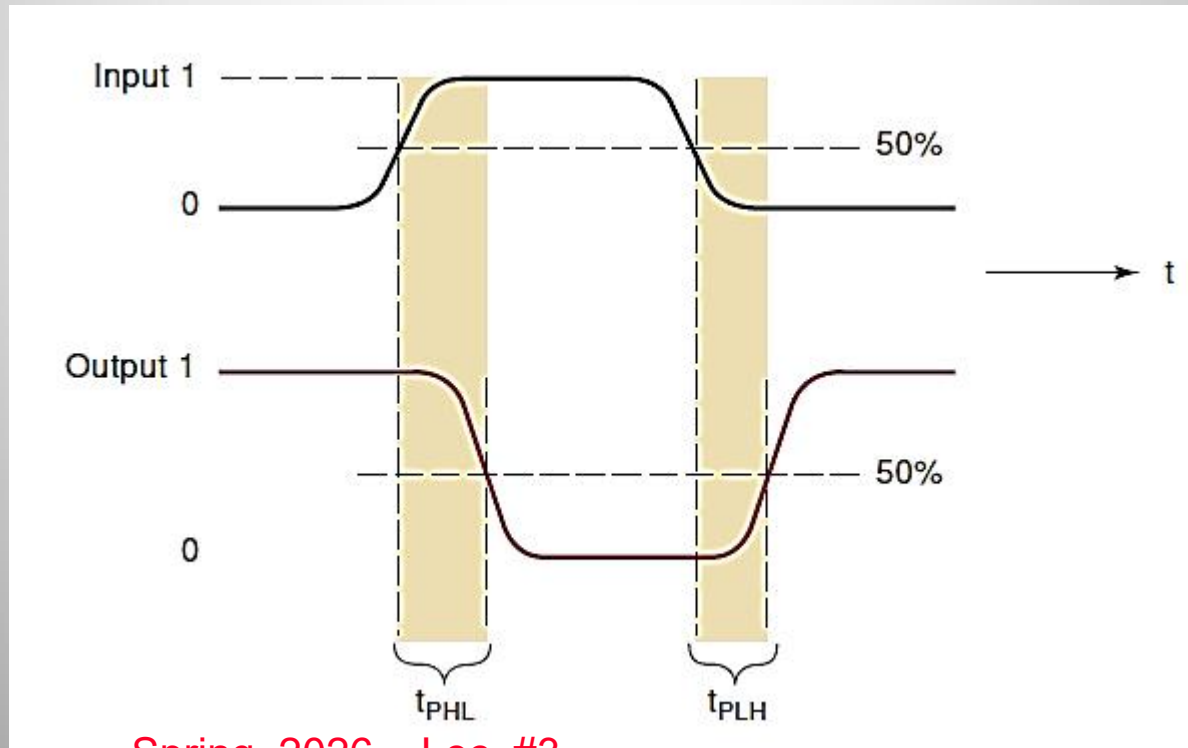
Propagation Delays

- t_{PLH}

- Delay time in going from logical 0 to logical 1 state (LOW to HIGH)

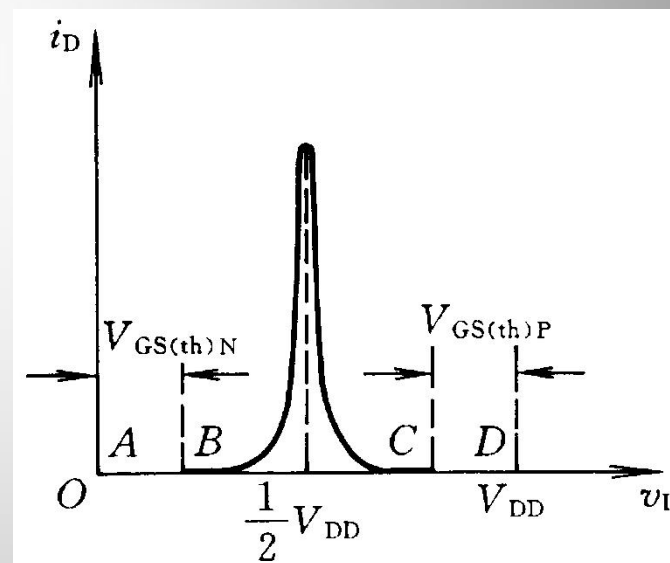
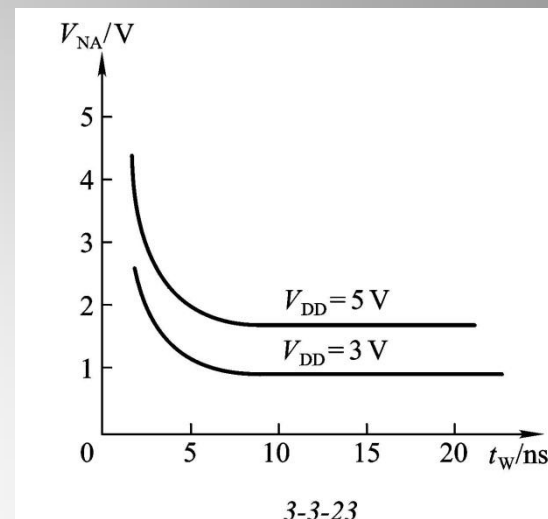
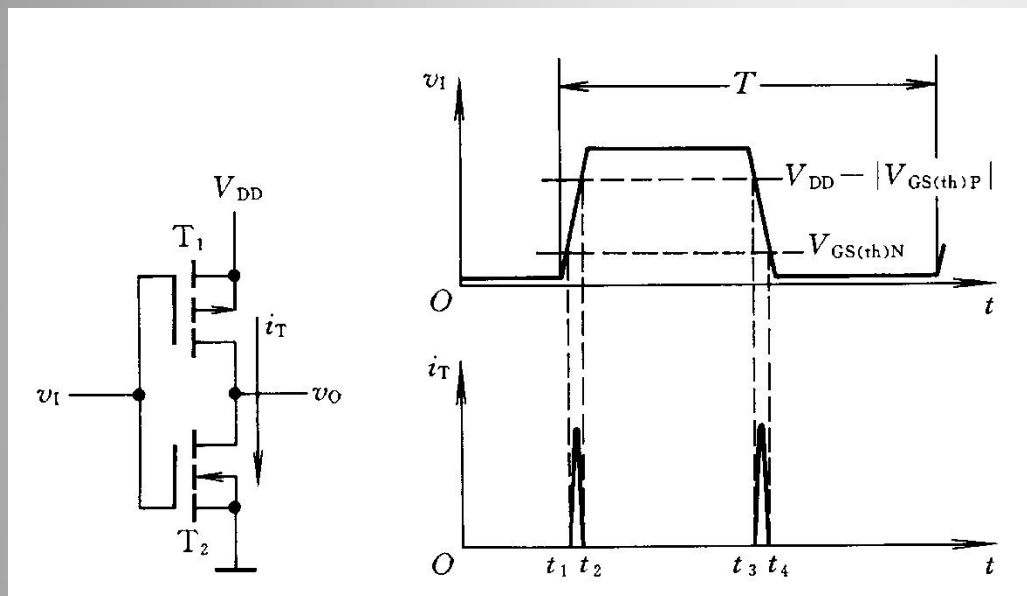
- t_{PHL}

- Delay time in going from logical 1 to logical 0 state (HIGH to LOW)



二、交流噪声容限

三、动态功耗

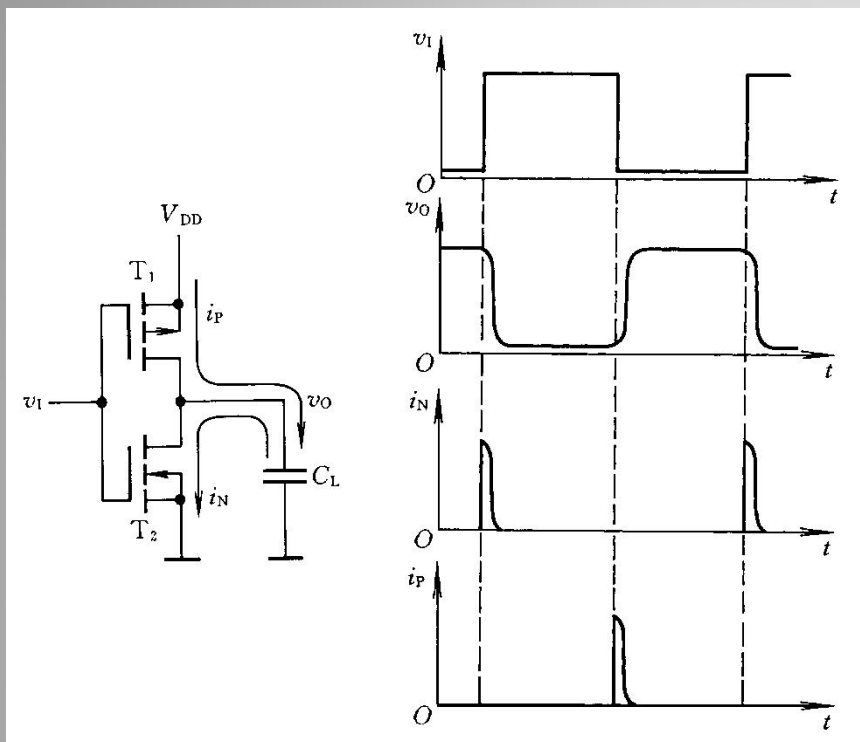


1. 导通功耗

$$P_T = V_{DD} I_{TAV}, \text{ 其中 } I_{TAV} = \frac{1}{T} \left(\int_{t_1}^{t_2} i_T dt + \int_{t_3}^{t_4} i_T dt \right)$$

静态功耗极小，与动态功耗相比，可以忽略

三、动态功耗



2. 负载电容充放电功耗 P_C

* 当 $V_I \downarrow$, V_{DD} 经 T_1 向 C_L 充电, 有 i_P

* 当 $V_I \uparrow$, C_L 经 T_2 放电, 有 i_N

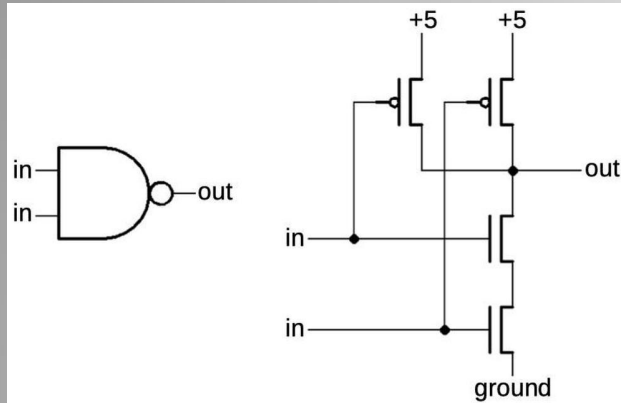
可得平均功耗

$$P_C = C_L f V_{DD}^2$$

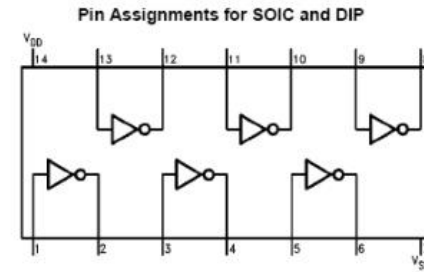
3. 总的动态功耗

$$P_D = P_T + P_C$$

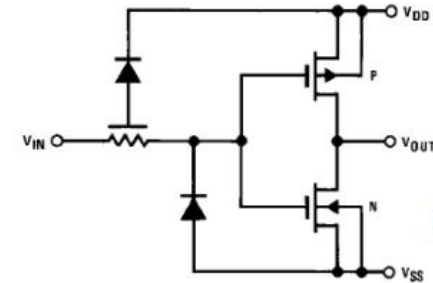
CD4069



Connection Diagram



Schematic Diagram



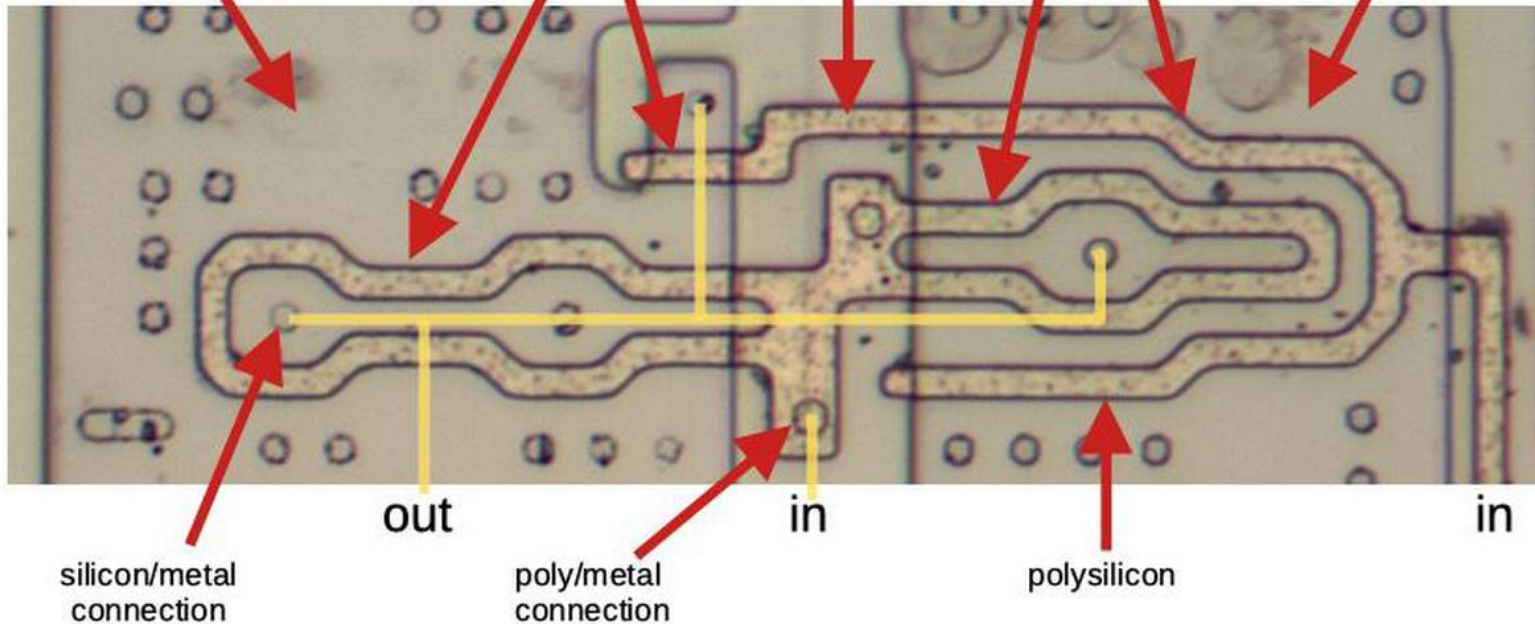
source region
(+5 V)

PMOS
transistors

polysilicon

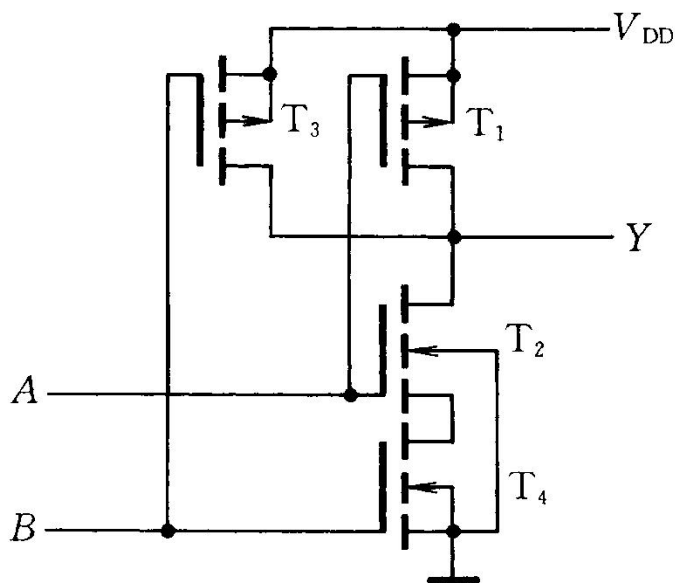
NMOS
transistors

source region
(ground)



带缓冲级的CMOS门

1、与非门



存在的缺点:

(1): 输出电阻 R_o 受输入状态影响

$$A = 1, B = 1 \text{ 则 } R_o = R_{ON2} + R_{ON4} = 2R_{ON}$$

$$A = 0, B = 0 \text{ 则 } R_o = R_{ON1} // R_{ON3} = \frac{1}{2} R_{ON}$$

$$A = 0, B = 1 \text{ 则 } R_o = R_{ON1} = R_{ON}$$

$$A = 1, B = 0 \text{ 则 } R_o = R_{ON3} = R_{ON}$$

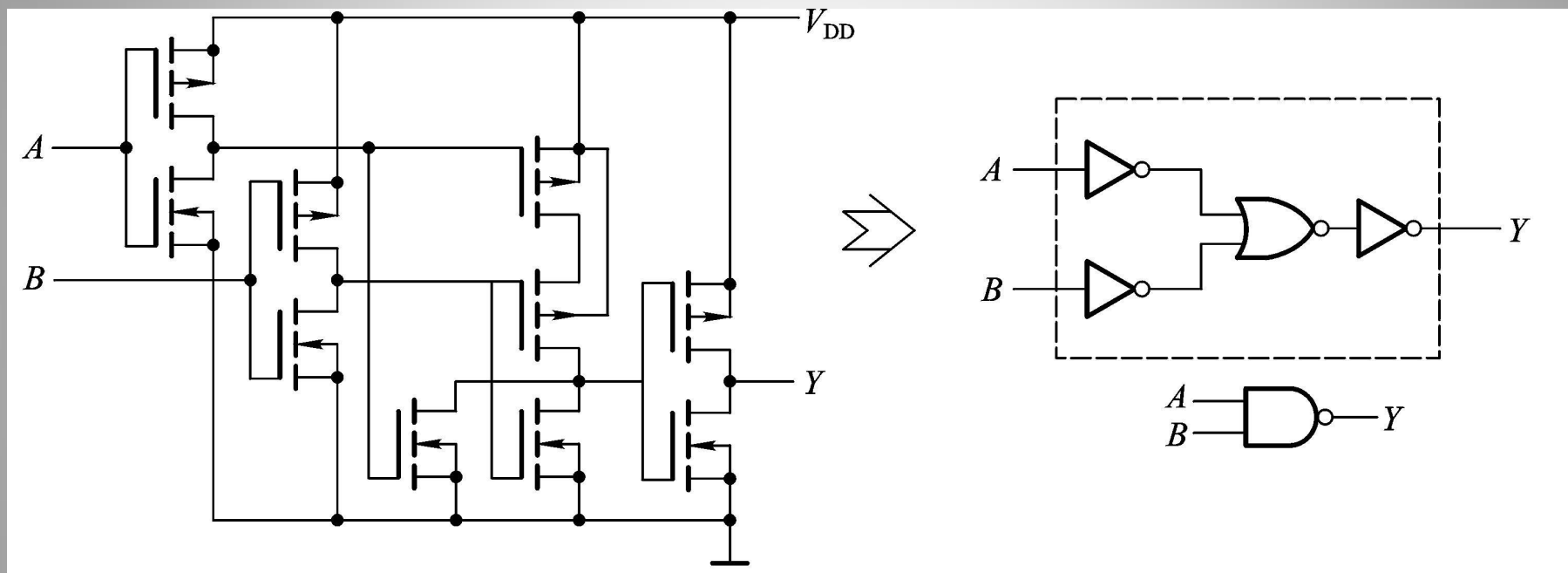
(2) 输出的高低电平受输入端数目的影响

输入端越多, V_{OL} 越高, V_{OH} 也更高

(3) 使 T_2 、 T_4 的 V_{GS} 达到开启电压时, 对应的 V_I 值不同

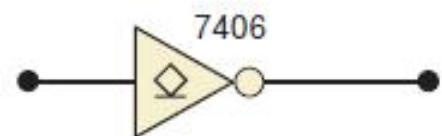
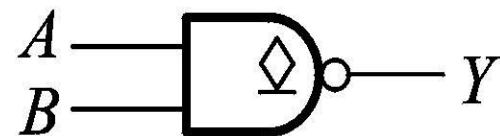
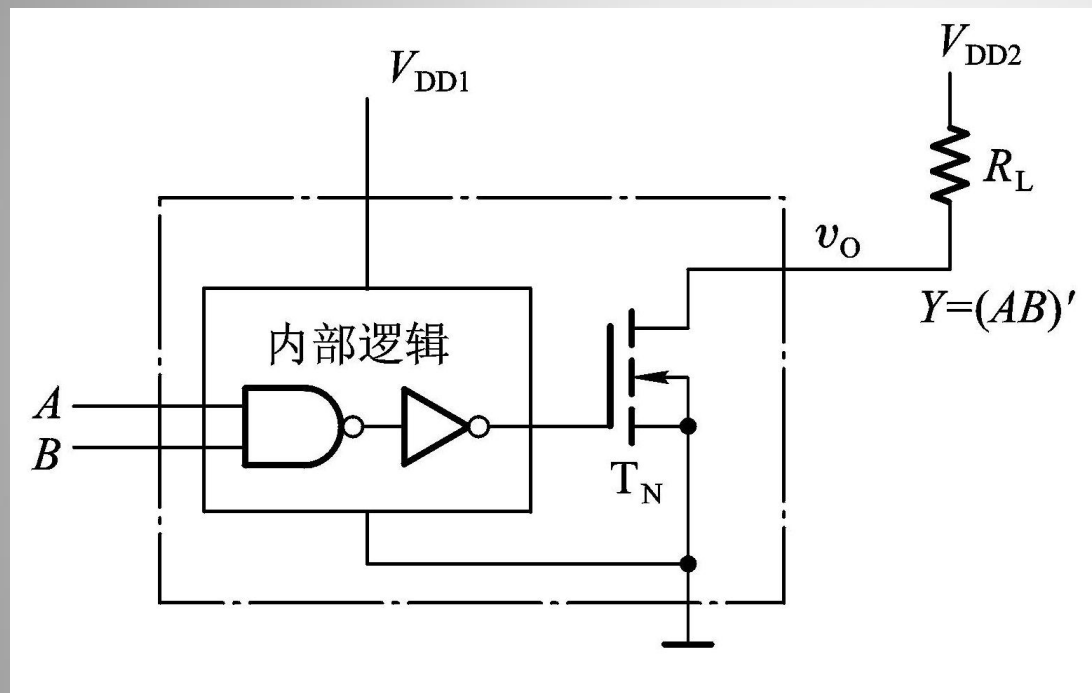
带缓冲级的CMOS门

2.解决方法



或非门+缓冲器 $\Rightarrow$ 与非门

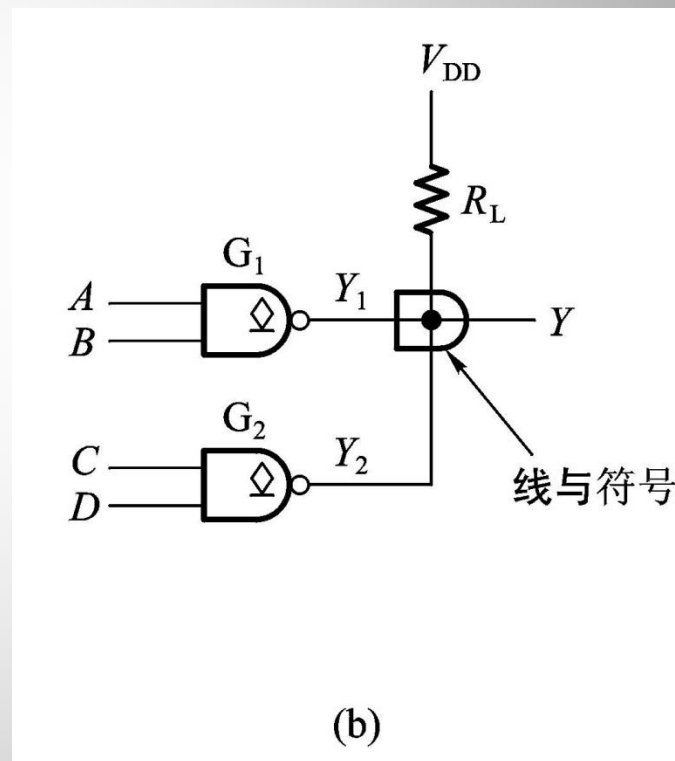
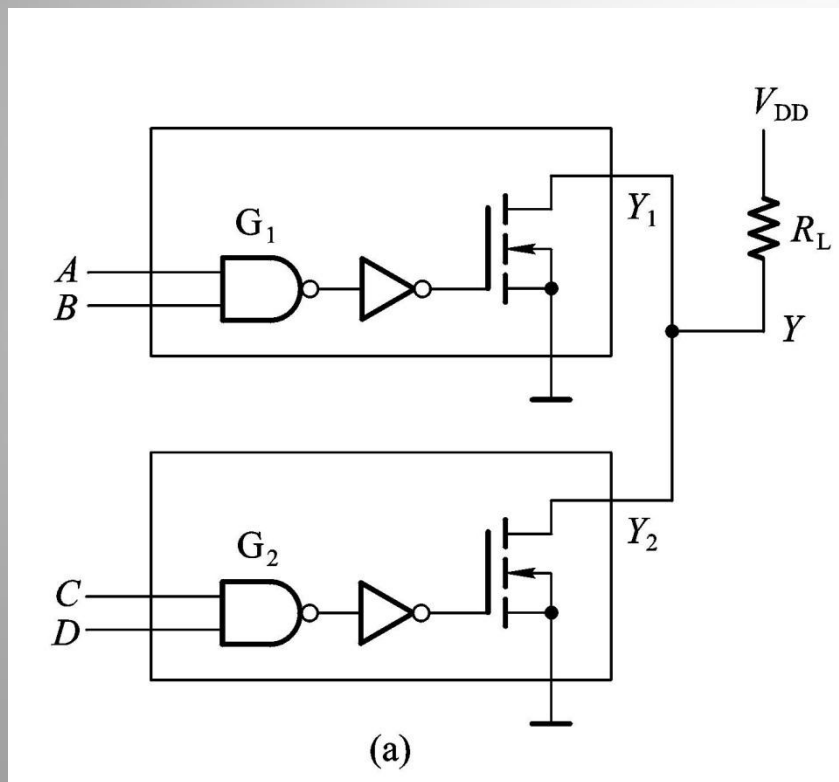
二、漏极开路的门电路（OD门）



(b)

1. 可将输出并联使用，实现线与
或用作电平转换、驱动器

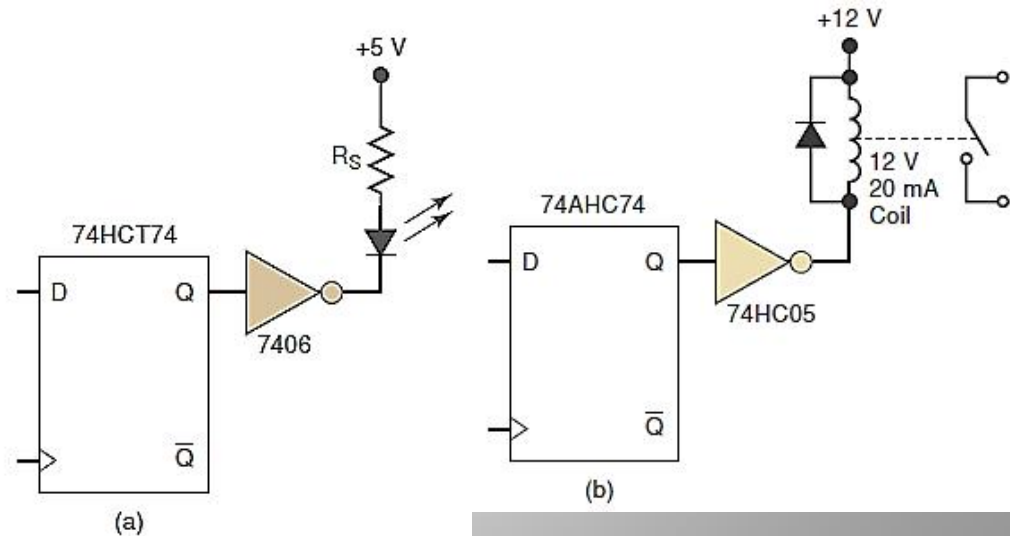
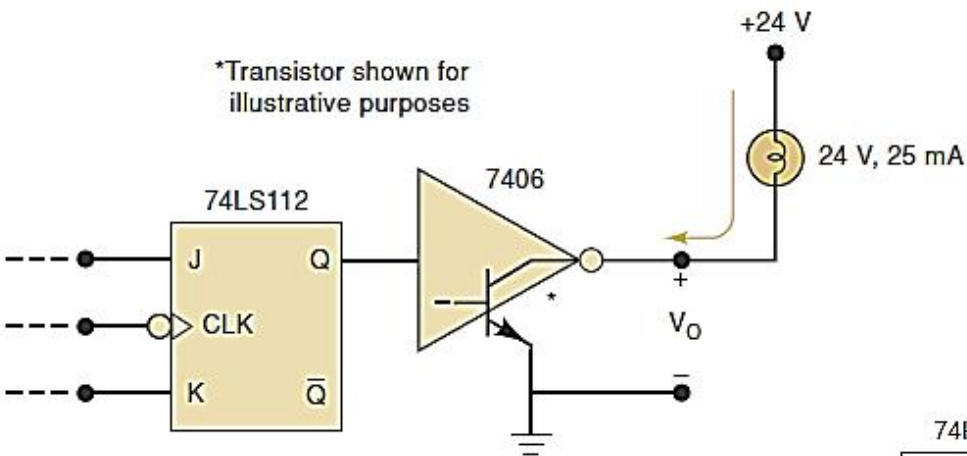
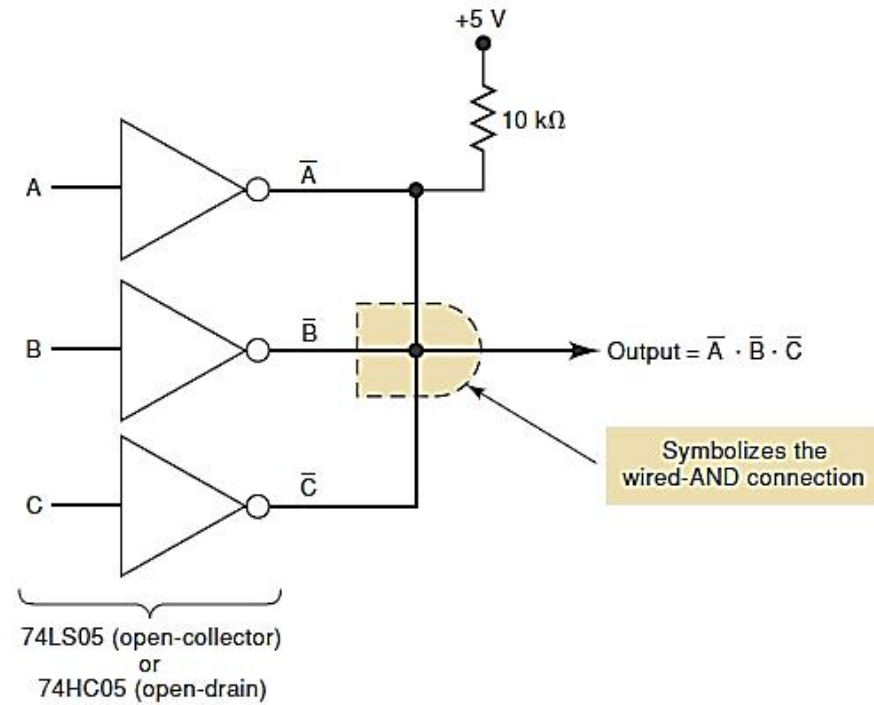
2. 使用时允许外接 R_L , V'_{DD} (V'_{DD} 可以不等于 V_{DD})



R_L 的计算方法

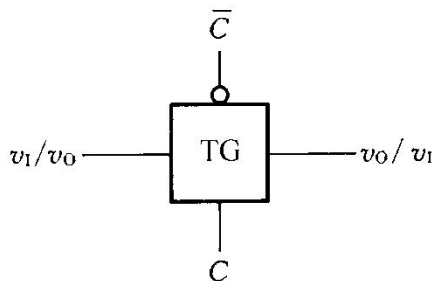
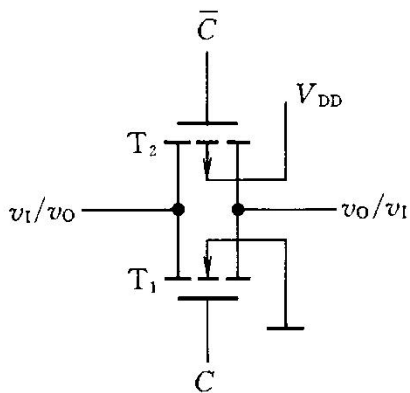
• RL的计算方法

- 前一级输出能力
- 后一级驱动要求

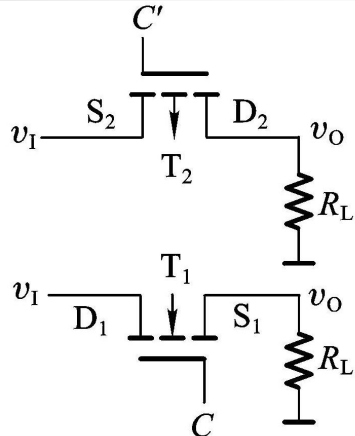
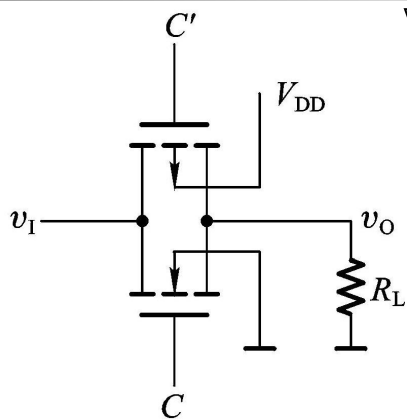


三、CMOS传输门及双向模拟开关

1. 传输门



设 V_I 为正, 另一端经 R_L 接地



设 $R_L \gg R_{ON}, V_{IH} = V_{DD}, V_{IL} = 0$

(1) 当 $C = 0, C' = 1$

则只要 $V_I = 0 \sim V_{DD}$, 则 T_1 、 T_2 均截止
相当于断开

(2) 当 $C = 1, C' = 0$

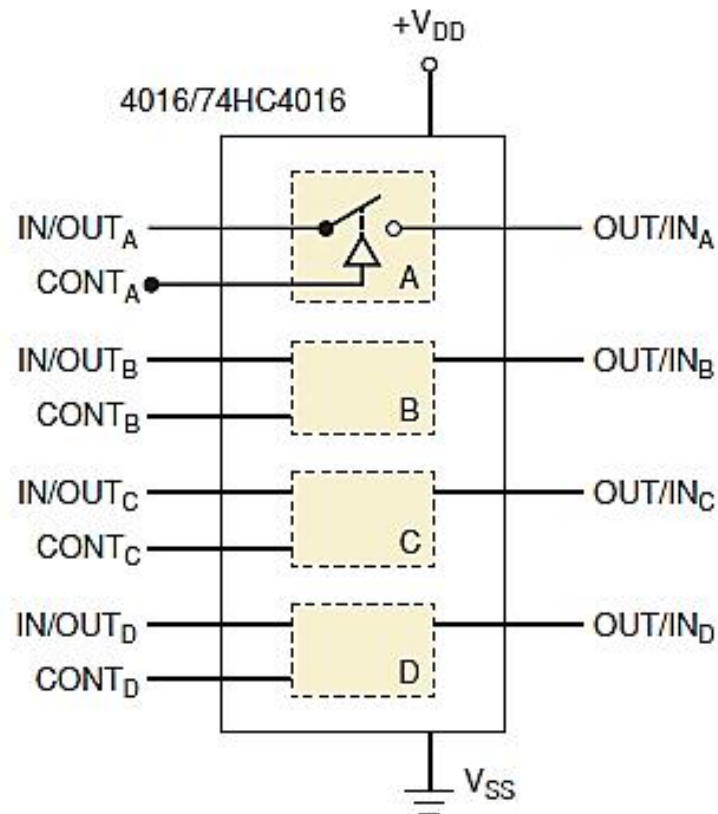
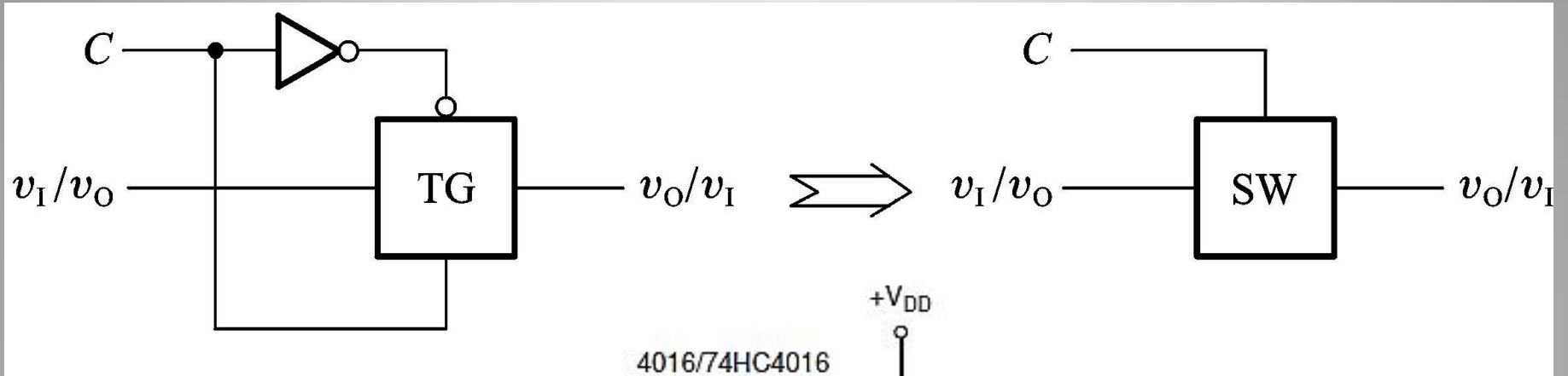
$V_I = 0 \sim V_{DD}$ 时

$0 < V_I < V_{DD} - V_{GS(th)N}, T_1$ 导通

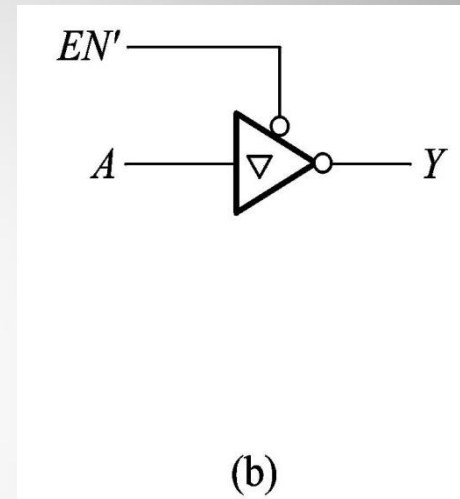
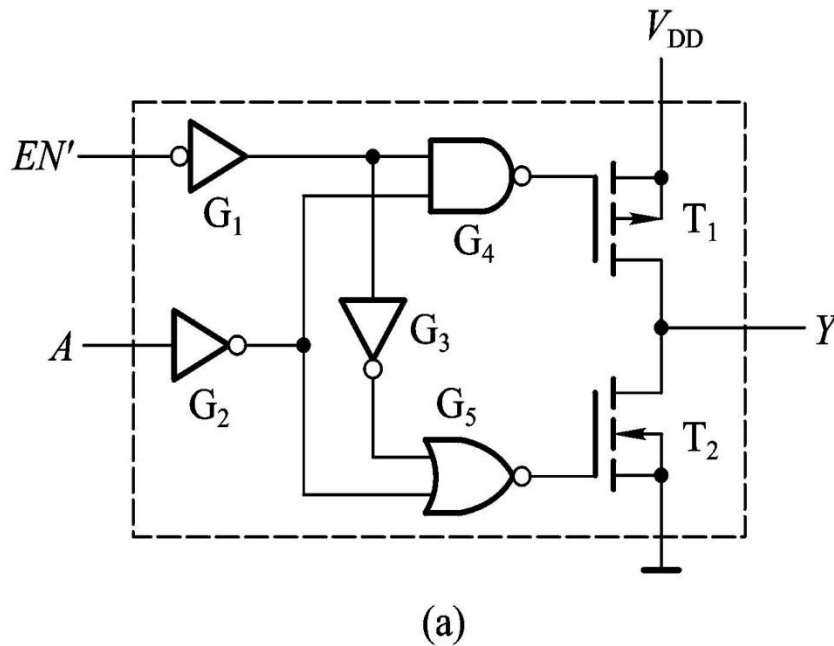
$|V_{GS(th)P}| < V_I < V_{DD}, T_2$ 导通

所以 V_I 在 $0 \sim V_{DD}$, T_1 和 T_2 至少一个导通
 $V_I - V_O$ 之间为低电阻

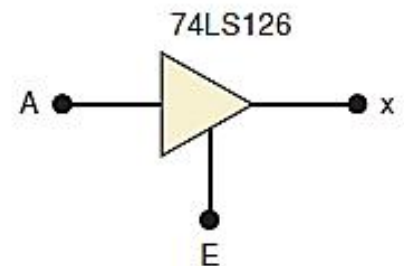
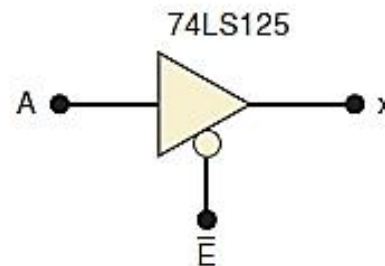
2. 双向模拟开关



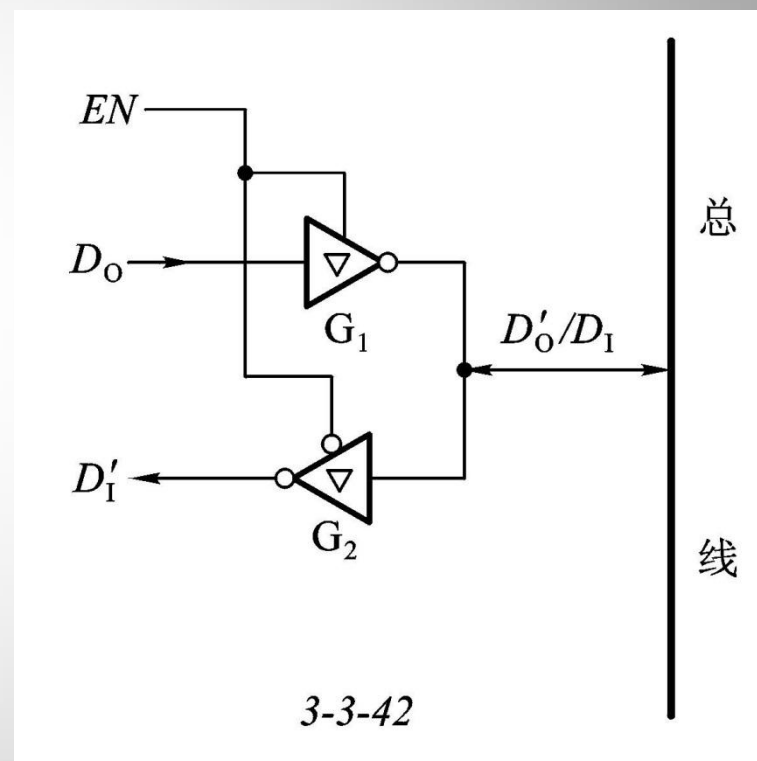
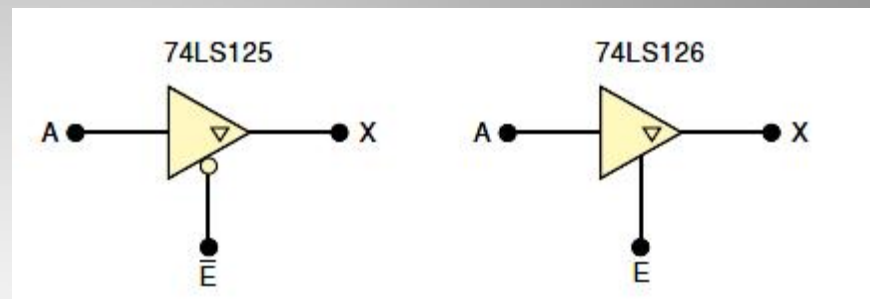
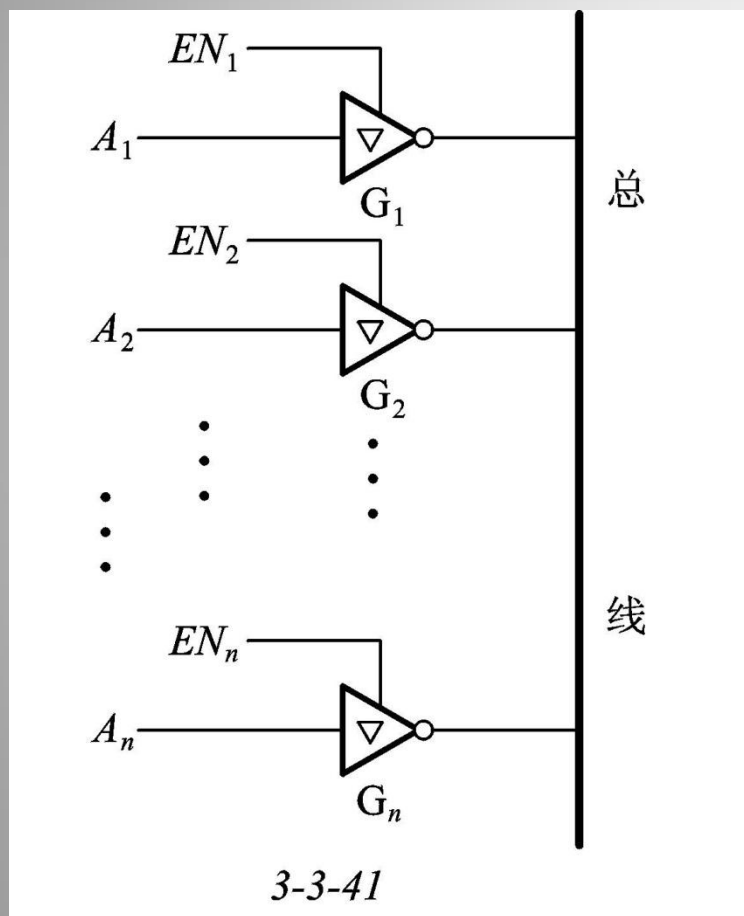
四、三态输出门



$EN' = 0$ 时, $Y = A'$
 $EN' = 1$ 时, $Y = Z$ (高阻)

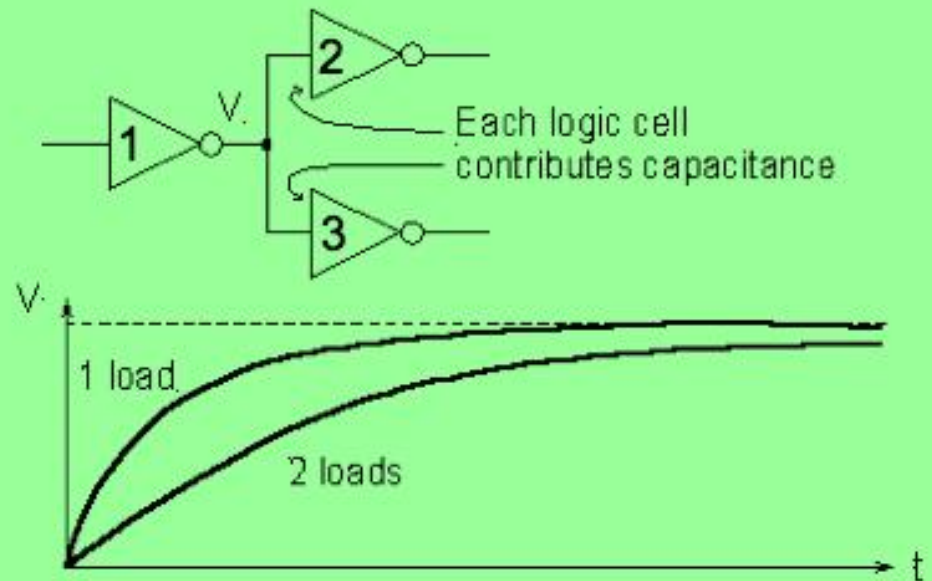


三态门的用途



Fan-Out扇出系数

- In general, a logic-circuit **output** is required to **drive** several logic **inputs**.
- The **fan-out** (also called *loading factor*) is defined as the **maximum number** of logic inputs that an output can drive reliably
- fan-out is assumed to refer to load devices of the **same family** as the driving output.



Unused Inputs

- Unused Inputs
 - **CMOS inputs** should **never be left** disconnected. All CMOS inputs must be tied either to a fixed voltage level (0 V or *VDD*) or to another input.
 - This rule applies even to the inputs of extra unused logic gates on a chip.
 - » An unconnected CMOS input is susceptible to noise and static charges that could easily bias both the P-channel and the N-channel MOSFETs in the conductive state, resulting in increased power dissipation and possible overheating.

CMOS SERIES CHARACTERISTICS

- **4000/14000 Series**

- The oldest CMOS series is the 4000 series first introduced by RCA,
 - » CD4001
- and its functionally equivalent 14000 series from Motorola.
 - » MC14001

- **74HC/HCT (High-Speed CMOS)**

- 74HC/HCT ICs are pin-compatible with and functionally equivalent to TTL ICs with the same device number.
 - » 74HC00
- 74HCT devices are electrically compatible with TTL
 - » 74HCT00

CMOS SERIES CHARACTERISTICS

- 74AC/ACT (**A**dvanced **C**MOS)
- 74AHC/AHCT (**A**dvanced **H**igh-Speed **C**MOS)
- **B**iCMOS 5-V Logic/74**B**CT
- The 74ABT (advanced BiCMOS technology) s

TABLE 8-9 Input/output voltage levels (in volts) with $V_{DD} = V_{CC} = +5\text{ V}$.

Parameter	CMOS							TTL			
	4000B	74HC	74HCT	74AC	74ACT	74AHC	74AHCT	74	74LS	74AS	74ALS
$V_{IH}(\text{min})$	3.5	3.5	2.0	3.5	2.0	3.85	2.0	2.0	2.0	2.0	2.0
$V_{IL}(\text{max})$	1.5	1.0	0.8	1.5	0.8	1.65	0.8	0.8	0.8	0.8	0.8
$V_{OH}(\text{min})$	4.95	4.9	4.9	4.9	4.9	4.4	3.15	2.4	2.7	2.7	2.5
$V_{OL}(\text{max})$	0.05	0.1	0.1	0.1	0.1	0.44	0.1	0.4	0.5	0.5	0.5
V_{NH}	1.45	1.4	2.9	1.4	2.9	0.55	1.15	0.4	0.7	0.7	0.7
V_{NL}	1.45	0.9	0.7	1.4	0.7	1.21	0.7	0.4	0.3	0.3	0.4

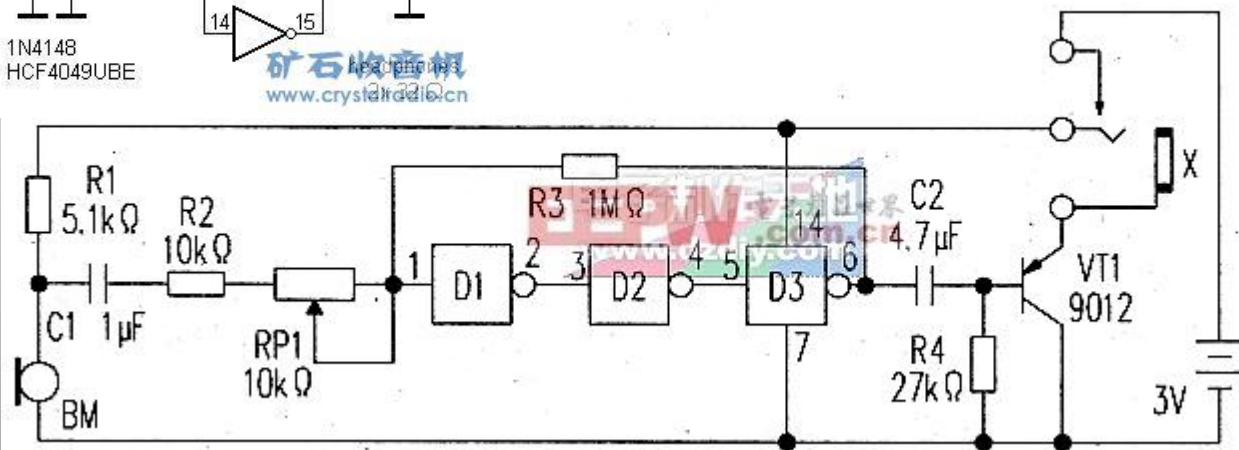
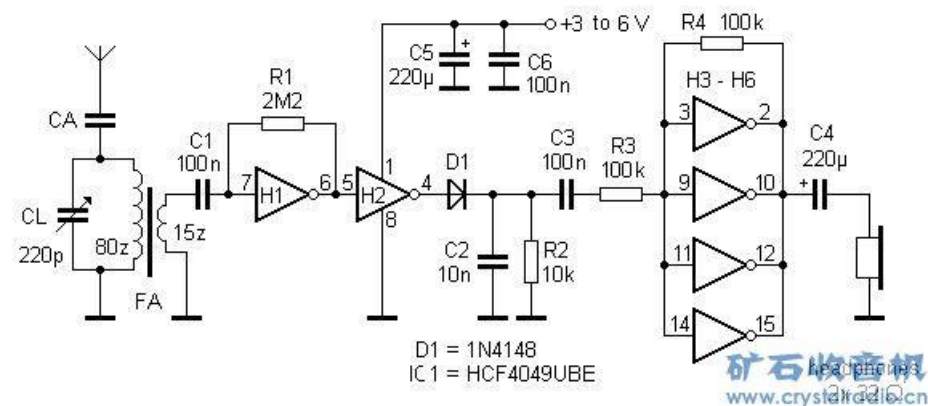
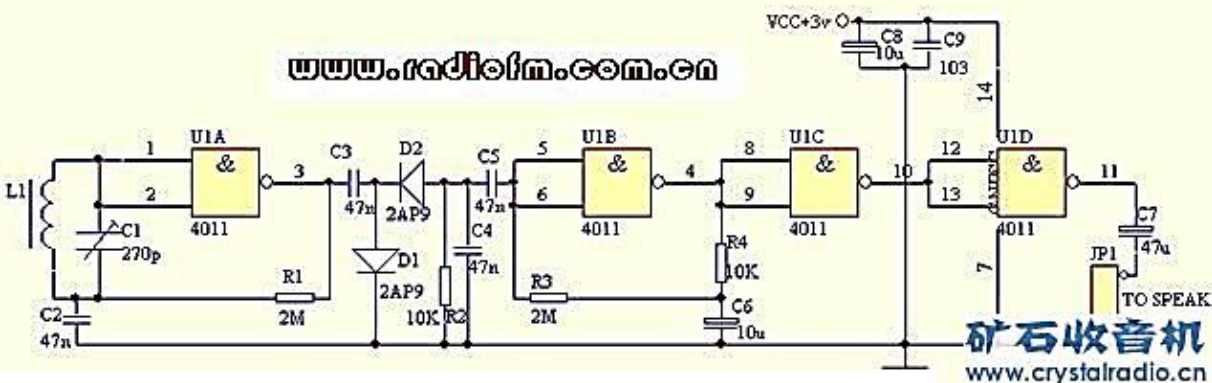
LOW-VOLTAGE TECHNOLOGY

- **CMOS Family**

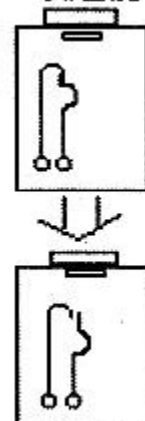
- The 74LVC (*Low-Voltage CMOS*) series
- The 74ALVC (*Advanced Low-Voltage CMOS*) series
- The 74LV (*Low-Voltage*) series
 - » It is intended to operate only with other 3.3-V devices.
- The 74AVC (*Advanced Very-Low-Voltage CMOS*)
 - » as low as 1.2 V or as high as 3.3 V.
- The 74AUC (*Advanced Ultra-Low-Voltage CMOS*)
 - » is optimized to operate at 1.8-V logic levels.
- The 74AUP (*Advanced Ultra-low Power*) series
- The 74CBT (*Cross Bar Technology*) series
- The 74CBTLV (*Cross Bar Technology Low Voltage*)
- The 74GTLP (*Gunning Transceiver Logic Plus*) series
- The 74SSTV (*Stub Series Terminated Logic*)

用CD4069数字集成电路搭了个收音机、助听器

CMOS Inverter as Analog Circuit: An Overview



改造前



改造后

三、TTL门电路

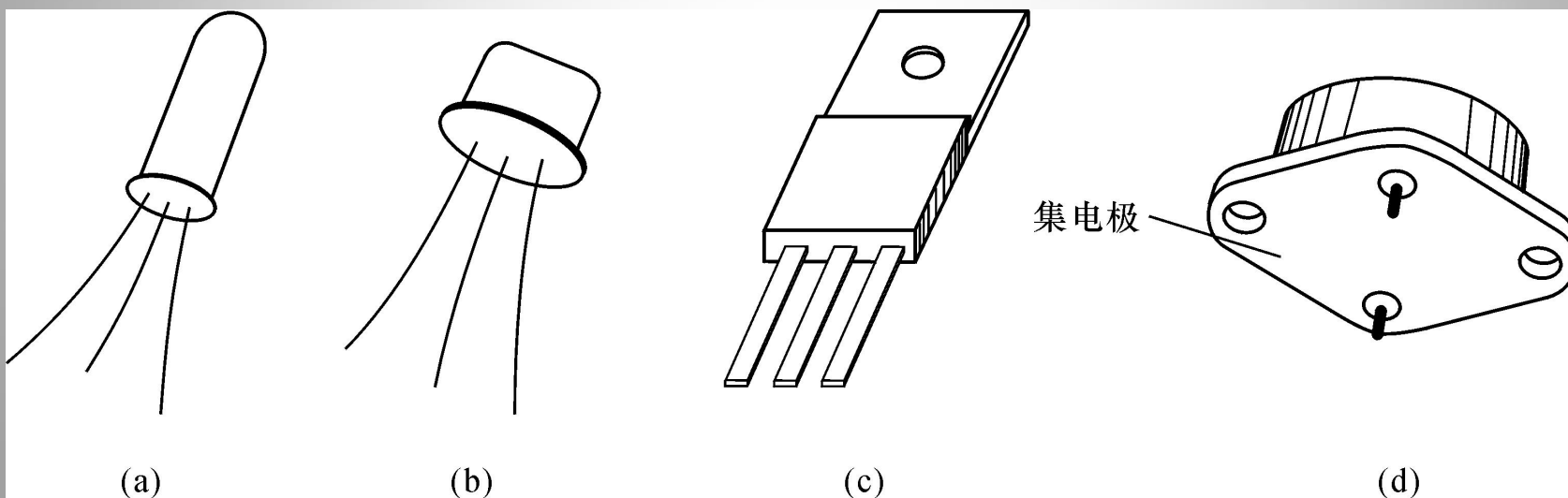
半导体三极管的开关特性

双极型三极管的开关特性

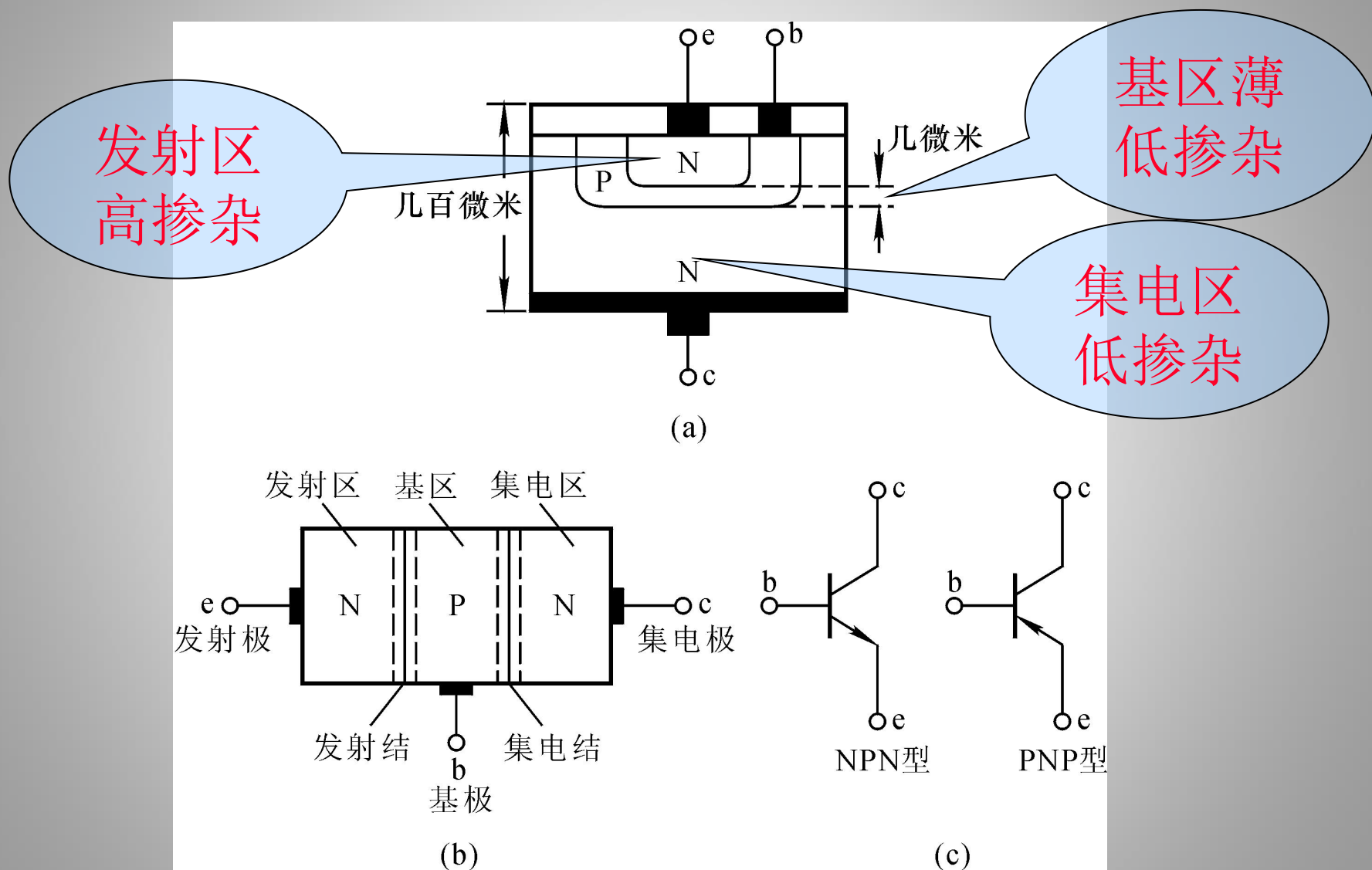
(BJT, Bipolar Junction Transistor)

一、双极型三极管的结构

管芯 + 三个引出电极 + 外壳

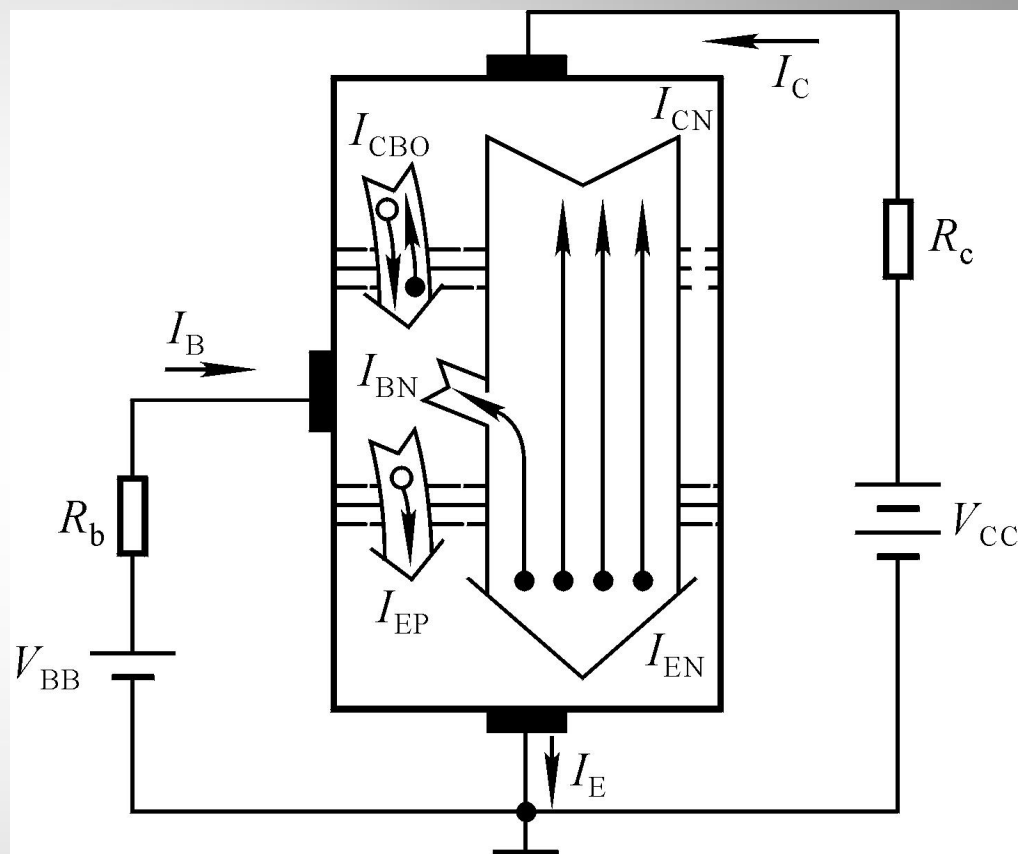


(BJT, Bipolar Junction Transistor)



以NPN为例说明工作原理:

- 当 $V_{CC} \gg V_{BB}$
 - be 结正偏, bc 结反偏
 - e 区发射大量的电子
 - b 区薄, 只有少量的空穴
 - bc 反偏, 大量电子形成 I_C



二、三极管的输入特性和输出特性

三极管的输入特性曲线 (NPN)

□ V_{ON} : 开启电压

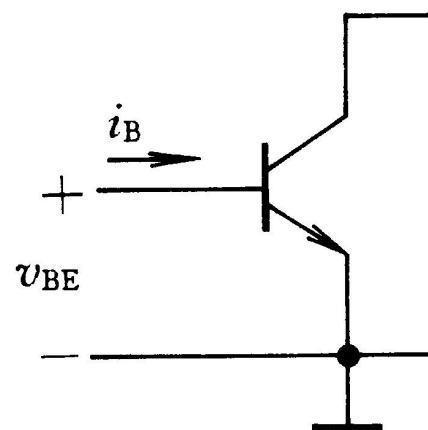
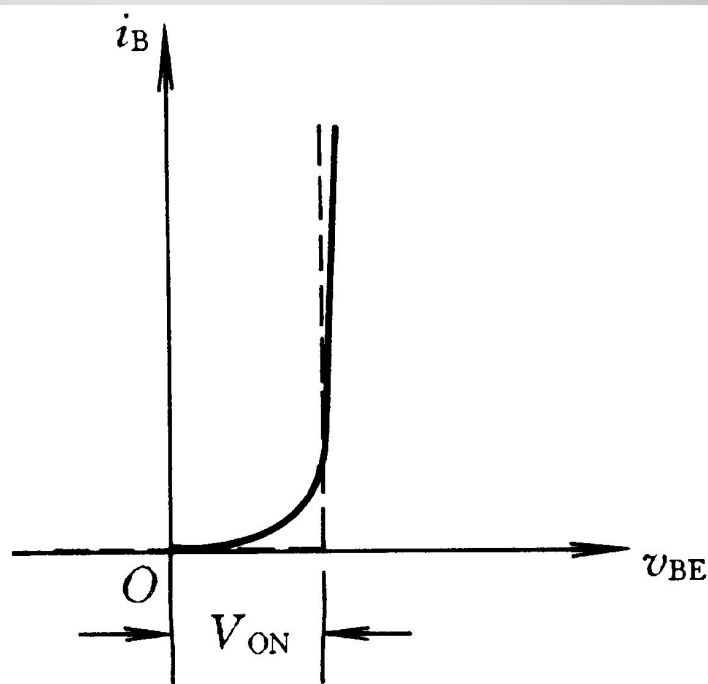
□ 硅管, $0.5 \sim 0.7V$

□ 锗管, $0.2 \sim 0.3V$

□ 近似认为:

□ $V_{BE} < V_{ON}$ $i_B = 0$

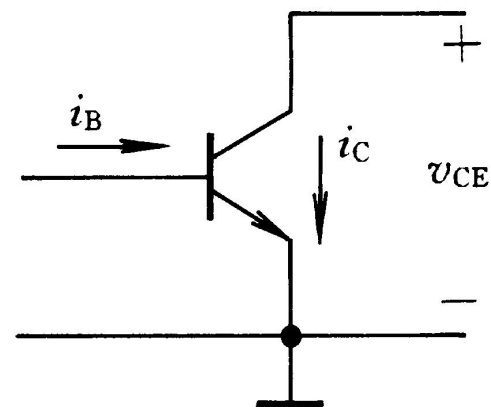
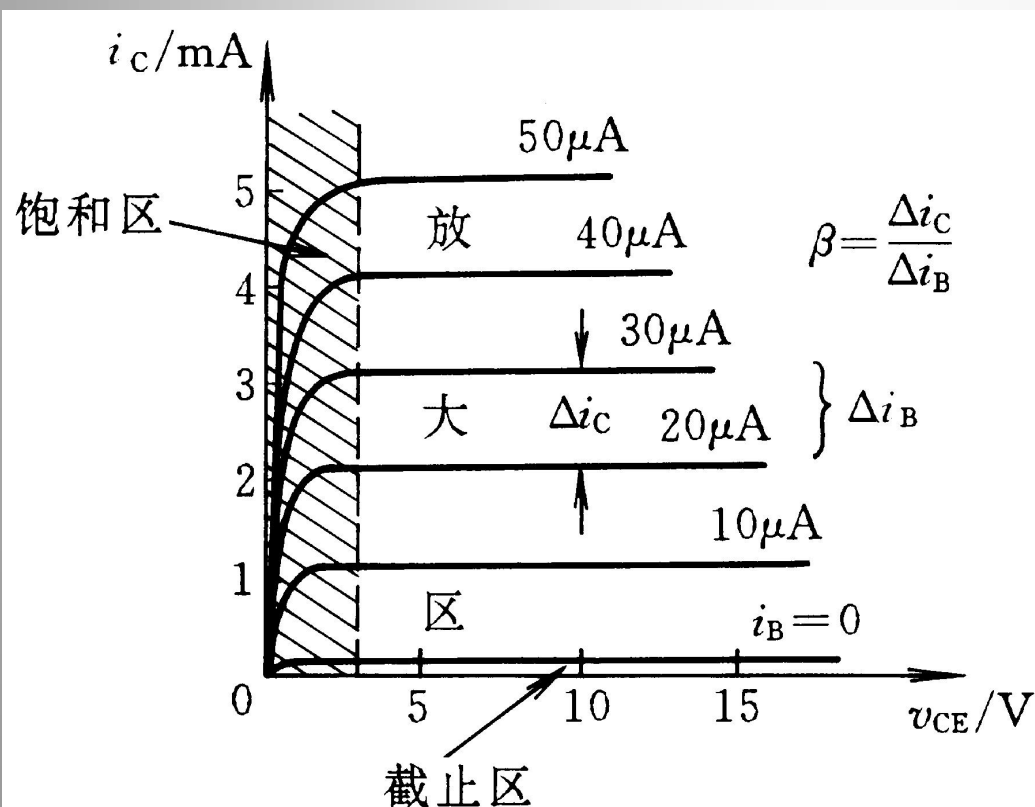
□ $V_{BE} \geq V_{ON}$ i_B 的大小由外电路电压, 电阻决定^(a)



$$i_B = \frac{V_{BB} - V_{BE}}{R_b}$$

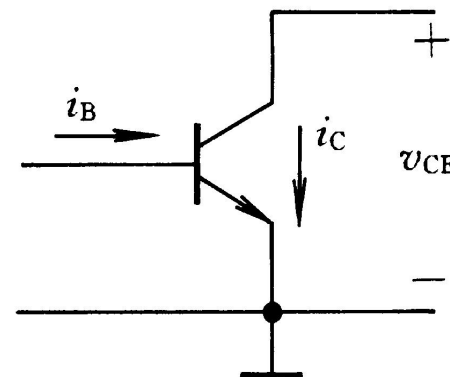
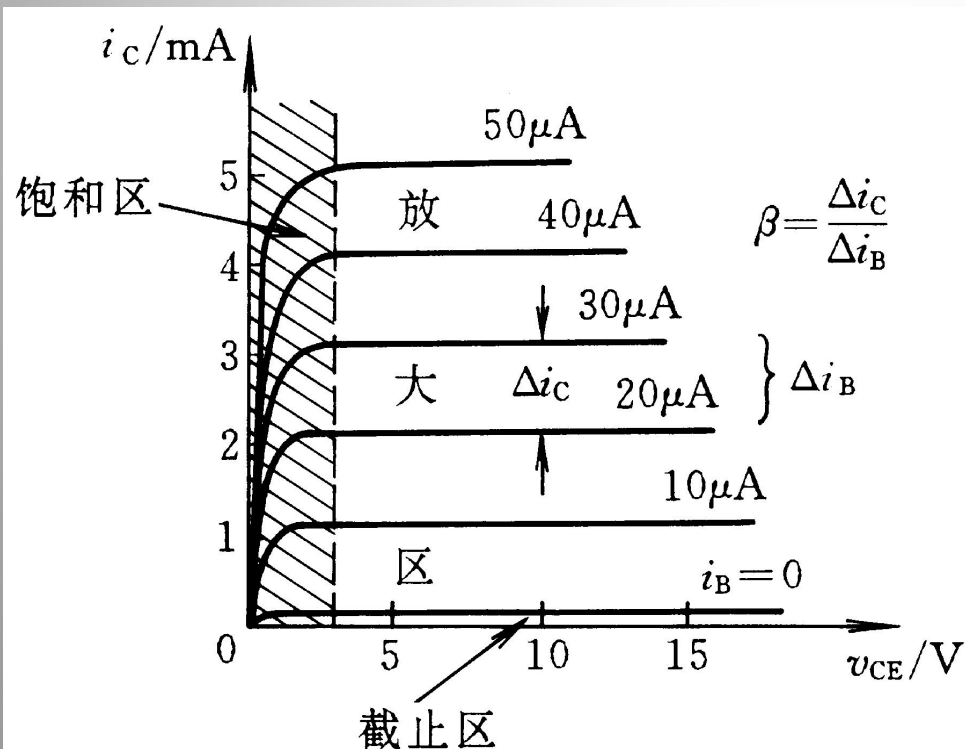
三极管的输出特性

- 固定一个 I_B 值，即得一条曲线，
在 $V_{CE} > 0.7V$ 以后，基本为水平直线 $i_C = f(V_{CE})$



特性曲线分三个部分

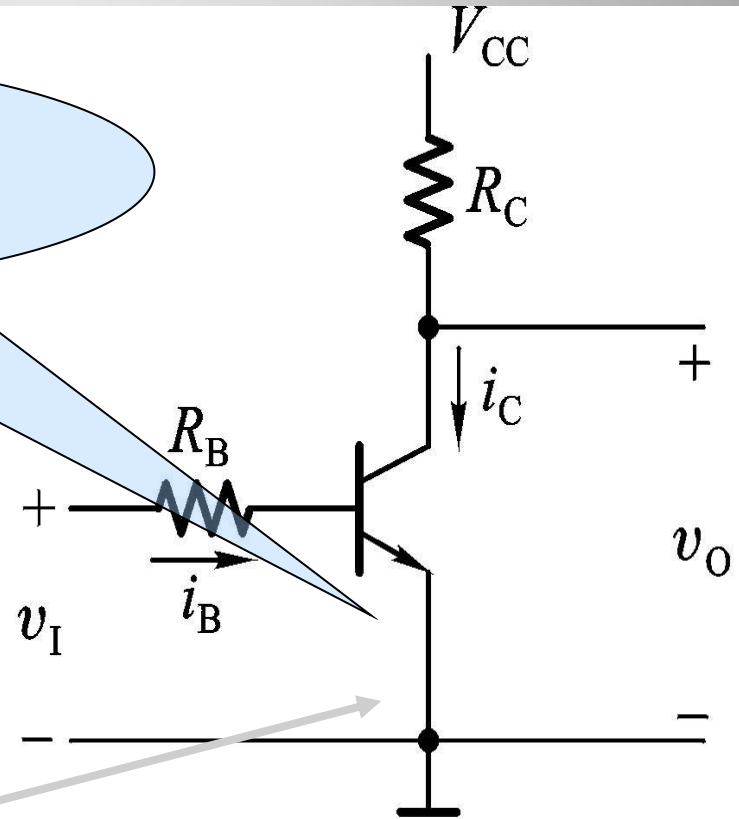
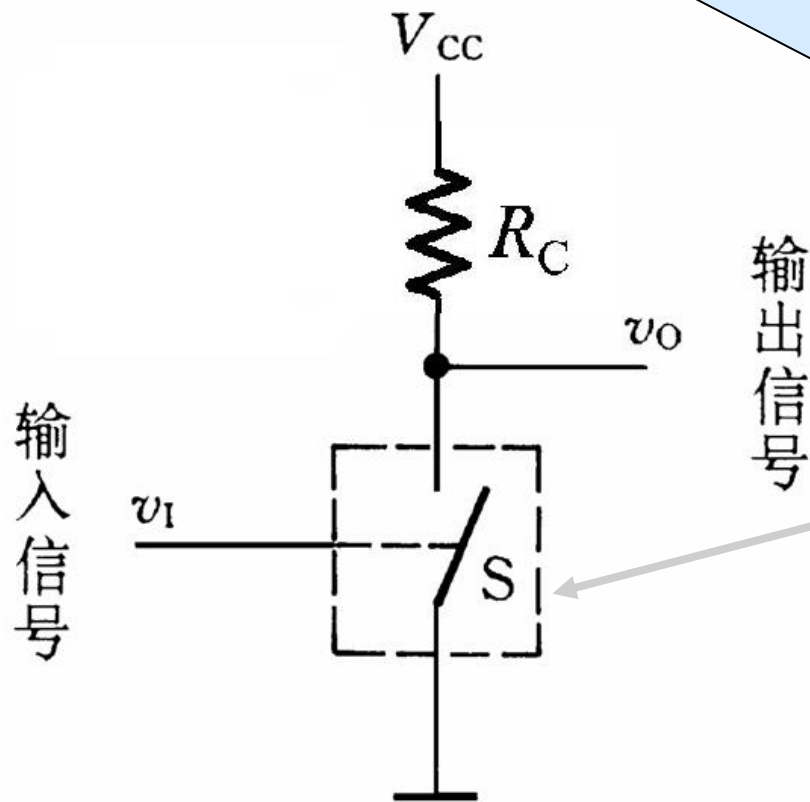
- ① 放大区：条件 $V_{CE} > 0.7V$, $i_B > 0$, i_C 随 i_B 成正比变化, $\Delta i_C = \beta \Delta i_B$ 。
- ② 饱和区：条件 $V_{CE} < 0.7V$, $i_B > 0$, V_{CE} 很低, Δi_C 随 Δi_B 增加变缓, 趋于“饱和”。
- ③ 截止区：条件 $V_{BE} = 0V$, $i_B = 0$, $i_C = 0$, c—e 间“断开”。



$$i_C = f(v_{CE})$$

三、双极型三极管的基本开关电路

只要参数合理：
 $V_I = V_{IL}$ 时，T截止， $V_O = V_{OH}$
 $V_I = V_{IH}$ 时，T导通， $V_O = V_{OL}$



工作状态分析:

(1) 设 $V_I = V_{IL} = 0$, 则 $V_{BE} < V_{ON}$ T 截止, i_B, i_C 近似为0。

(2) 当 V_I 上升至 $> V_{ON}$ 后, 有 i_B 产生, $i_B = \frac{V_I - V_{ON}}{R_B}$, 并有对应的 $i_C = \beta i_B$ 流过 R_C

于是得到 $V_O = V_{CE} = V_{CC} - i_C R_C = V_{CC} - \beta i_B R_C$ 。

所以 $V_I \uparrow \rightarrow i_B \uparrow \rightarrow i_C \uparrow \rightarrow V_O \downarrow$, 三极管工作在放大区

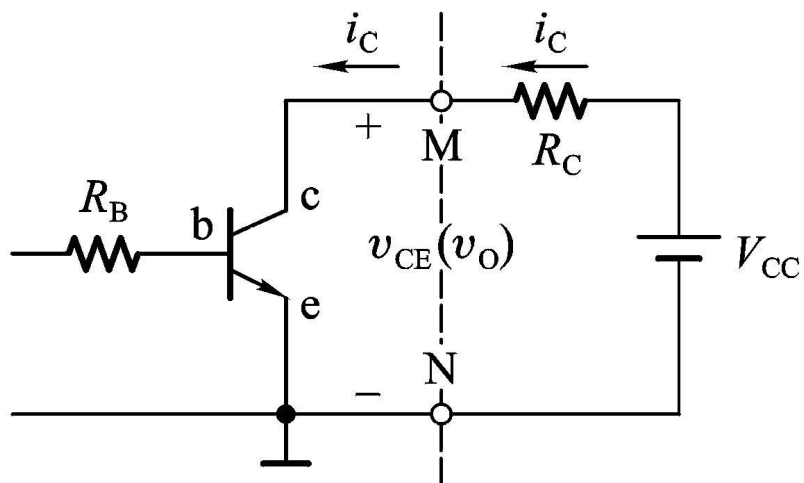
$$A_V = - \frac{\Delta V_O}{\Delta V_I}$$

(3) 当 V_I 继续上升, i_B 继续上升, V_O 继续下降。

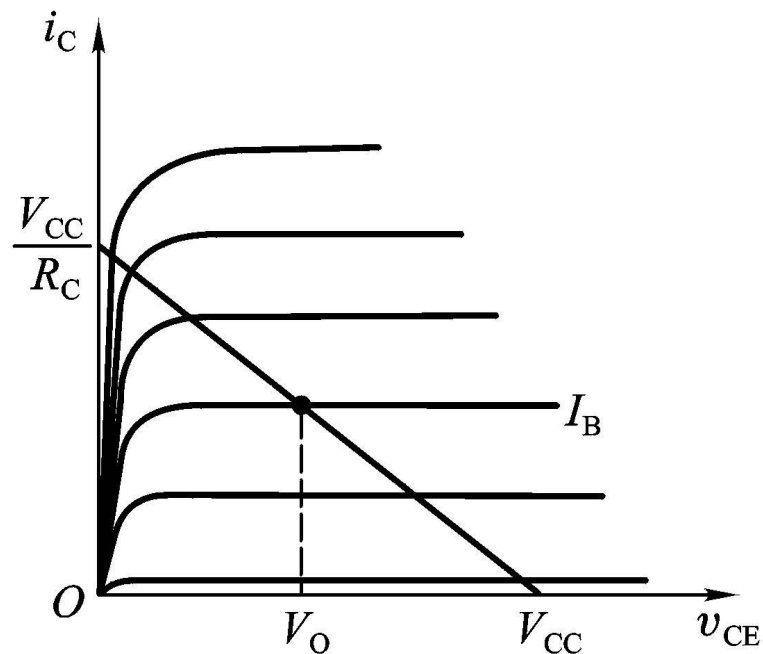
当 R_C 上压降接近于 V_{CC} 时, $V_O \approx 0$ 。

三极管工作在深饱和状态 $V_O = V_{OL} = V_{CE(sat)} \approx 0$ 。

图解分析法:



(a)



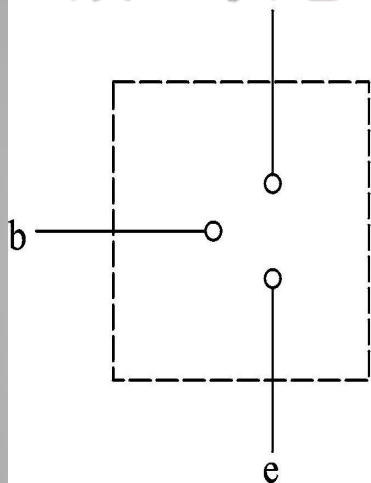
(b)

$$V_{CE} = V_{CC} - R_C i_C = V_{CC} - R_C \beta i_B$$

$$V_{RC} = R_C i_C$$

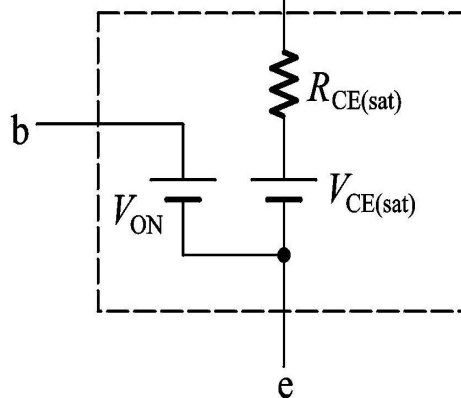
四、三极管的开关等效电路

截止状态

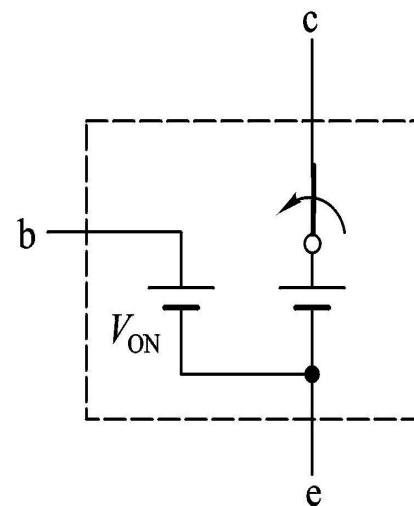


(a)

饱和导通状态

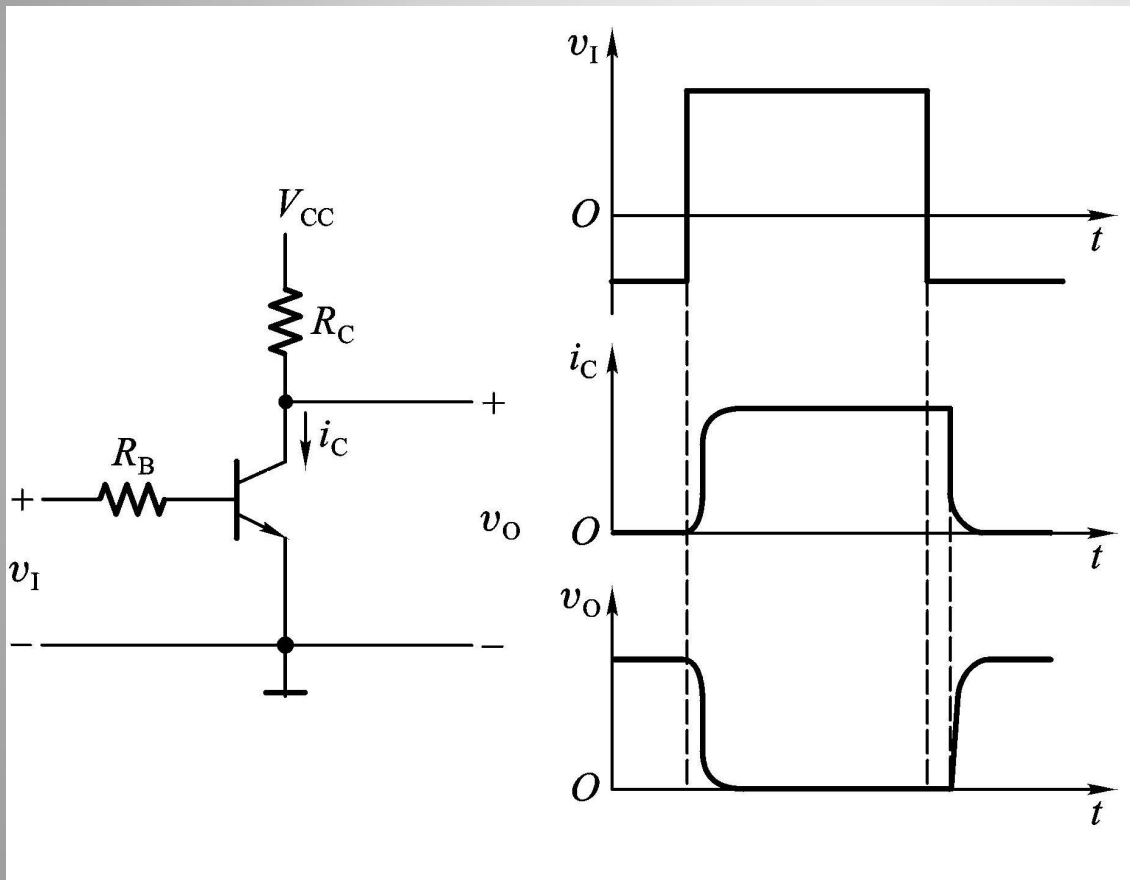


(b)



(c)

五、动态开关特性



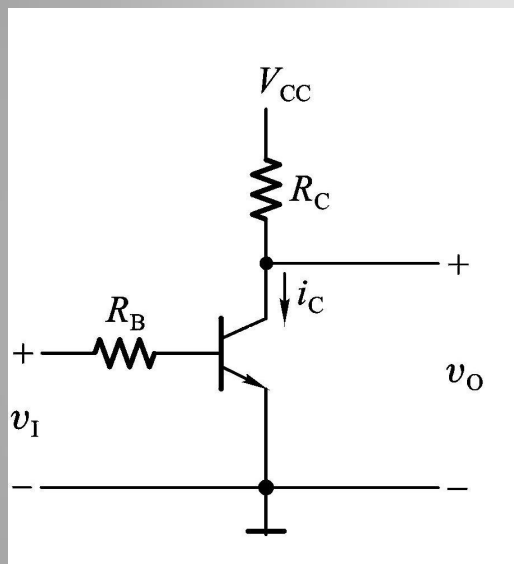
从二极管已知，PN结存在电容效应。

在饱和与截止两个状态之间转换时， i_C 的变化将滞后于 V_I ，则 V_O 的变化也滞后于 V_I 。

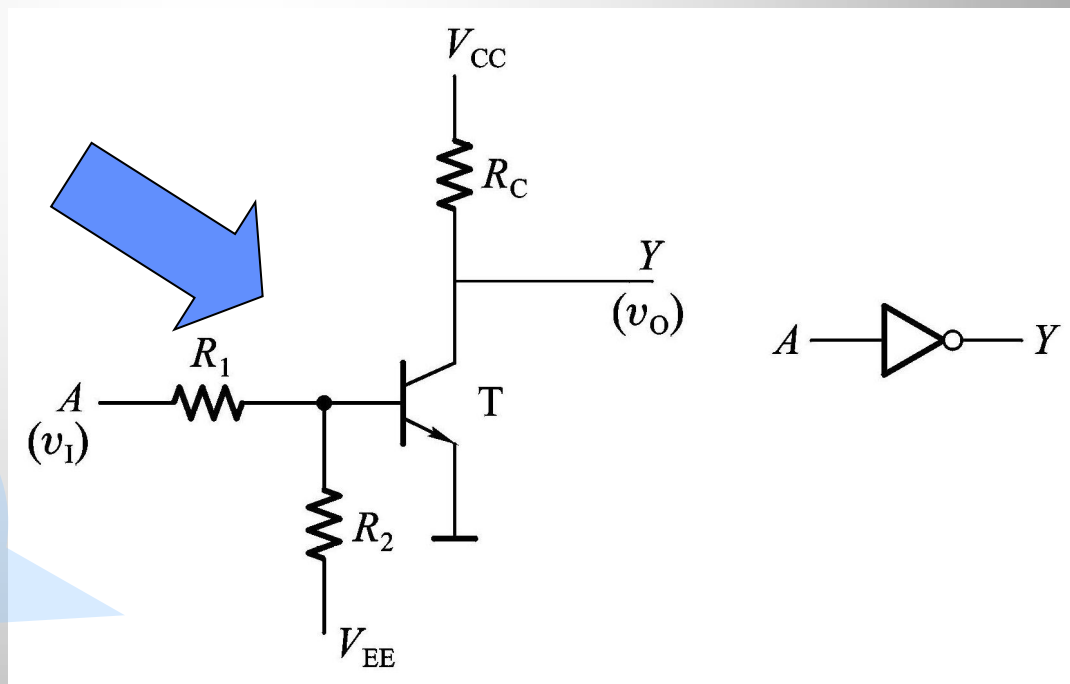
六、三极管反相器

- 三极管的基本开关电路就是非门

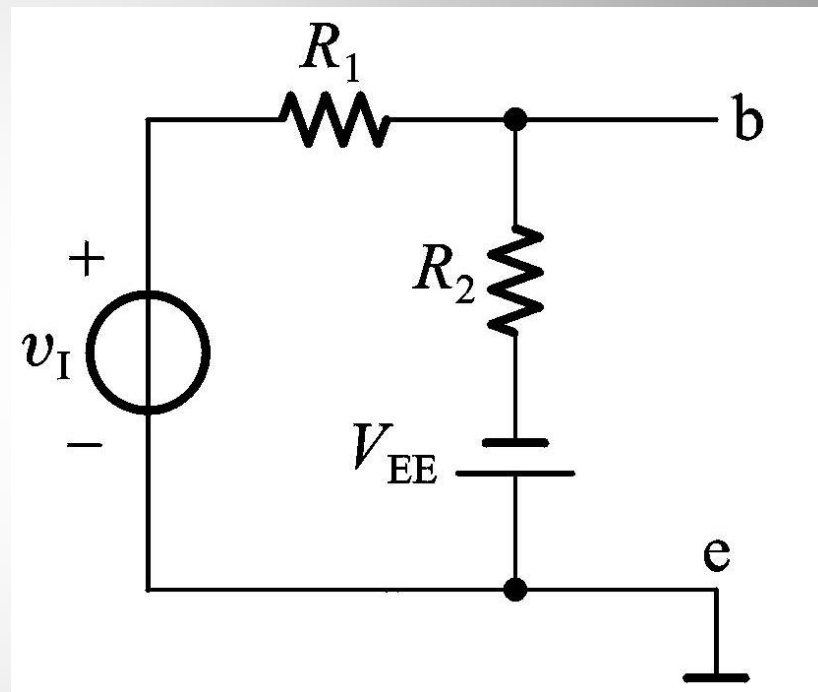
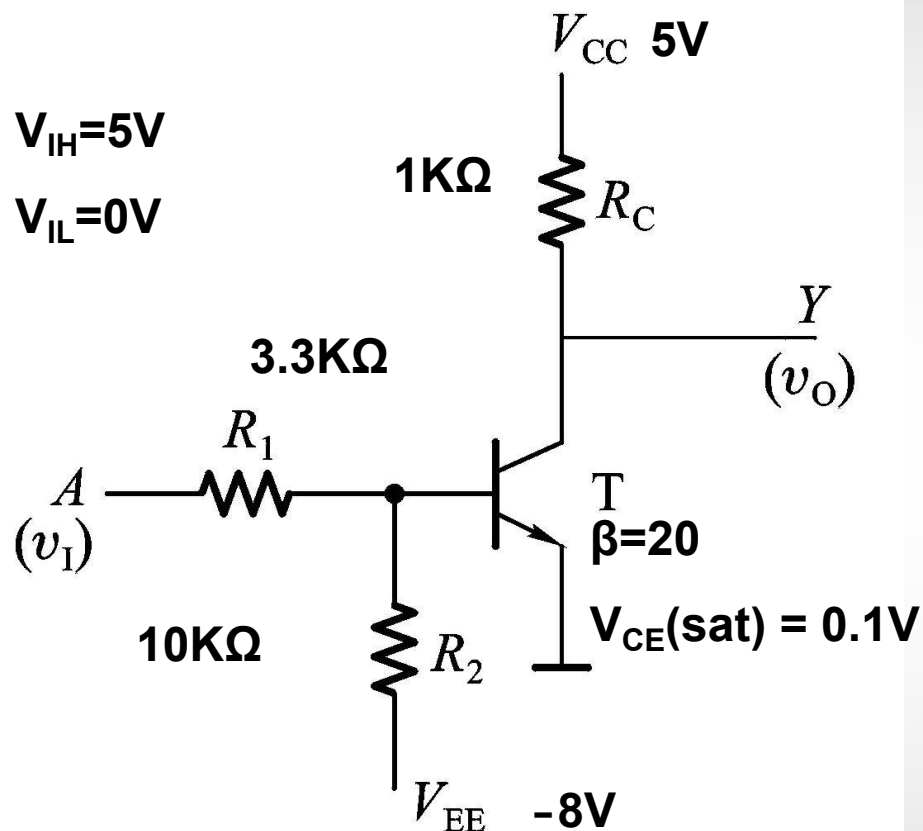
实际应用中，为保证 $V_I = V_{IL}$ 时 T 可靠截止，常在输入接入负压。



参数合理?
 $V_I = V_{IL}$ 时, T 截止,
 $V_O = V_{OH}$
 $V_I = V_{IH}$ 时, T 截止,
 $V_O = V_{OL}$

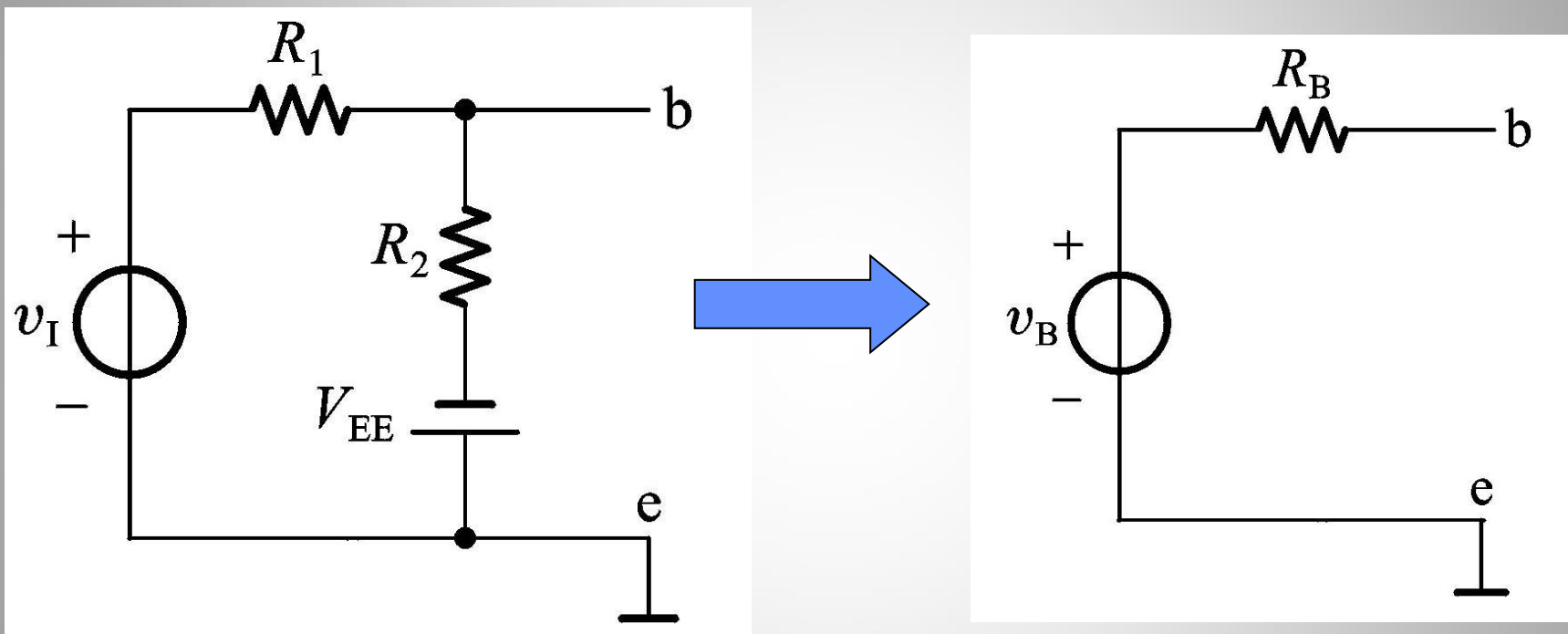


例3.5.1：计算参数设计是否合理



例3.5.1：计算参数设计是否合理

① 将发射极外接电路化为等效的 V_B 与 R_B 电路



$$R_B = R_1 // R_2 = 2.5 K\Omega$$

$$V_B = V_I - \frac{V_I - V_{EE}}{R_1 + R_2} R_1 = V_I - \frac{V_I + 8}{13.3} \times 3.3$$

② 当 $V_I = V_{IL} = 0$ 时, $V_B = 0 - \frac{8}{13.3} 3.3 = -2.0V$

所以 T 截止, $i_C = 0, V_O = V_{OH} = V_{CC} = 5.0V$

③ 当 $V_I = V_{IH} = 5V$ 时, $V_B = 5 - \frac{5+8}{13.3} 3.3 = 1.8V$ 所以 T 导通

如果用折线等效电路, 认为 $V_{BE} = 0.7V$

④ 又 则得: $i_B = \frac{V_B - V_{BE}}{R_B} = 0.44mA$

深度饱和时 I_{BS} 为:

$$I_{BS} = \frac{I_{CS}}{\beta} = \frac{V_{CC} - V_{CE}(sat)}{\beta R_C} = 0.25mA$$

故 $i_B > i_{BS}$, T 饱和, $V_O = V_{CE}(sat) \approx 0V$

⑤ 因此, 参数设计合理

3.5.2 TTL反相器的电路结构和工作原理(重点)

一、电路结构

设

$$V_{CC} = 5V$$

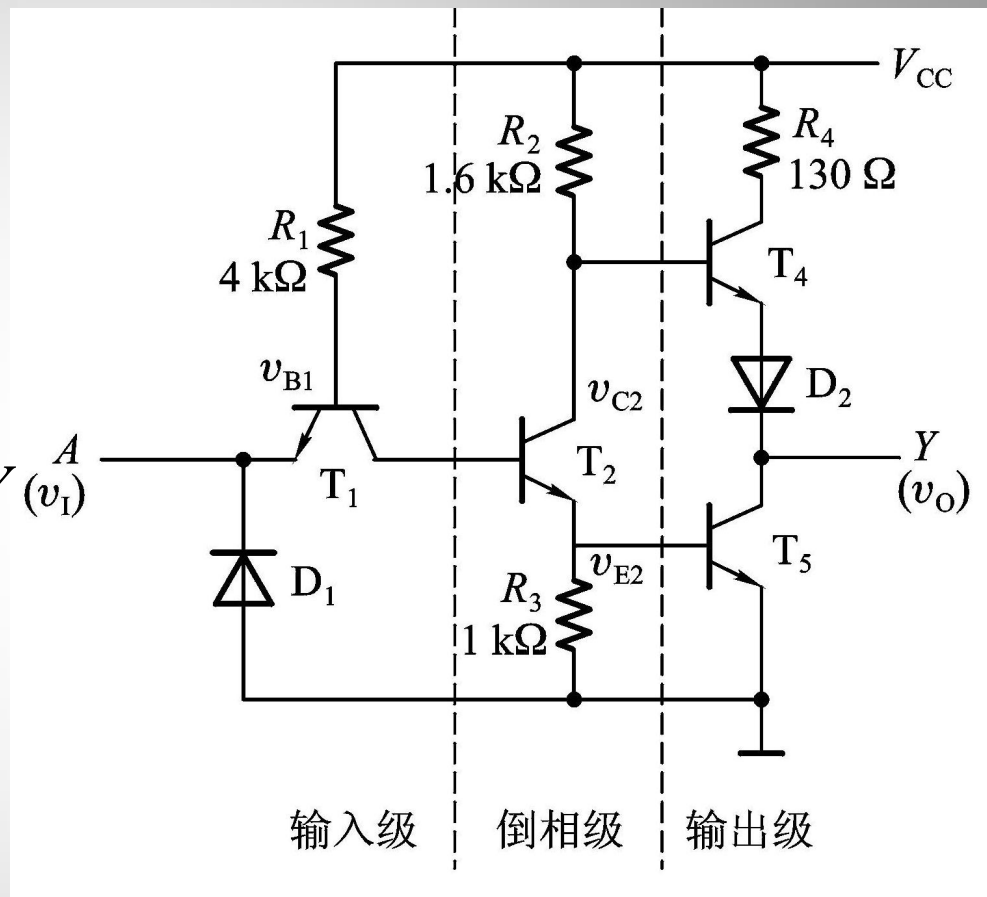
$$V_{IH} = 3.4V$$

$$V_{IL} = 0.2V$$

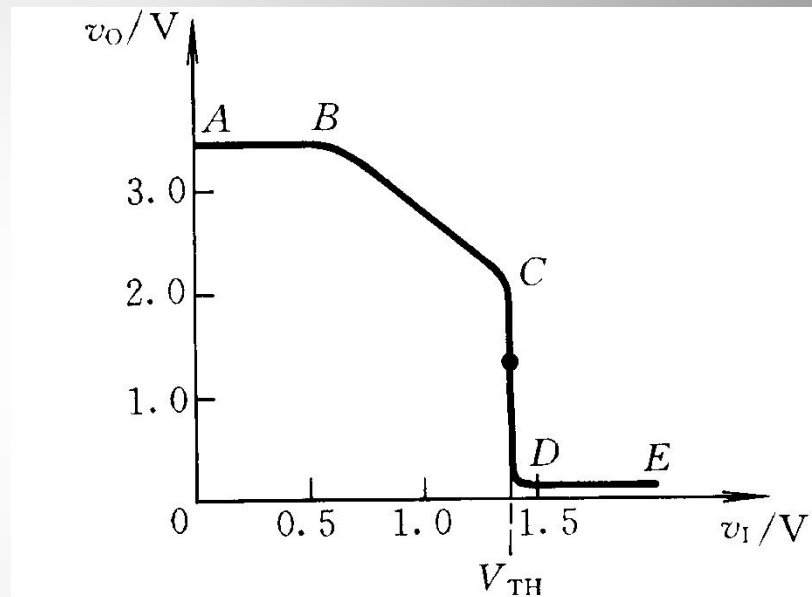
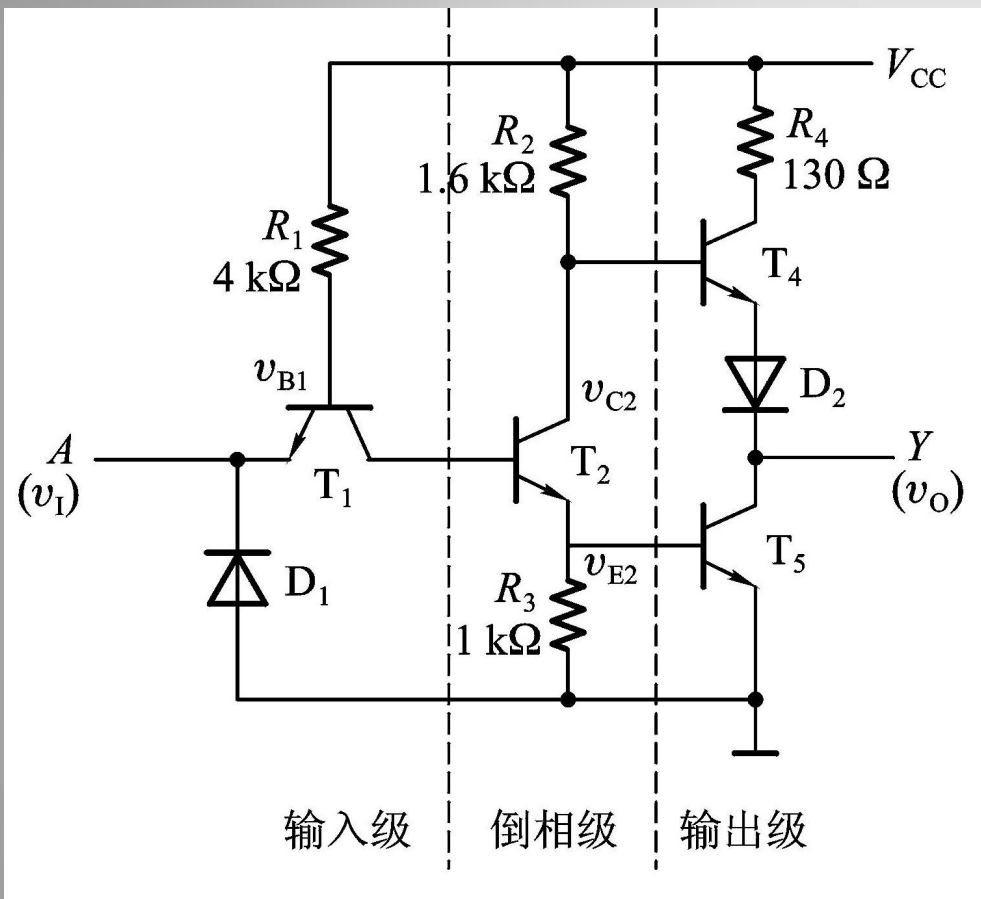
① PN 结导通压降 $V_{ON} = 0.7V$ (v_i)

② $V_i = V_{IL} = 0.2V$ ($A = 0$)
 $V_o = V_{OH}$ ($Y = 1$)

$V_i = V_{IH} = 3.4V$ ($A = 1$)
 $V_o = V_{OL}$ ($Y = 0$)



二、电压传输特性



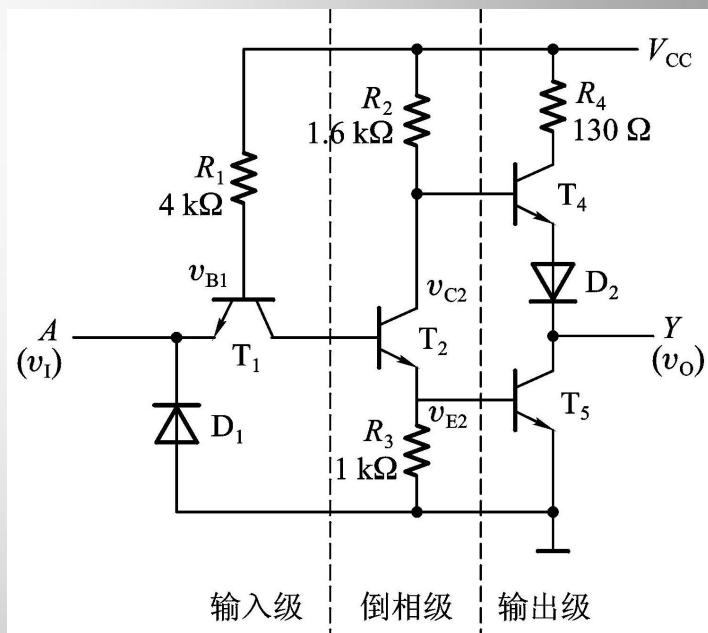
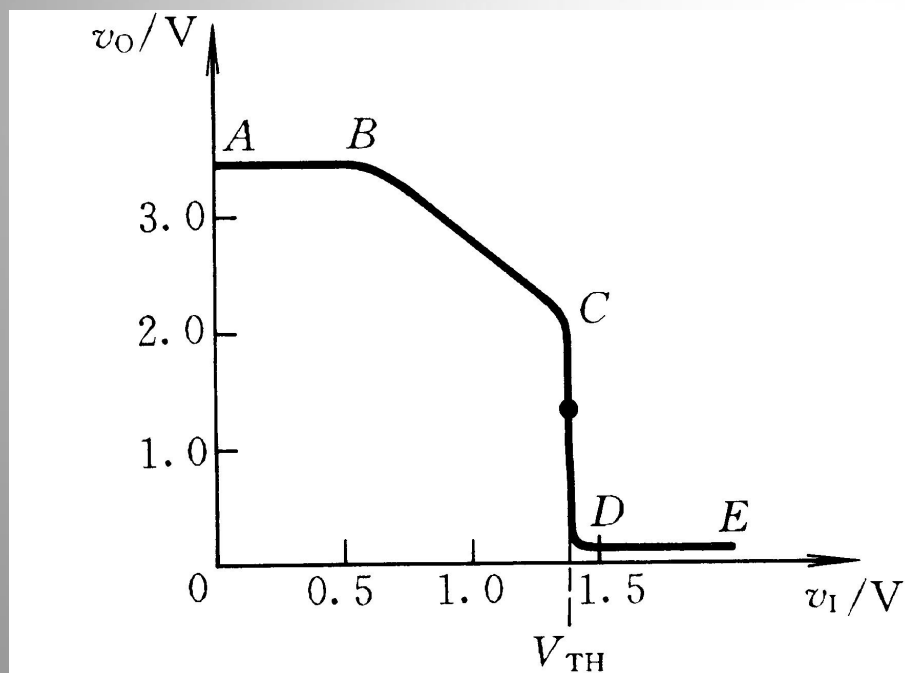
二、电压传输特性

* **AB段**: 截止区 $V_I < 0.6V$, $\therefore V_{B1} < 1.3V$

T_1 导通, T_2, T_5 截止, T_4 导通 $\Rightarrow V_{OH} = V_{CC} - V_{R2} - V_{BE4} - V_{D2} = 3.4V$

* **BC段**: 线性区 $0.7V < V_I < 1.3V$

T_2 导通且工作在放大区, T_5 截止, T_4 导通, $V_I \uparrow \Rightarrow V_O \downarrow$



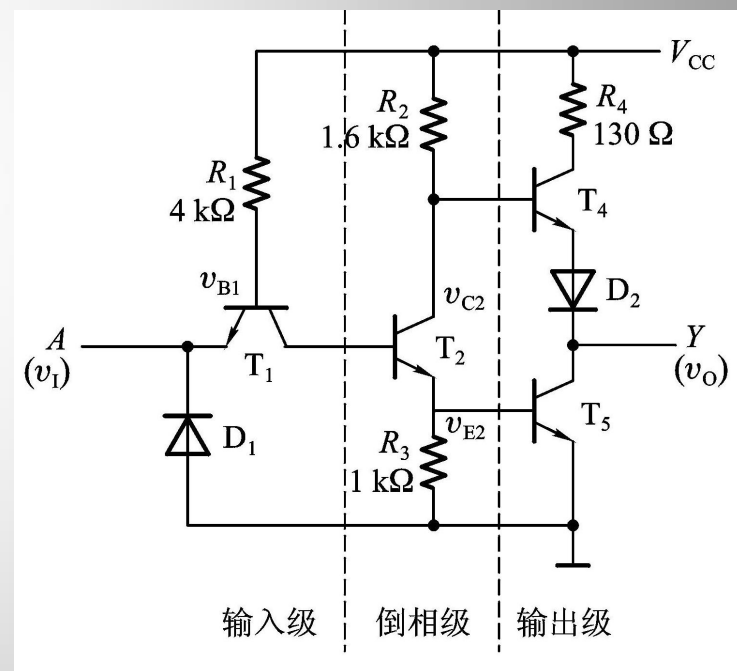
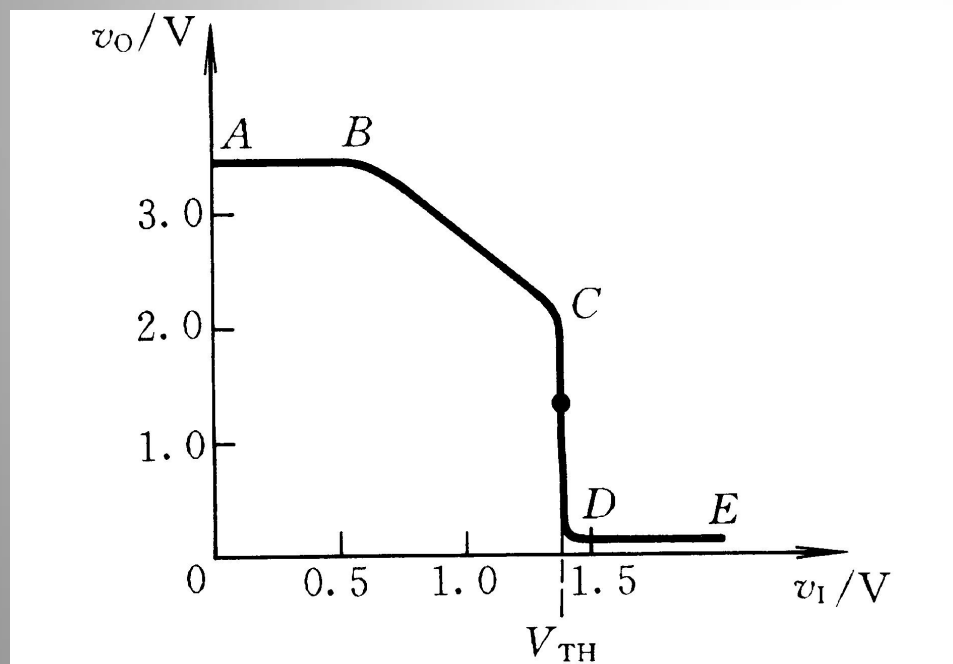
二、电压传输特性

* **CD**段：转折区 $V_I = V_{TH} \approx 1.4V$ ，所以 $V_{B1} \geq 2.1V$

T_2, T_5 同时导通， T_4 截止，所以 V_O 迅速 $\downarrow \Rightarrow V_{OL} \approx 0$

* **DE**段：饱和区 V_I 继续 $\uparrow$ ，而 V_O 不变

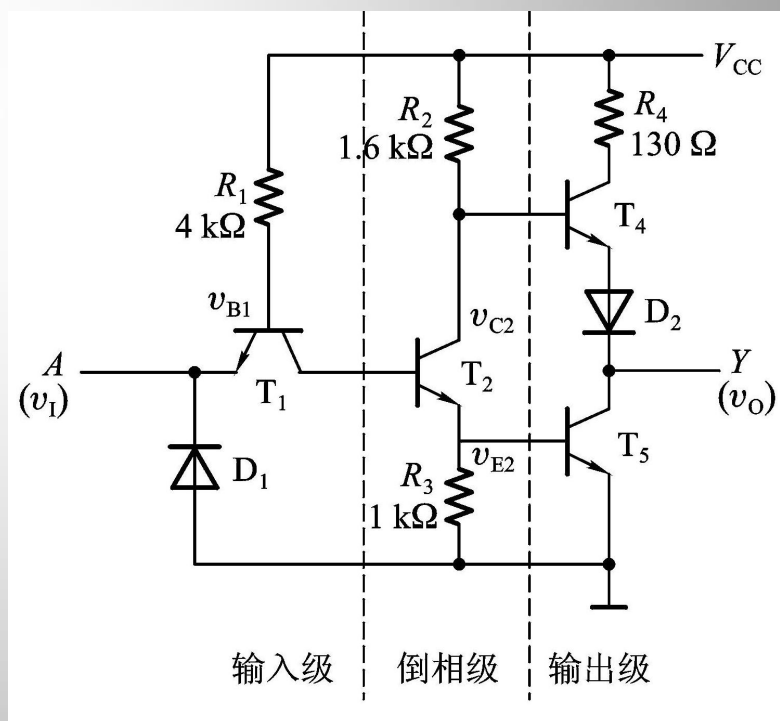
$$V_O = V_{OL}$$



- 需要说明的几个问题：

- ① T_2 的输出 V_{C2} 和 V_{e2} 变化方向相反, 故称倒相级。
- ② 输出级在稳态下, T_4 和 T_5 总有一个导通、一个截止。
既能降低功耗又提高了带负载能力, 称推拉式。

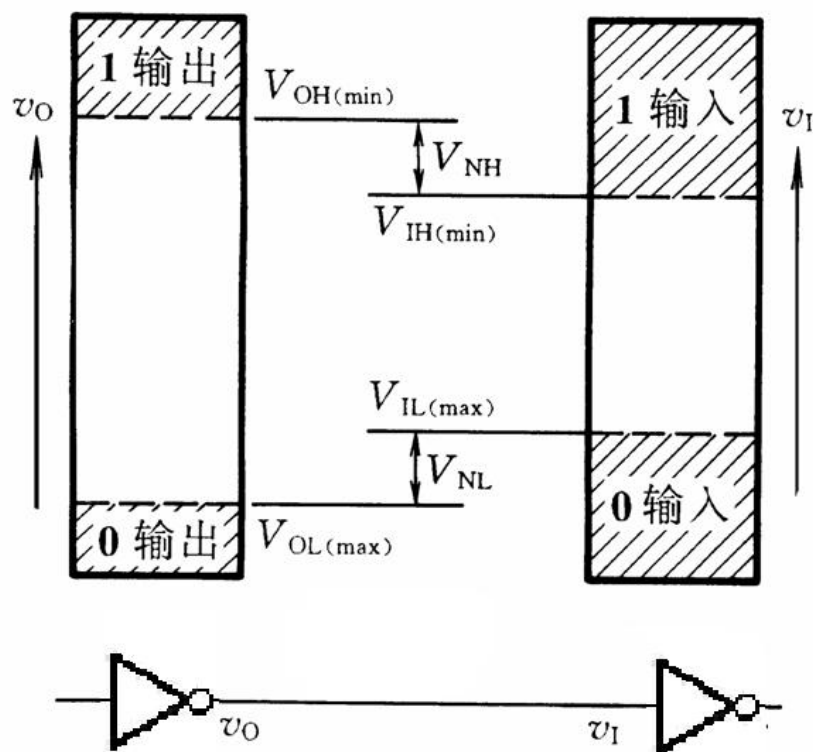
- ③ D_1 抑制负向干扰
 D_2 保证 T_2 导通时 T_4 可靠地截止。



三、输入噪声容限

在 V_I 偏离 V_{IH} 和 V_{IL} 的一定范围内， V_O 基本不变；

在输出变化允许范围内，允许输入的变化范围称为输入噪声容限



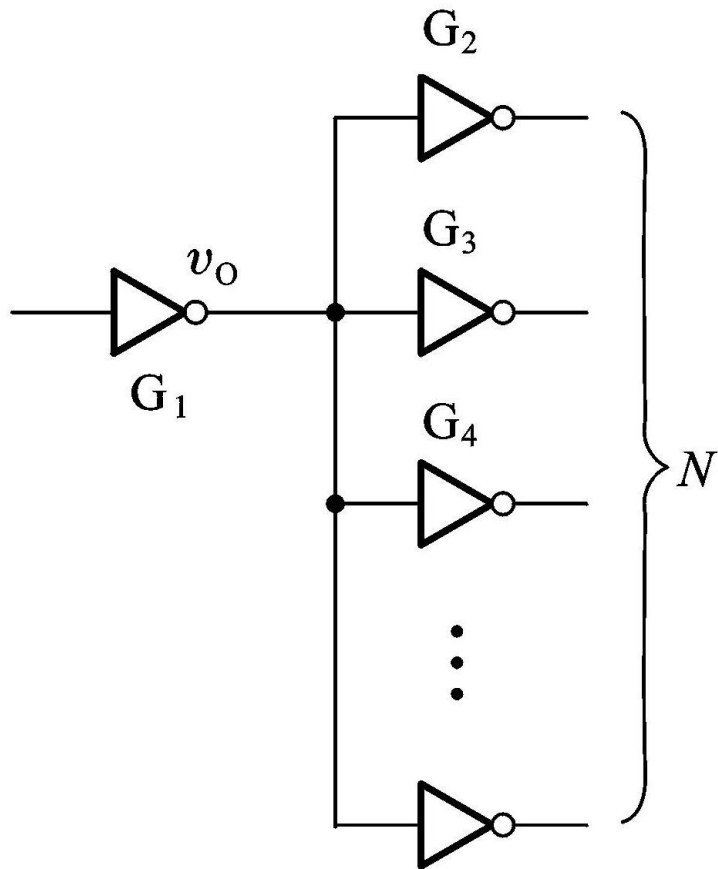
$$V_{NH} = V_{OH(min)} - V_{IH(min)}$$

$$V_{NL} = V_{IL(max)} - V_{OL(max)}$$

3.5.3 TTL反相器的静态输入特性和输出特性

例：扇出系数（**Fan-out**），

试计算门G1能驱动多少个同样的门电路负载。



要求保证 $V_I = V_{IL}$ 时，

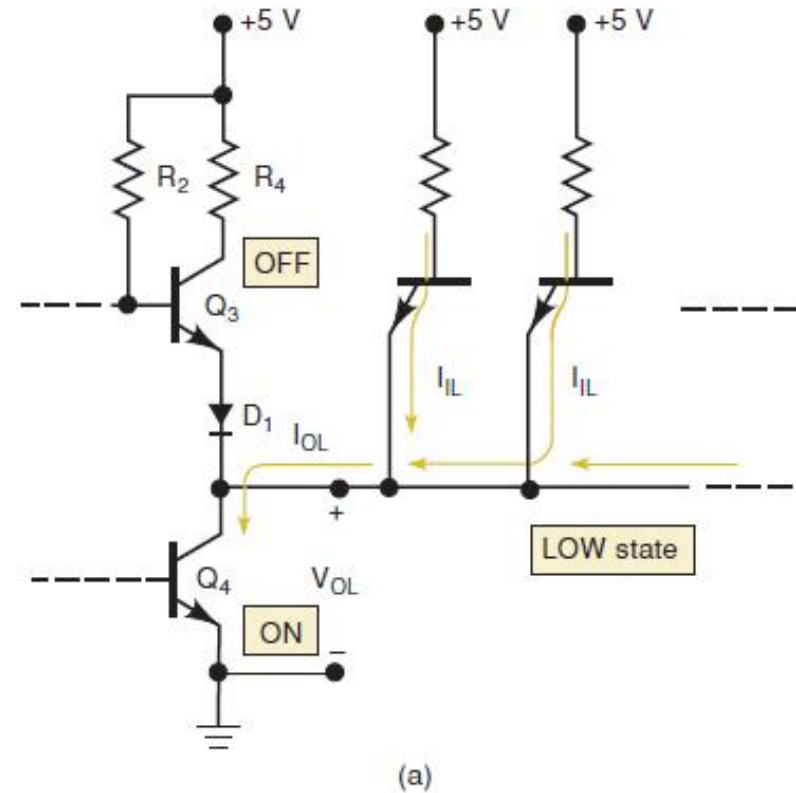
$V_{OH} \geq 3.2V$ （查得 $|i_{OH}| \leq 0.4mA$ ）；

$V_I = V_{IH}$ 时，

$V_{OL} \leq 0.2V$ （查得 $|i_{OL}| \leq 16mA$ ）

TTL LOADING AND FAN-OUT

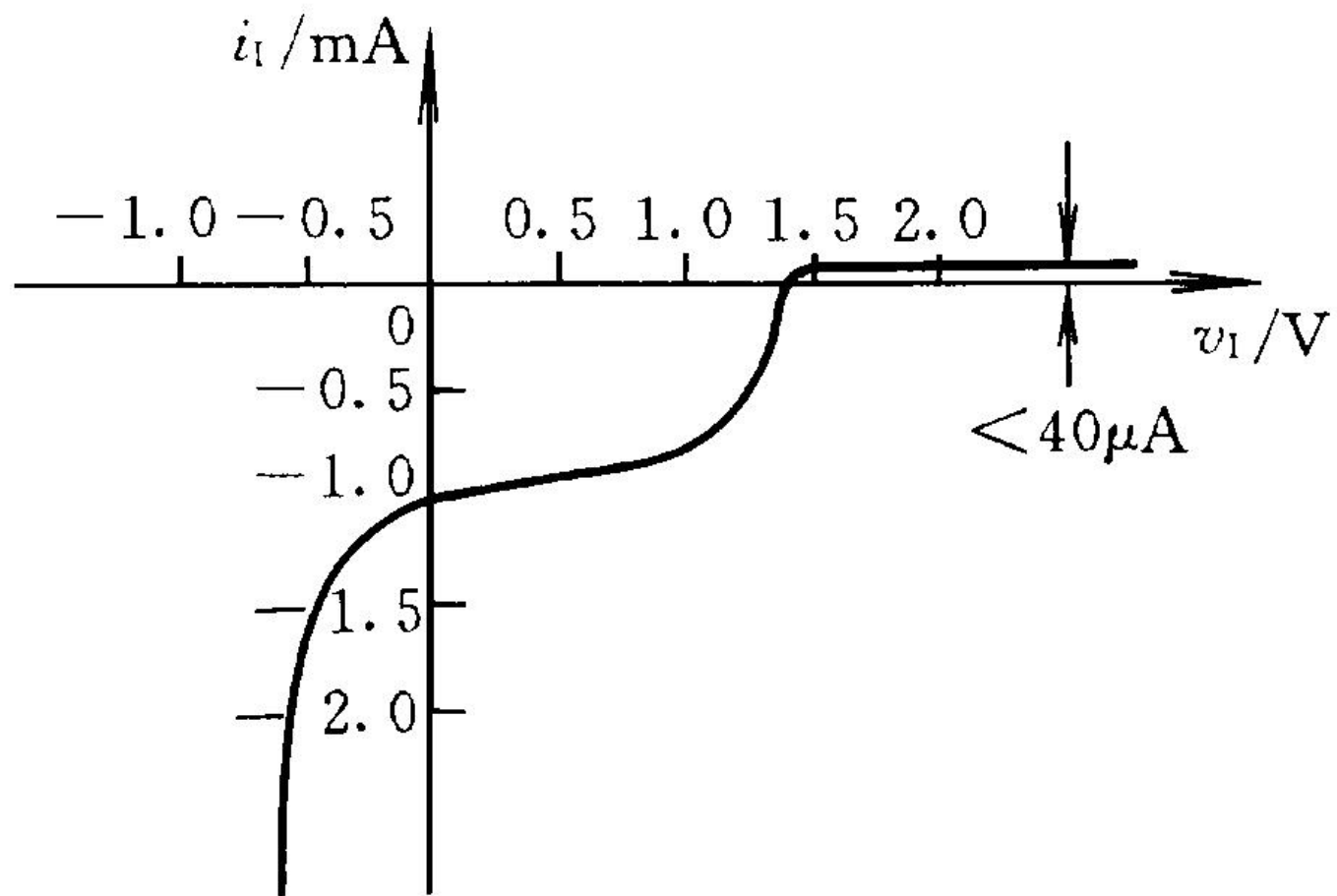
- Currents when a TTL output is driving several inputs



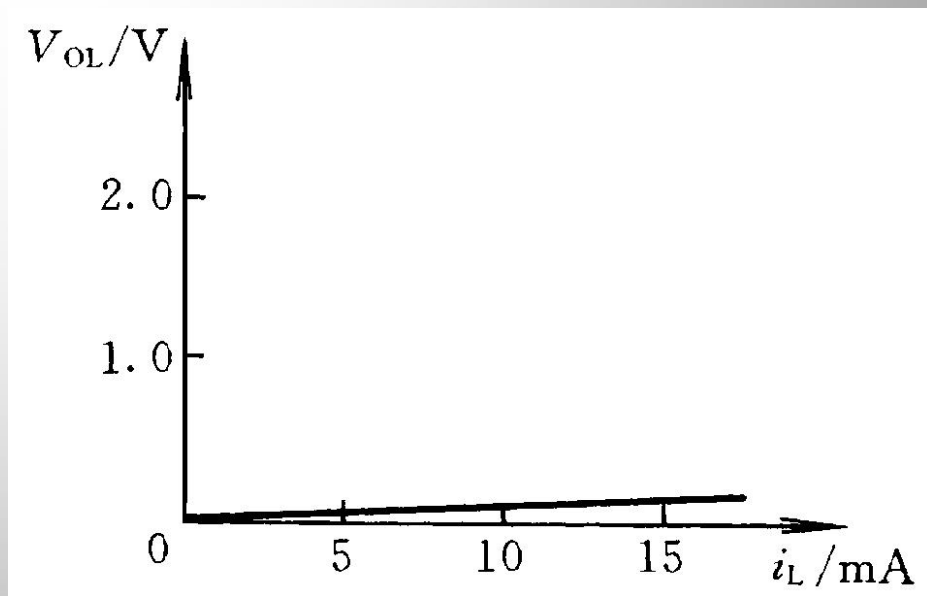
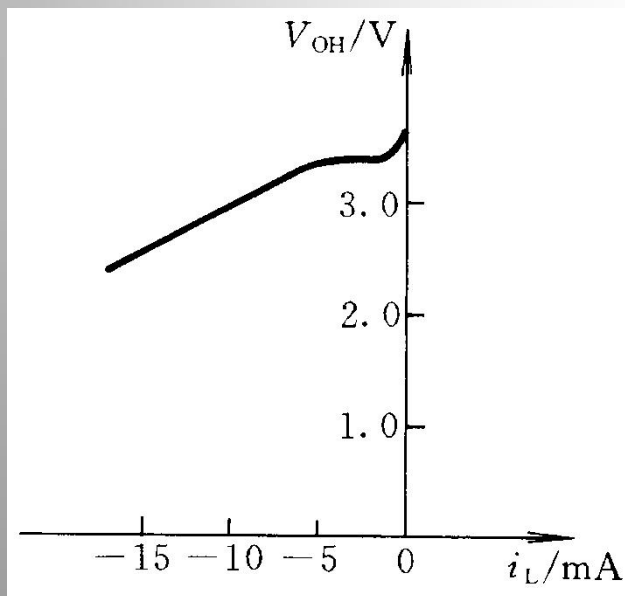
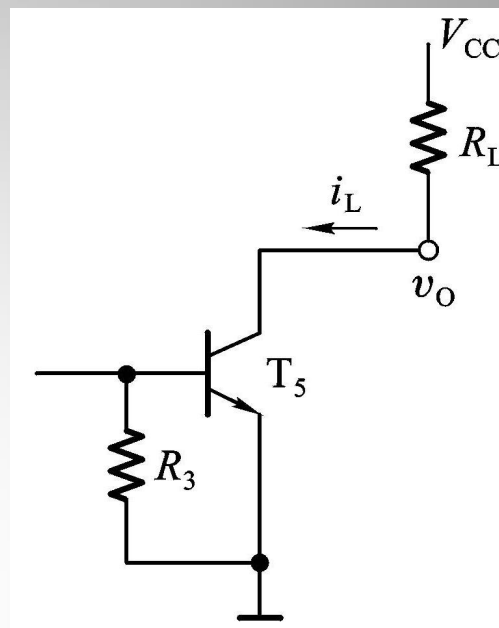
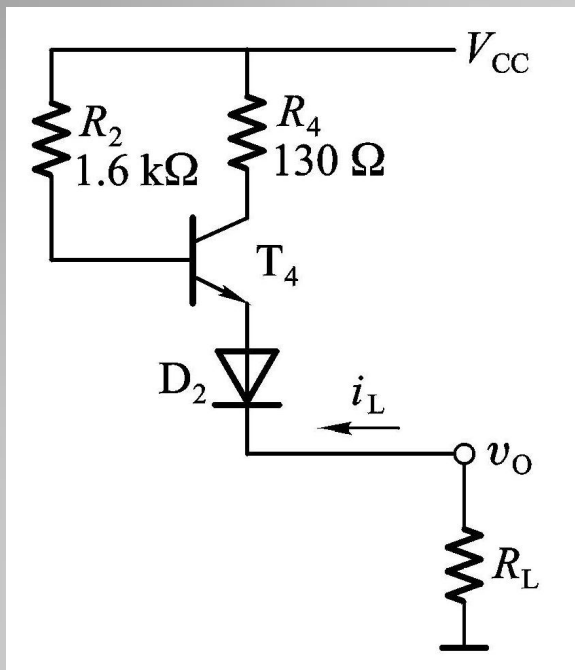
TTL Series	Outputs		Inputs	
	I_{OH}	I_{OL}	I_{IH}	I_{IL}
74	-0.4 mA	16 mA	40 μ A	-1.6 mA
74S	-1 mA	20 mA	50 μ A	-2 mA
74LS	-0.4 mA	8 mA	20 μ A	-0.4 mA
74AS	-2 mA	20 mA	20 μ A	-0.5 mA
74ALS	-0.4 mA	8 mA	20 μ A	-0.1 mA
74F	-1 mA	20 mA	20 μ A	-0.6 mA

*Some devices may have different input or output current ratings. Always consult the data sheet.

输入特性



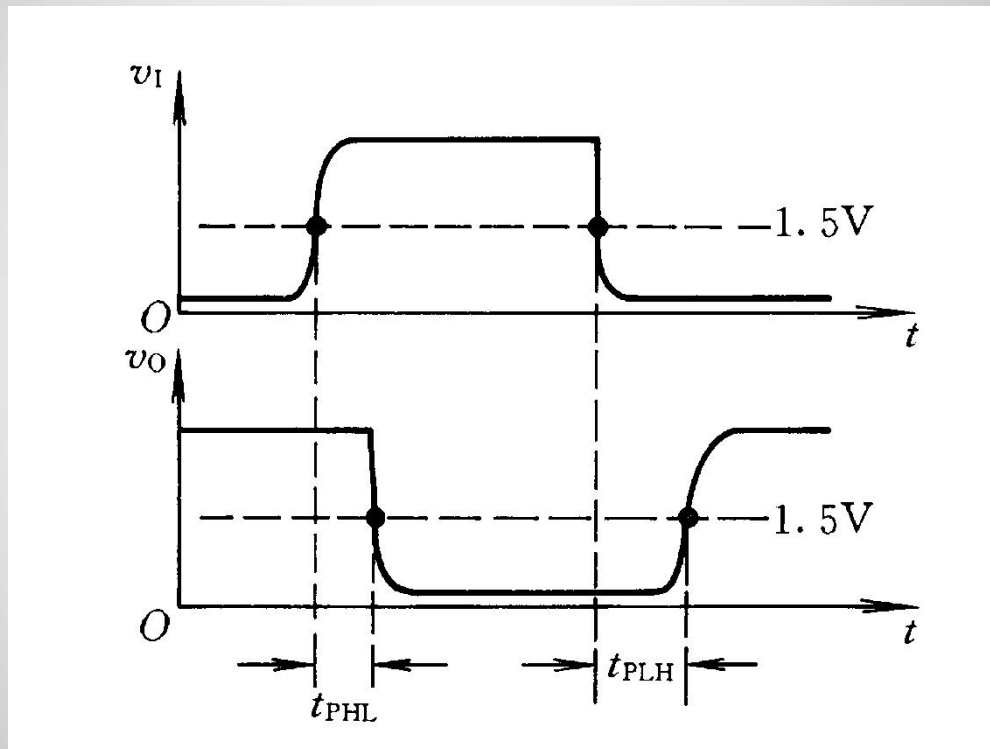
输出特性



3.5.4 TTL反相器的动态特性

一、传输延迟时间

1、现象

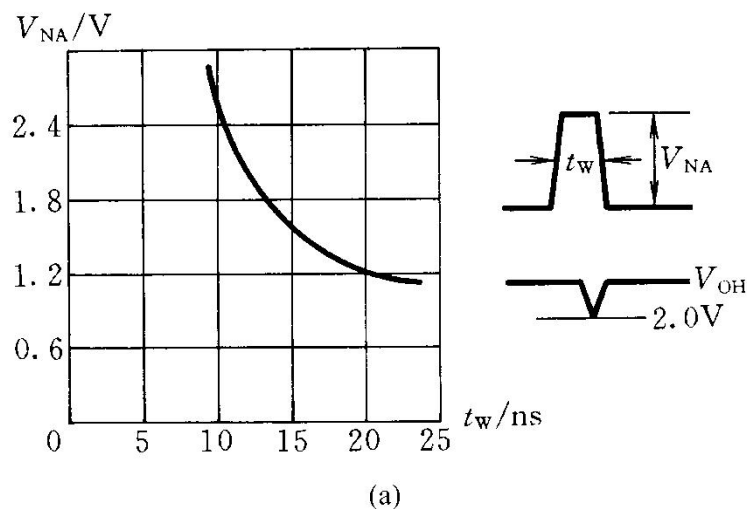


2、原因

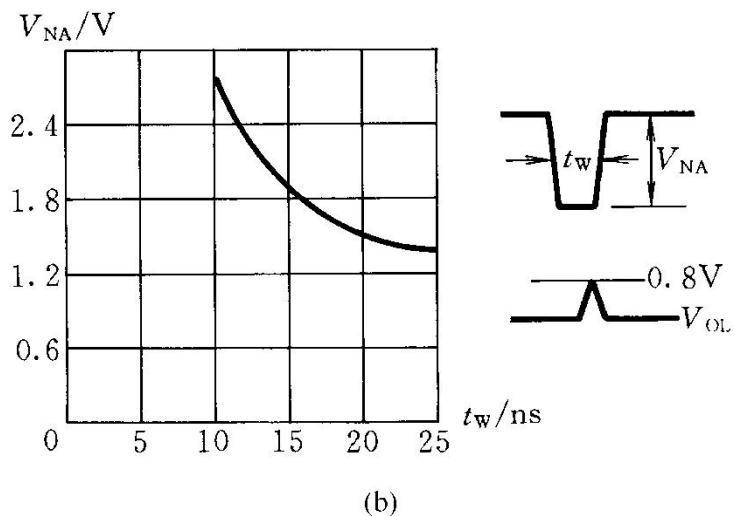
结电容 (D 和 T) 的存在,分布电容的影响

二、交流噪声容限

当输入信号为窄脉冲，且接近于 t_{pd} 时，输出变化跟不上，变化很小，因此交流噪声容限远大于直流噪声容限。



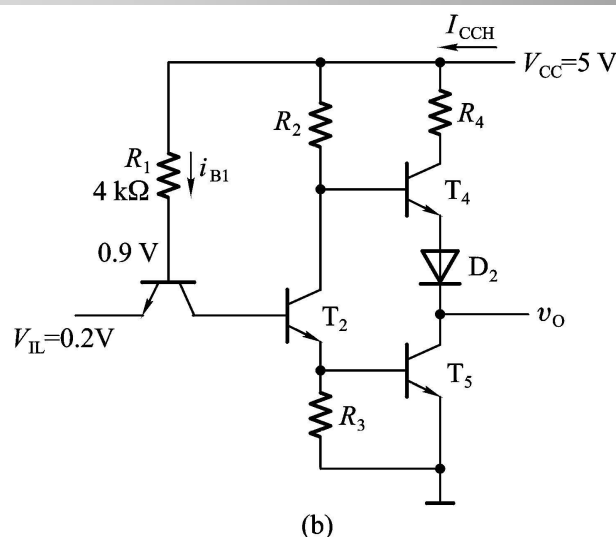
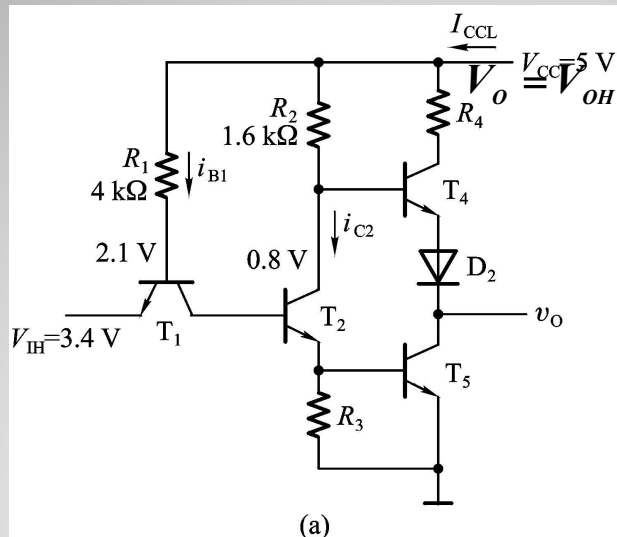
(a) 正脉冲噪声容限



(b) 负脉冲噪声容限

三、电源的动态尖峰电流

$$V_O = V_{OL}$$



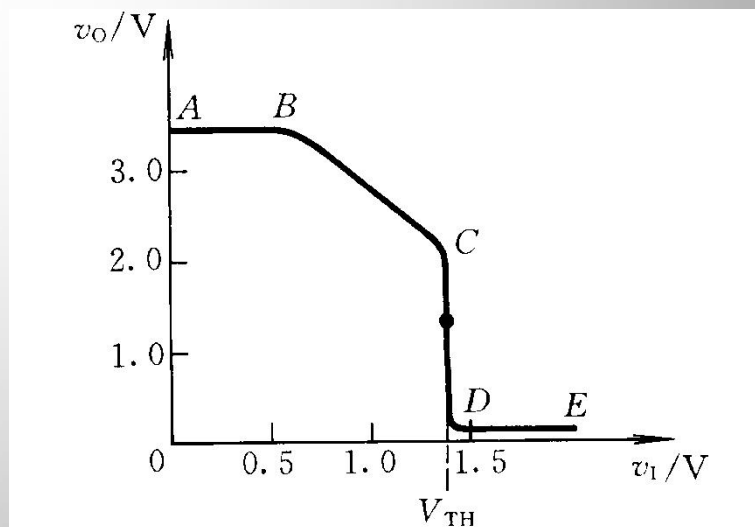
1、两种静态下的电源负载电流不等
空载条件下：

* $V_O = V_{OL}$ 时, $T_{2,5}$ 导通, T_4 截止

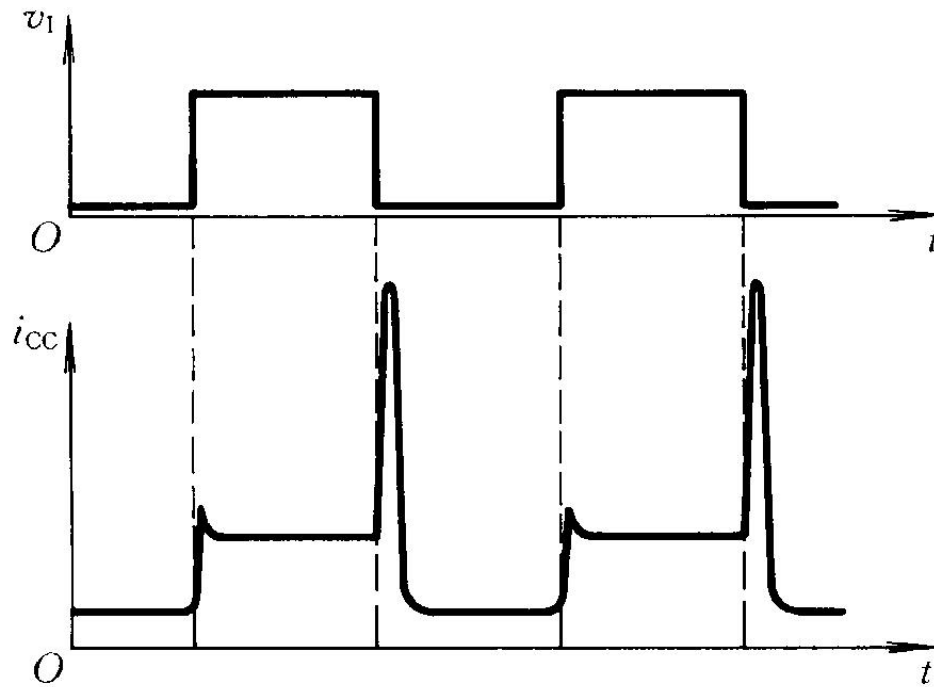
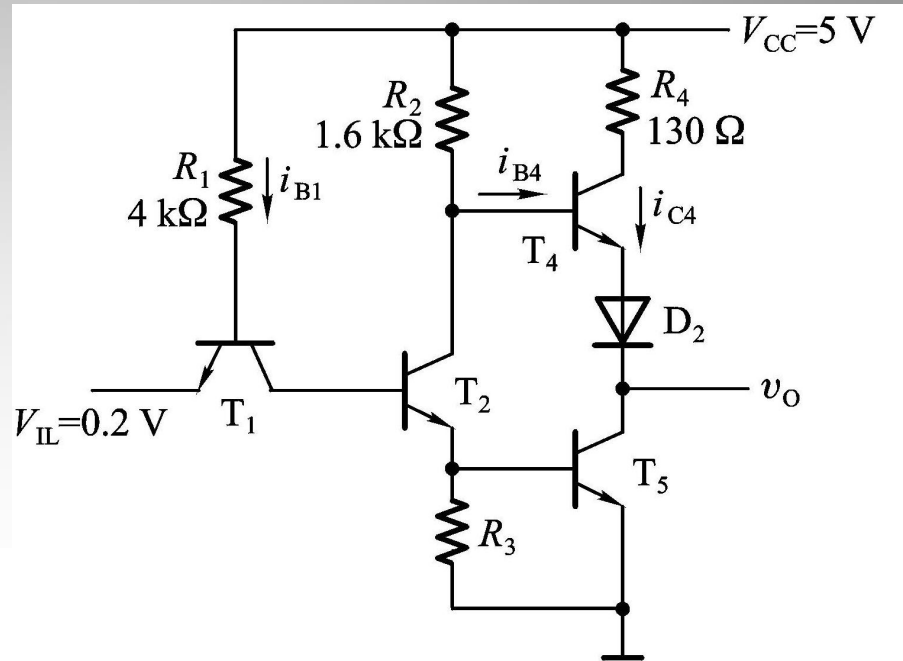
$$I_{CCL} = \frac{V_{CC} - V_{BE1}}{R_1} + \frac{V_{CC} - V_{C2}}{R_2} = 3.4 \text{ mA}$$

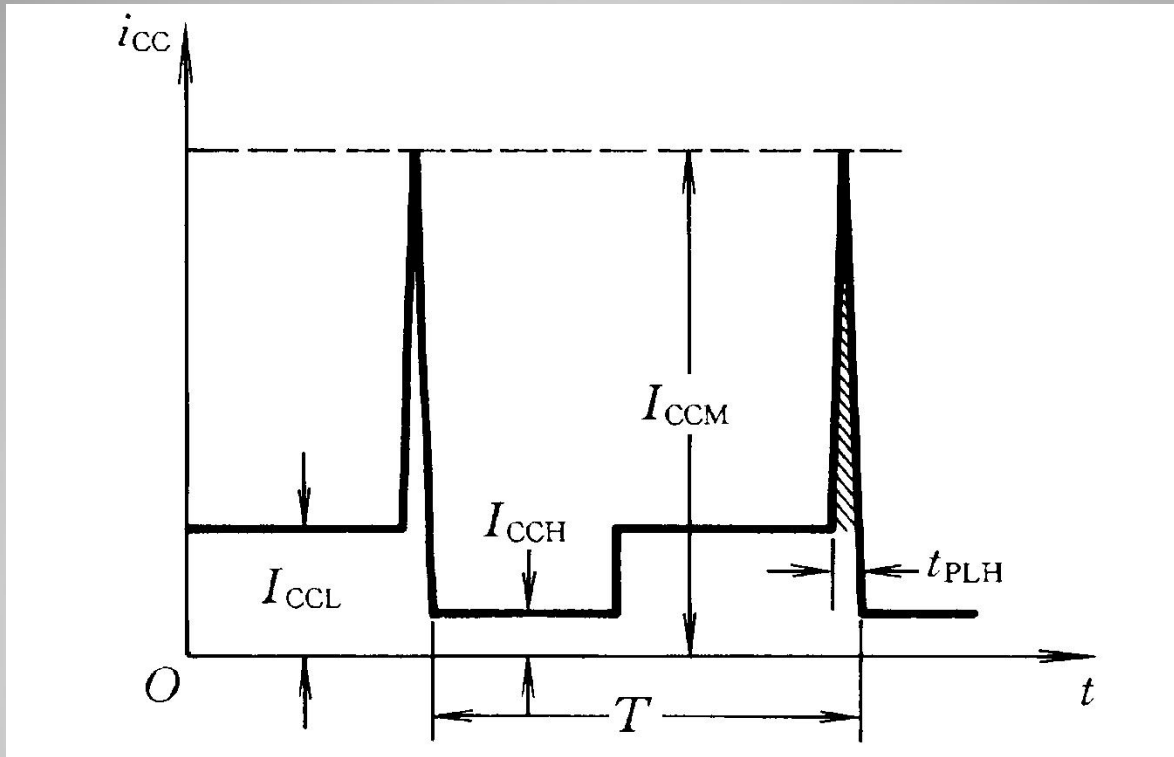
* $V_O = V_{OH}$ 时, 仅 T_1 导通

$$I_{CCH} = \frac{V_{CC} - V_{BE1}}{R_1} = 1.1 \text{ mA}$$



2、动态尖峰电流

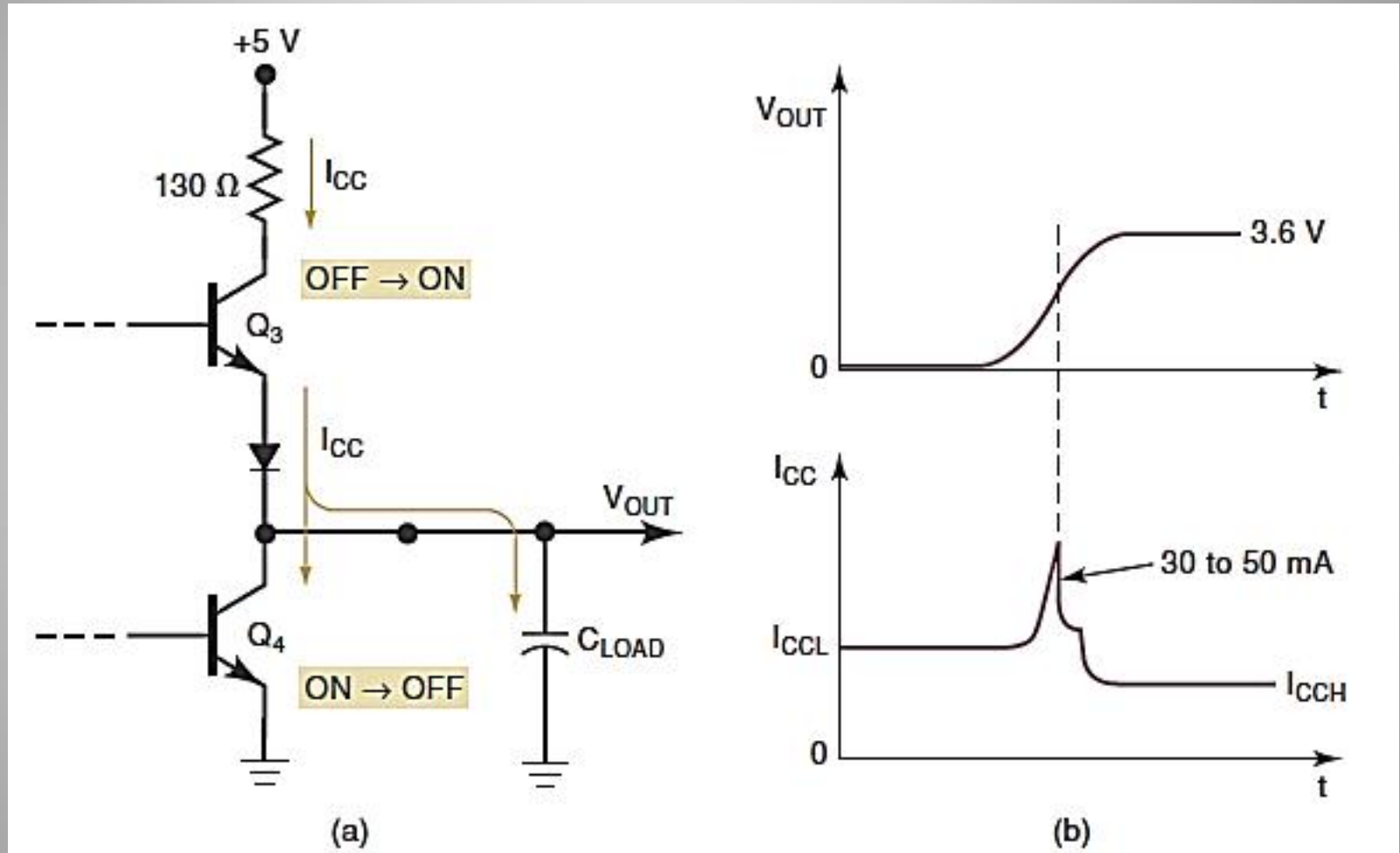




成因： $V_I \uparrow$ 和 $\downarrow$ 过程中，瞬时 T_4 、 T_5 同时导通。

影响： $f \uparrow \Rightarrow \overline{I_{CC}} \uparrow$ ；通过电源线和地线形成干扰源。

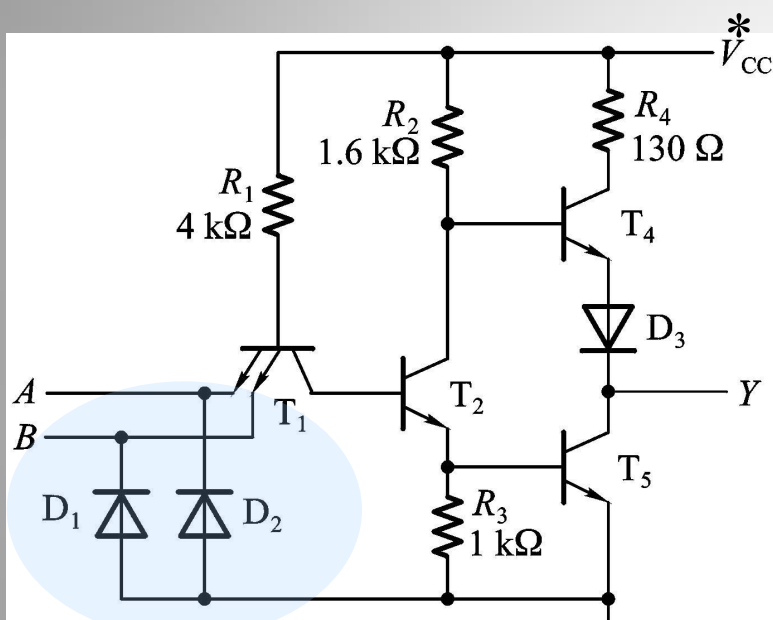
Current Transients



3.5.5其他类型的TTL门电路

一、其他逻辑功能的门电路

1. 与非门



$A \bullet B$ 由多发射极三极管实现

当 A 和 B 有一个为 $0.2V$ 时, $V_{B1} = 0.9V$,

T_5 截止, T_4 导通, $V_O = V_{OH} = 1$

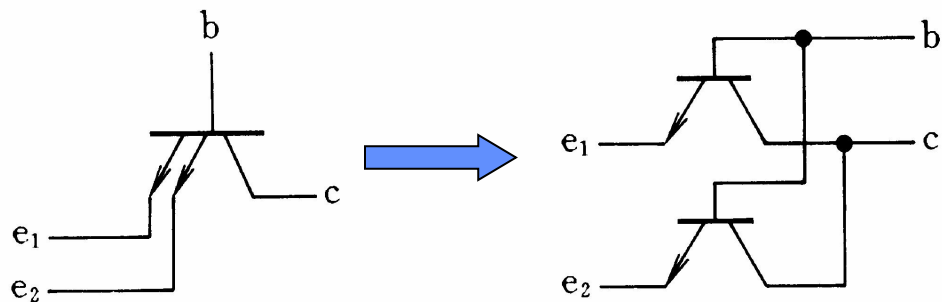
当 A 和 B 同为高电平时, $V_{B1} = 2.1V$,

T_4 截止, T_2 和 T_5 导通, $V_O = V_{OL} = 0$

* 输入电流计算:

I_{IL} : 并联后与仅一个接地 时相同

I_{IH} : 每个值相同, 并联后 加倍



SN7400二输入与非门

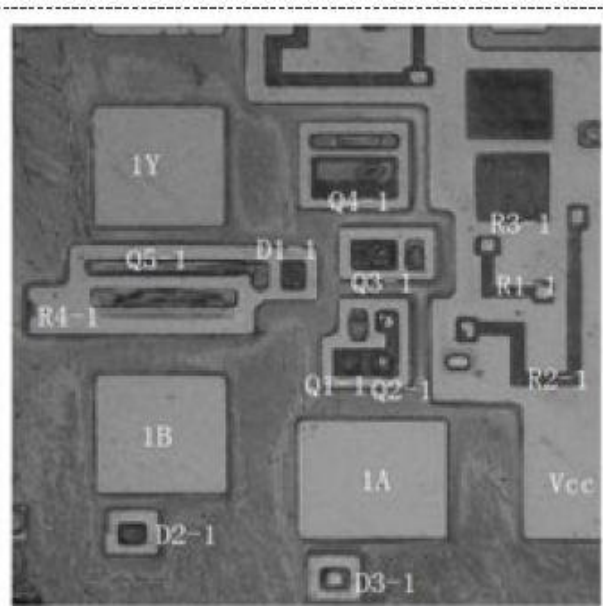


图1 SN7400去铝层版图局部照片

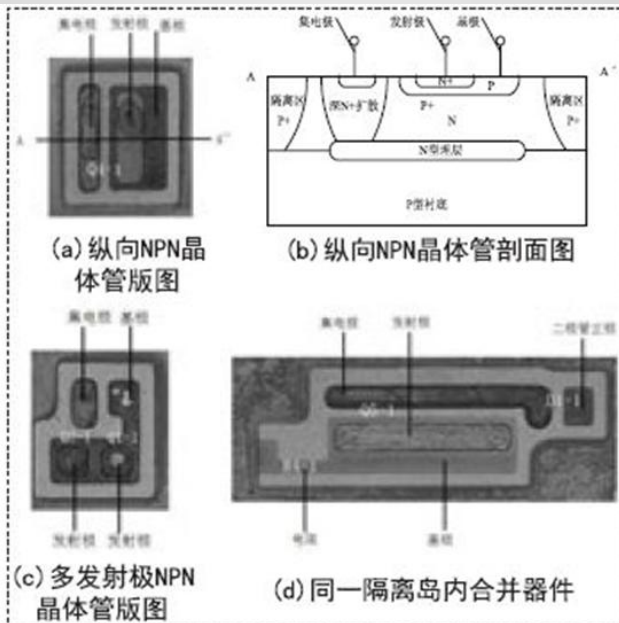


图2 SN7400芯片三极管版图

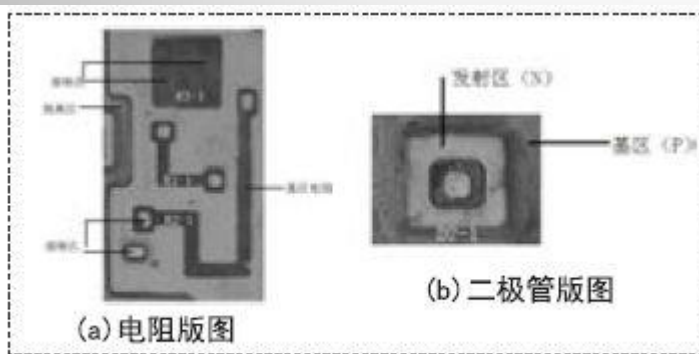


图3 SN7400芯片二极管和电阻版图

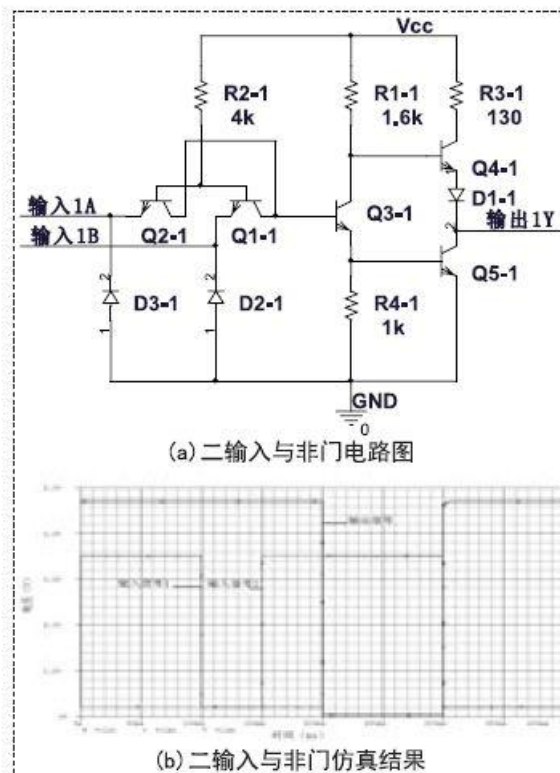
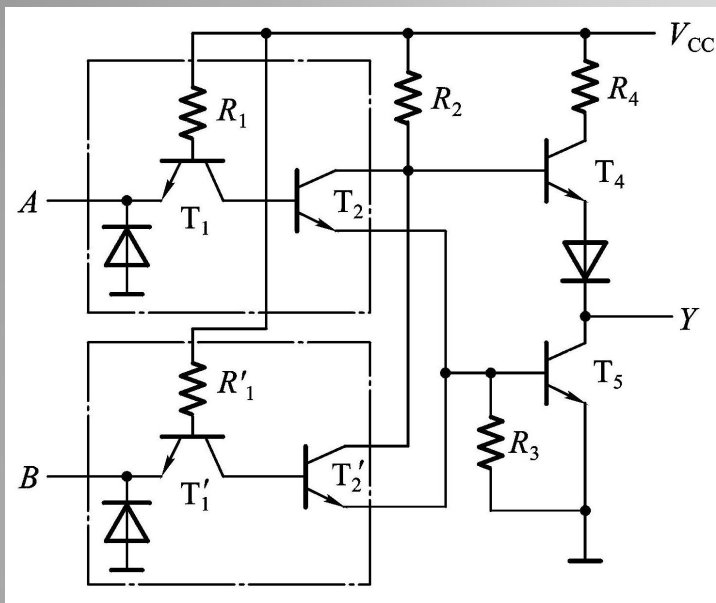
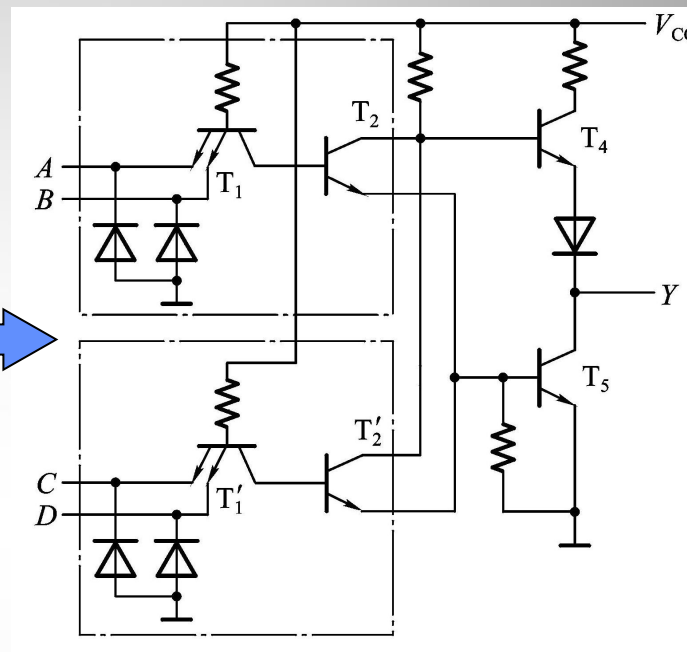


图4 SN7400芯片二输入与非门电路图和仿真结果

2. 或非门



3. 与或非门



*两个完全一样的输入电路

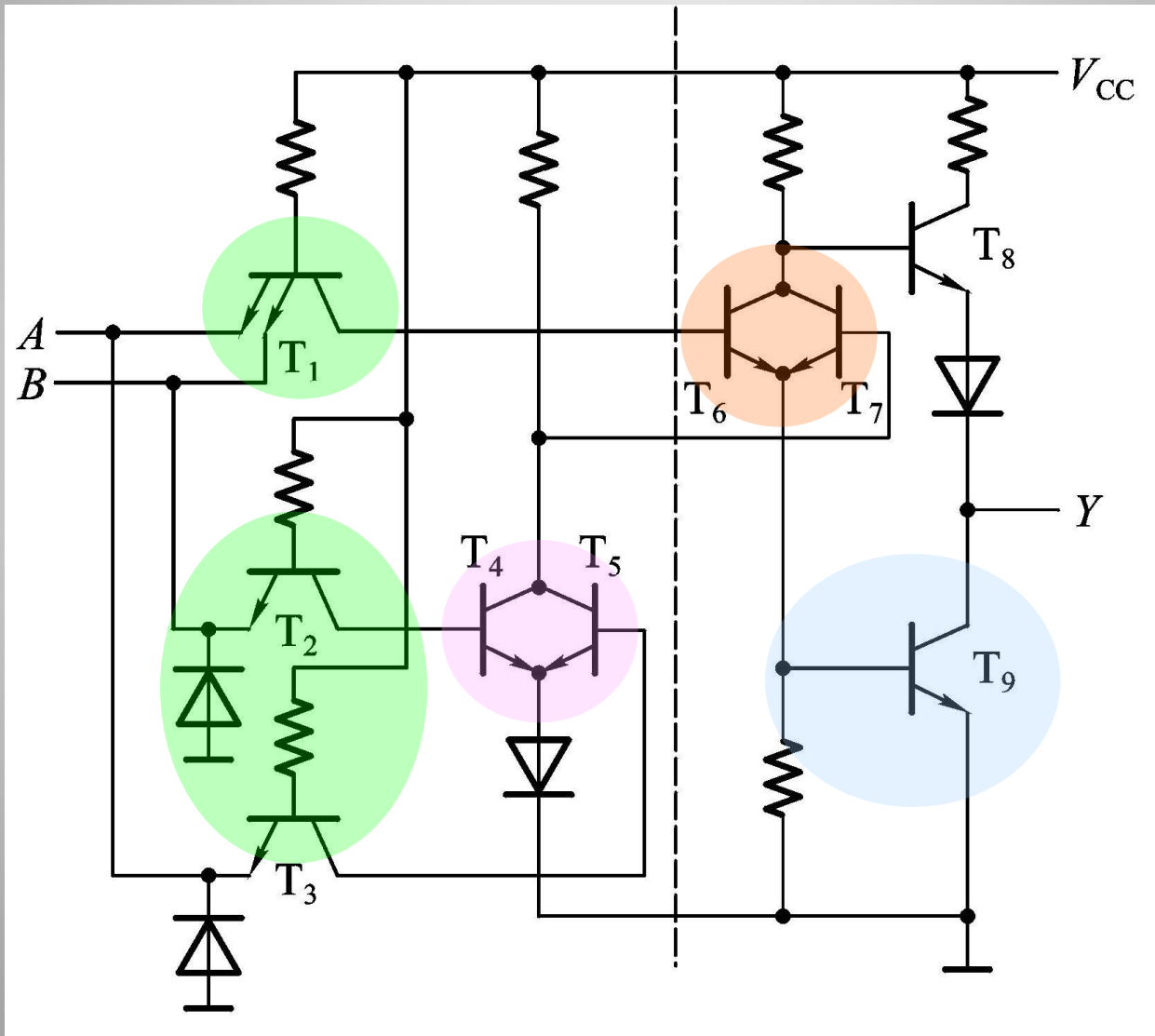
*因为 T_2 和 T_2' 的输出并联

所以 A 、 B 任何一个为1均使 T_5 导通， T_4 截止 $\Rightarrow V_O = V_{OL}$

只有 A 、 B 同为0，才有 T_5 截止， T_4 导通 $\Rightarrow V_O = V_{OH}$

*输入电流计算时， I_{IH} 和 I_{IL} 均加倍

4. 异或门



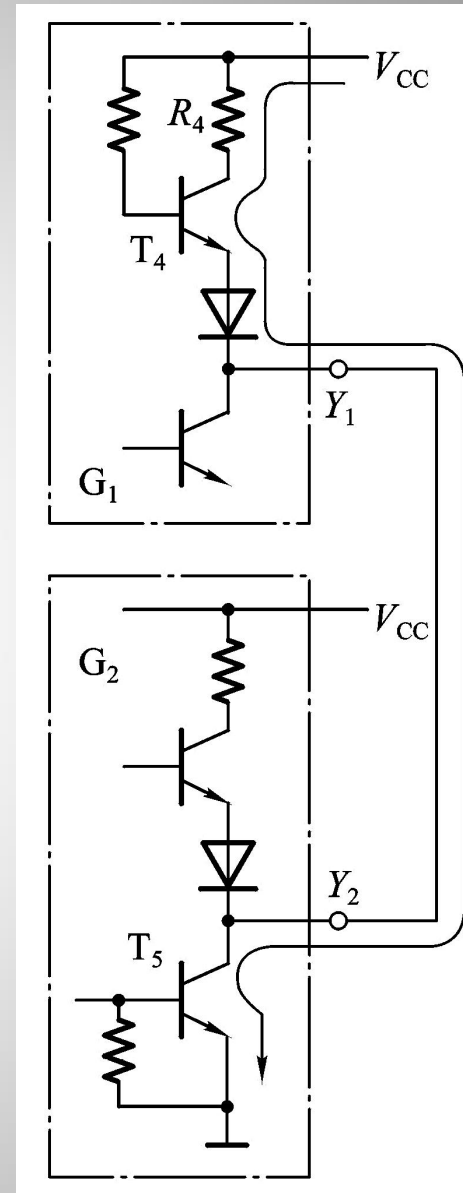
二、集电极开路的门电路

1、推拉式输出电路结构的局限性

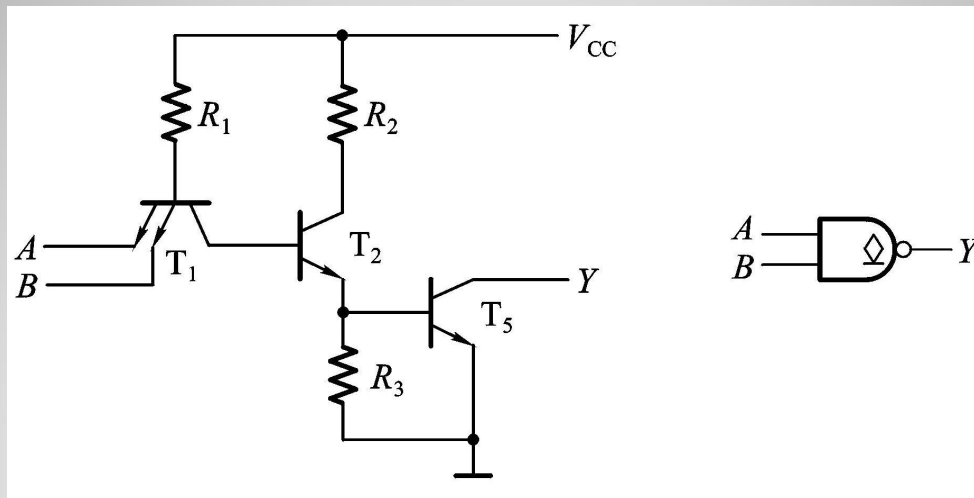
- ① 输出电平不可调；
- ② 负载能力不强，尤其是高电平输出；
- ③ 输出端不能并联使用。



OC门

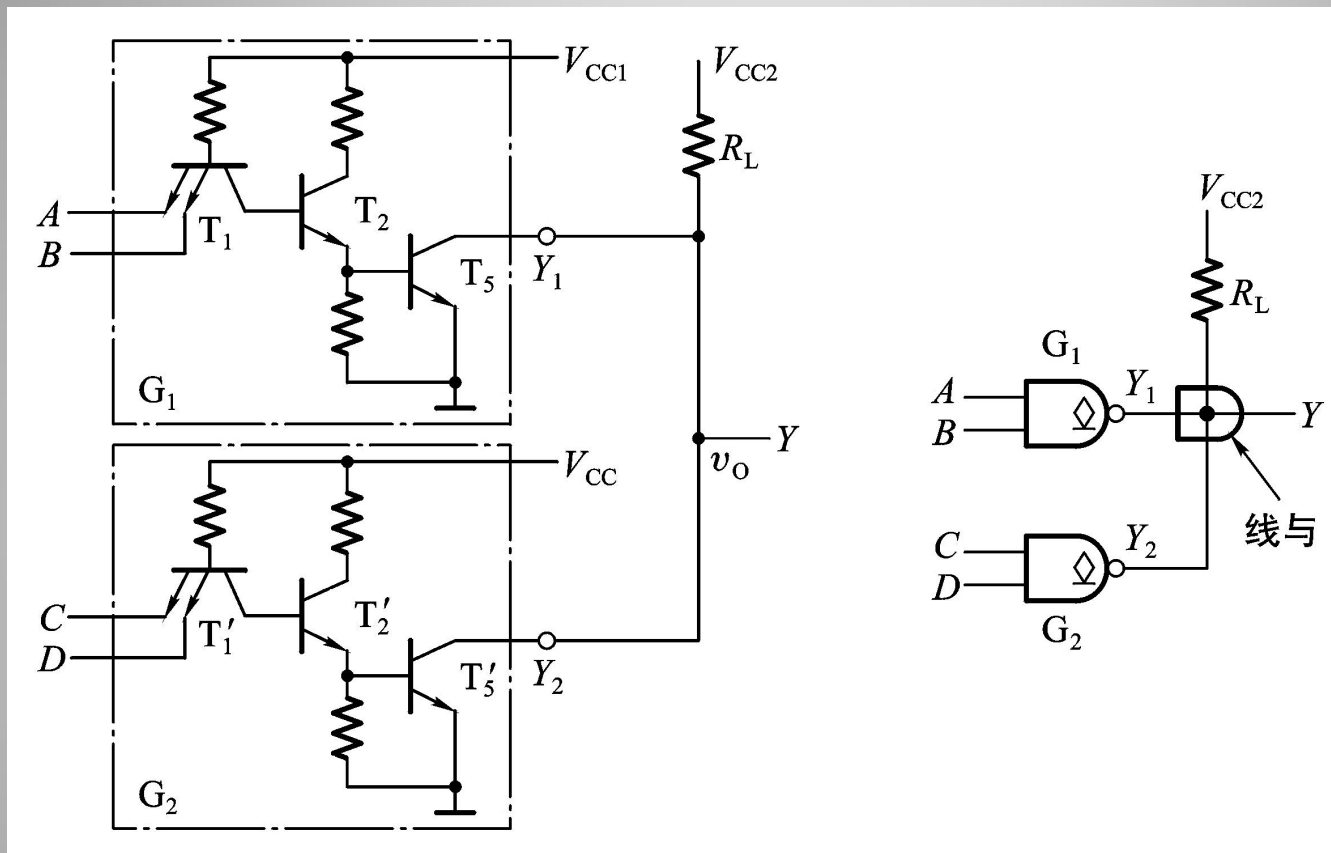


2、OC门的结构特点



- * 输出端为 **OC** 三极管 T_5 ， T_5 可承受较大电压、电流，
如 **SN 7407** : $40mA / 30V$
- * 工作时需要外接 R_L, V_{CC}' ；只要 R_L, V_{CC}' 取值合适，定可使
 A, B 同为高时， T_5 饱和 $V_{OL} \approx 0$
 A 或 B 为 0 时， T_5 截止 $V_O \approx V_{CC}'$ (V_{CC}' 可以不等于 V_{CC})
- * 输出端并联可实现 “线 与”

OC门实现的线与

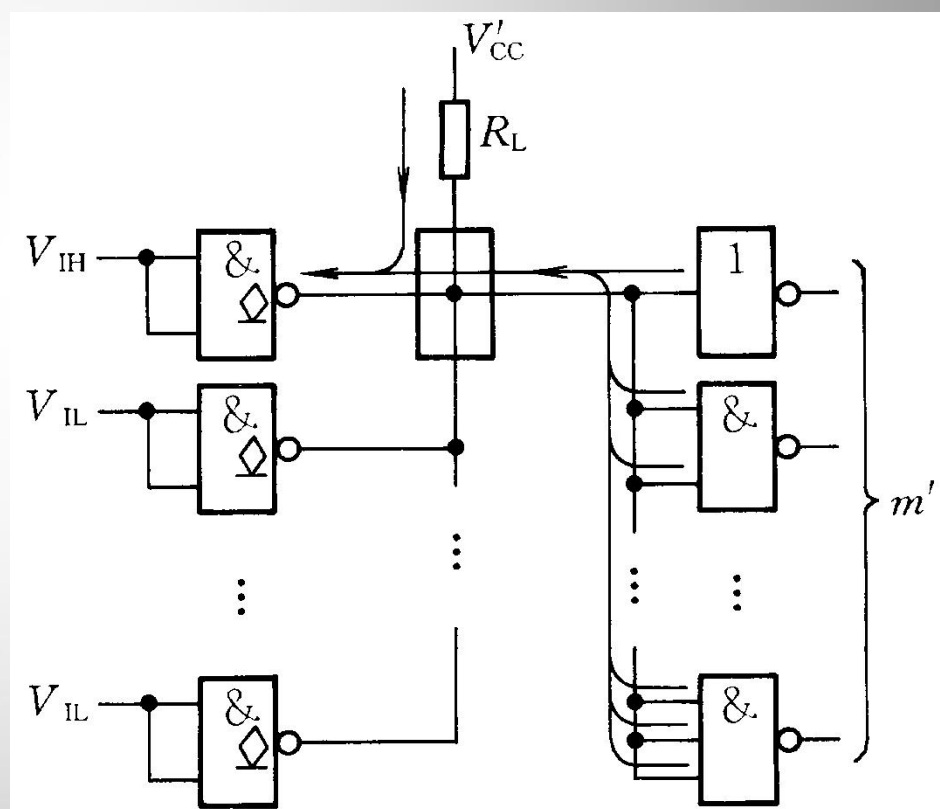
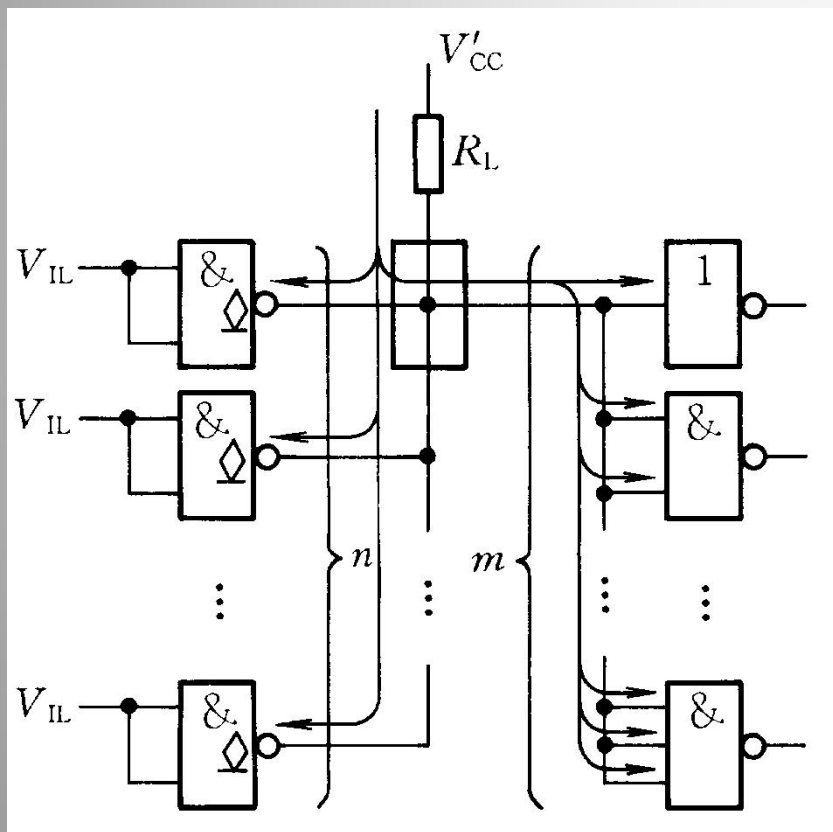


因为 Y_1 、 Y_2 有一个低， Y 即为低，只有两者同高， Y 才为高，
所以 $Y = Y_1 Y_2 = (AB)' \cdot (CD)' = (AB + CD)'$

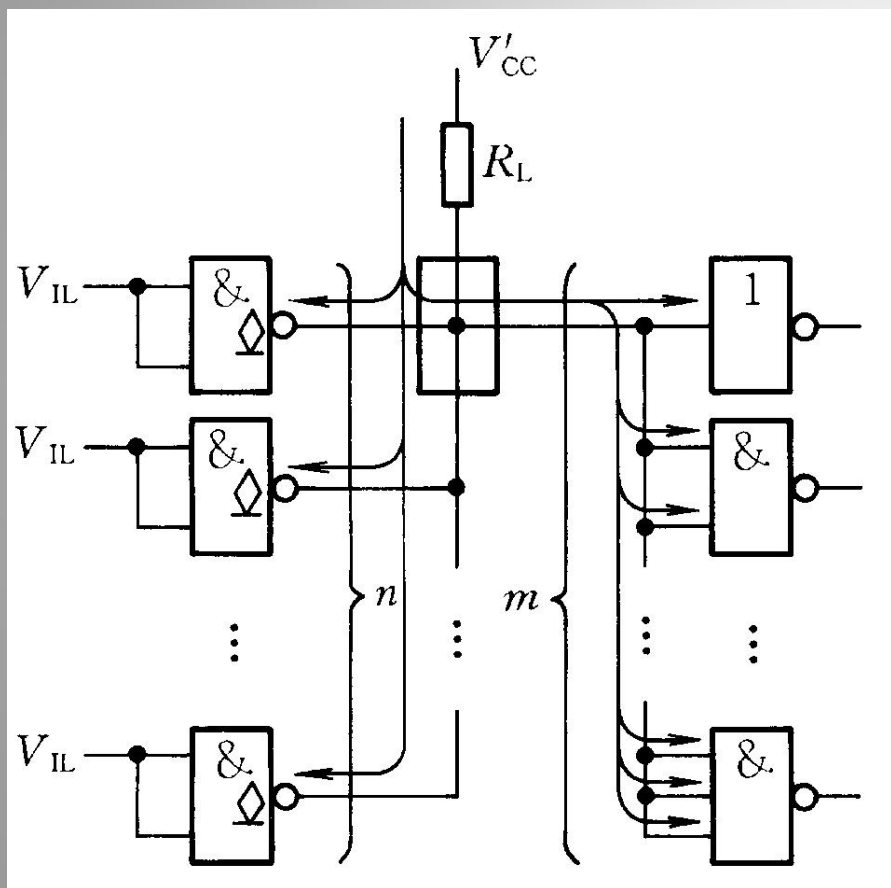
3、外接负载电阻 R_L 的计算(重视)

要求:

- 1、**OC**门输出管全部截止时, $V_o > V_{OH} \Rightarrow R_L$ 不能太大 (得 R_L 的上限)
- 2、**OC**门仅一个输出管道通, $V_o > V_{OL} \Rightarrow R_L$ 不能太小, 否则 i_L 过大 (得 R_L 下限)



3、外接负载电阻 R_L 的计算(重视)



OC门同时截止,

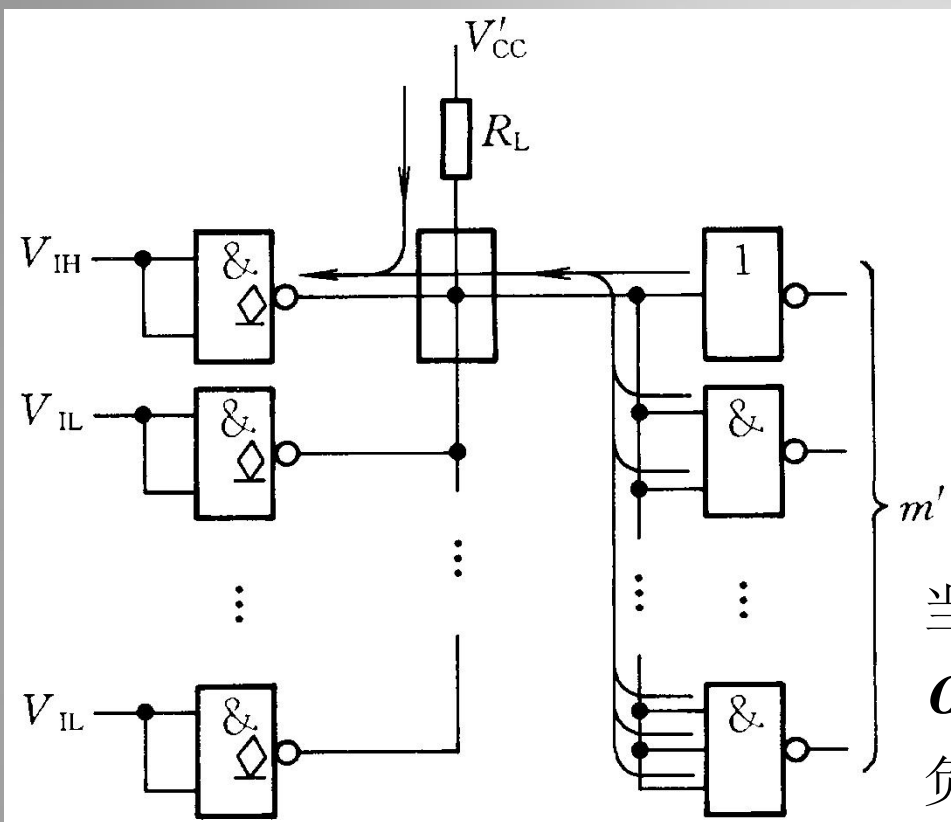
截止漏电流为 I_{OH} ,负载输入电流为 I_{IH}

为保证 $V_O \geq V_{OH}$,

则 $V_{CC} - R_L(nI_{OH} + mI_{IH}) \geq V_{OH}$

所以 $R_L \leq \frac{V'_{CC} - V_{OH}}{nI_{OH} + mI_{IH}} = R_{L(\max)}$

3、外接负载电阻 R_L 的计算



当仅一个 **OC** 门导通，

OC 门饱和时允许的最大电流为 I_{LM}

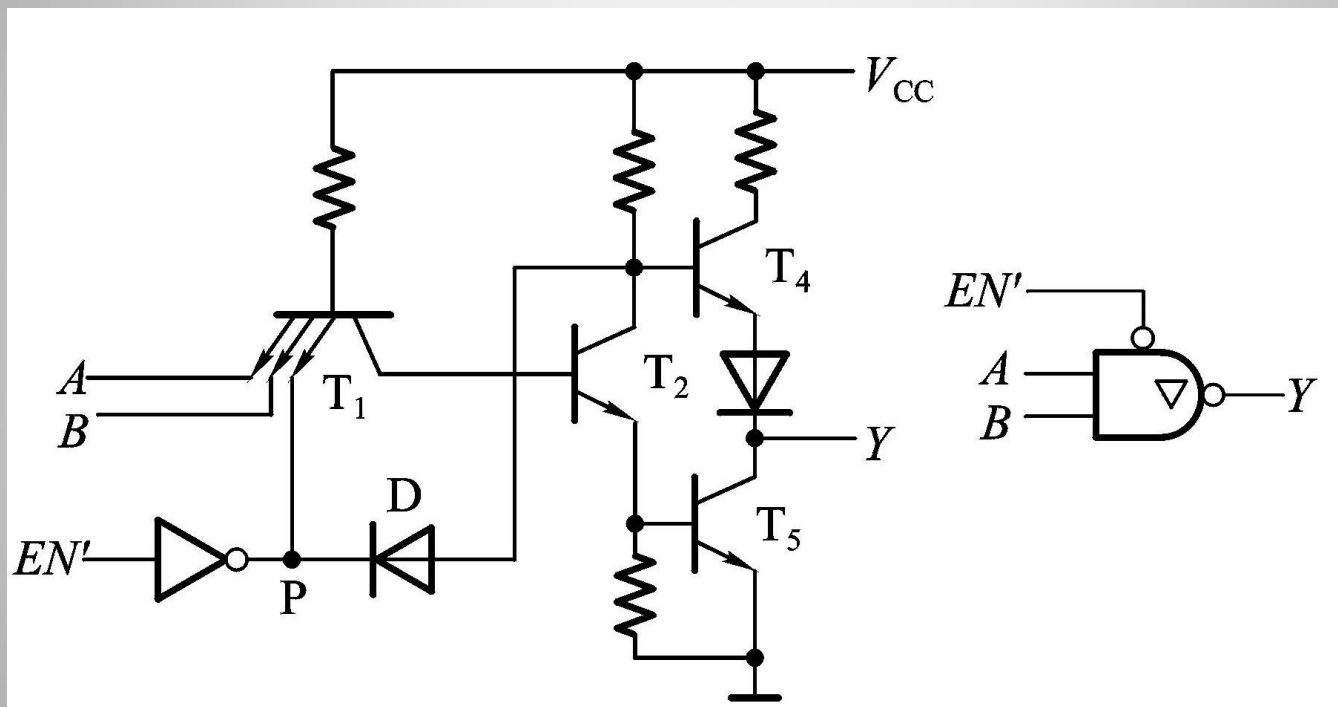
负载门输入电流为 I_{IL}

为保证 $V_o \leq V_{OL}$ ，且 I_L 不过大， R_L 不能太小

$$\text{所以 } R_L \geq \frac{V'_{CC} - V_{OL}}{I_{LM} - m' I_{IL}} = R_{L(\min)}$$

三、三态输出门 (Three state Output Gate ,TS)

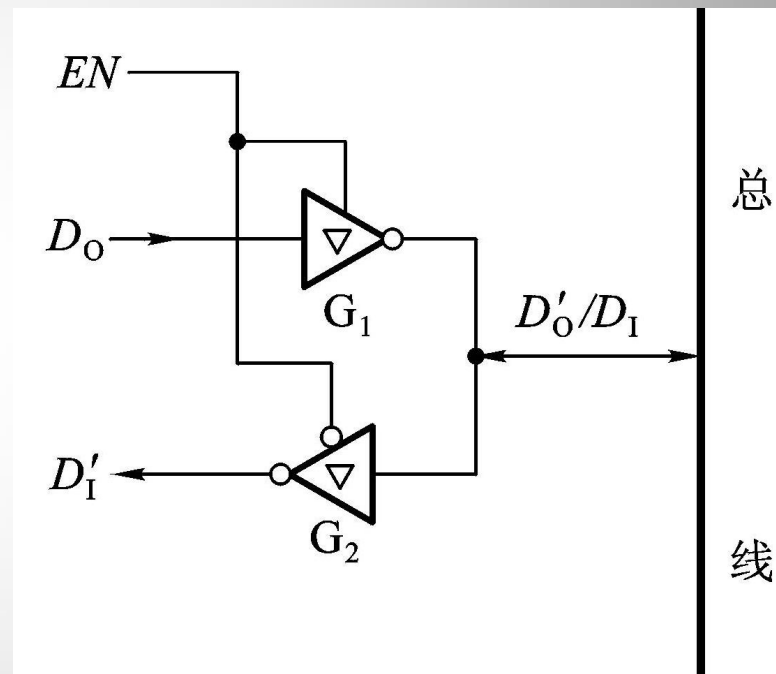
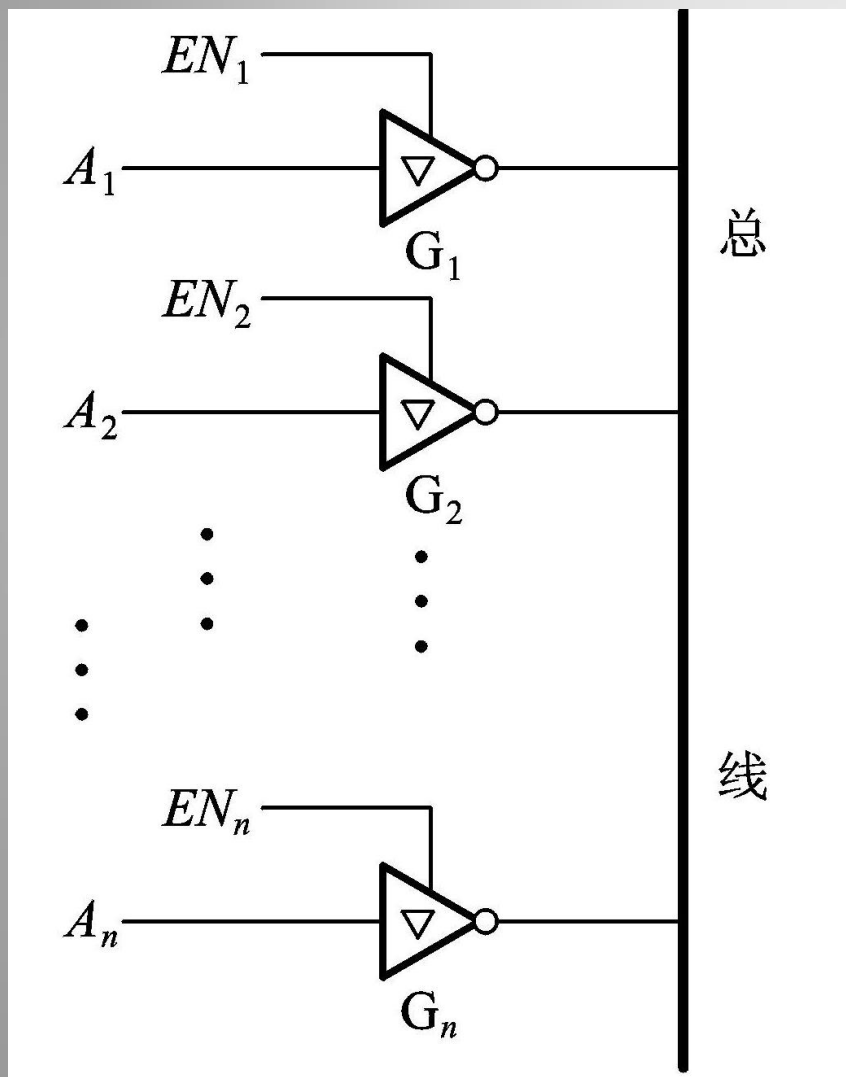
输出有三个状态: V_{OL}, V_{OH} , 高阻(Z)



(1) $EN = 0, P = 1, D$ 截止, 为“工作状态” $\Rightarrow Y = (AB)'$

(2) $EN = 1, P = 0, D$ 导通, 为“高阻状态” $\Rightarrow Y = Z$

三态门的用途



2.4.5 TTL电路的改进系列

(改进指标: $dp = t_{pd}(ns) \cdot P(mw) \downarrow$)

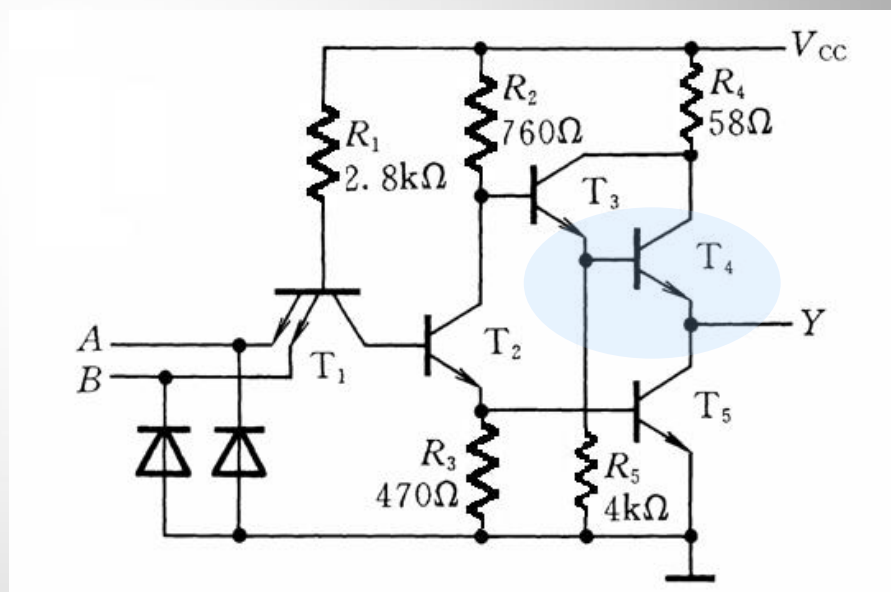
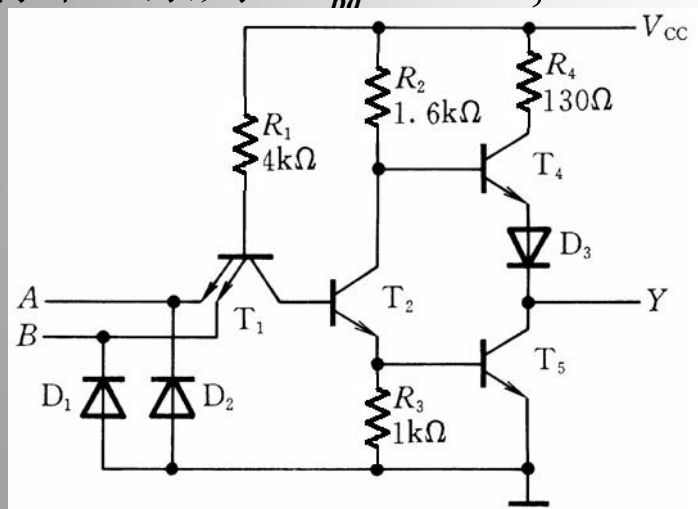
一、高速系列74H/54H (High-Speed TTL)

1. 电路的改进

(1)输出级采用复合管 (减小输出电阻 R_o)

(2)减少各电阻值

标准74系列: $t_{nd} = 10ns, P = 10mW$



2. 性能特点

速度提高 $t_{pd}(ns) \downarrow \frac{1}{2}$ 的同时功耗也增加 $P(mw) \uparrow 2$ 倍

二、肖特基系列74S/54S (Schottky TTL)

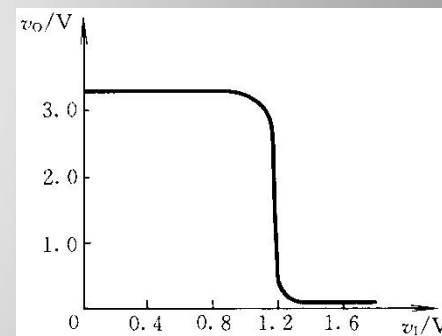
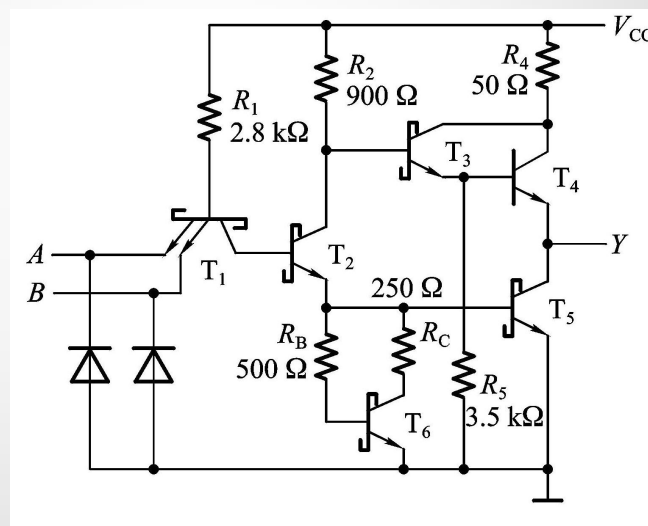
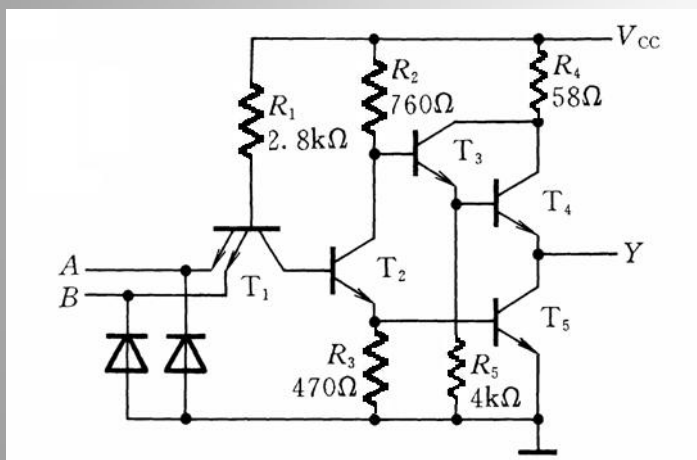
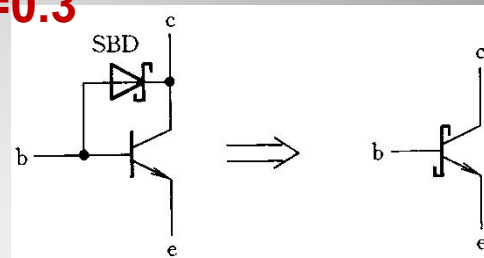
1. 电路改进

(1) 采用抗饱和三极管

(2) 用有源泄放电路代替74H系列中的R3

(3) 减小电阻值

$V_{pn}=0.3$



2. 性能特点

速度进一步提高，电压传输特性没有线性区，功耗增大

V_{TH} 下降到1.1V左右， V_{OL} $\uparrow$ 至0.4V左右

三、低功耗肖特基系列

74LS/54LS (Low-Power Schottky TTL)

- there are several series in the TTL family of logic devices (74, 74LS, 74S, etc.).

四、74AS, 74ALS (Advanced Low-Power Schottky TTL)

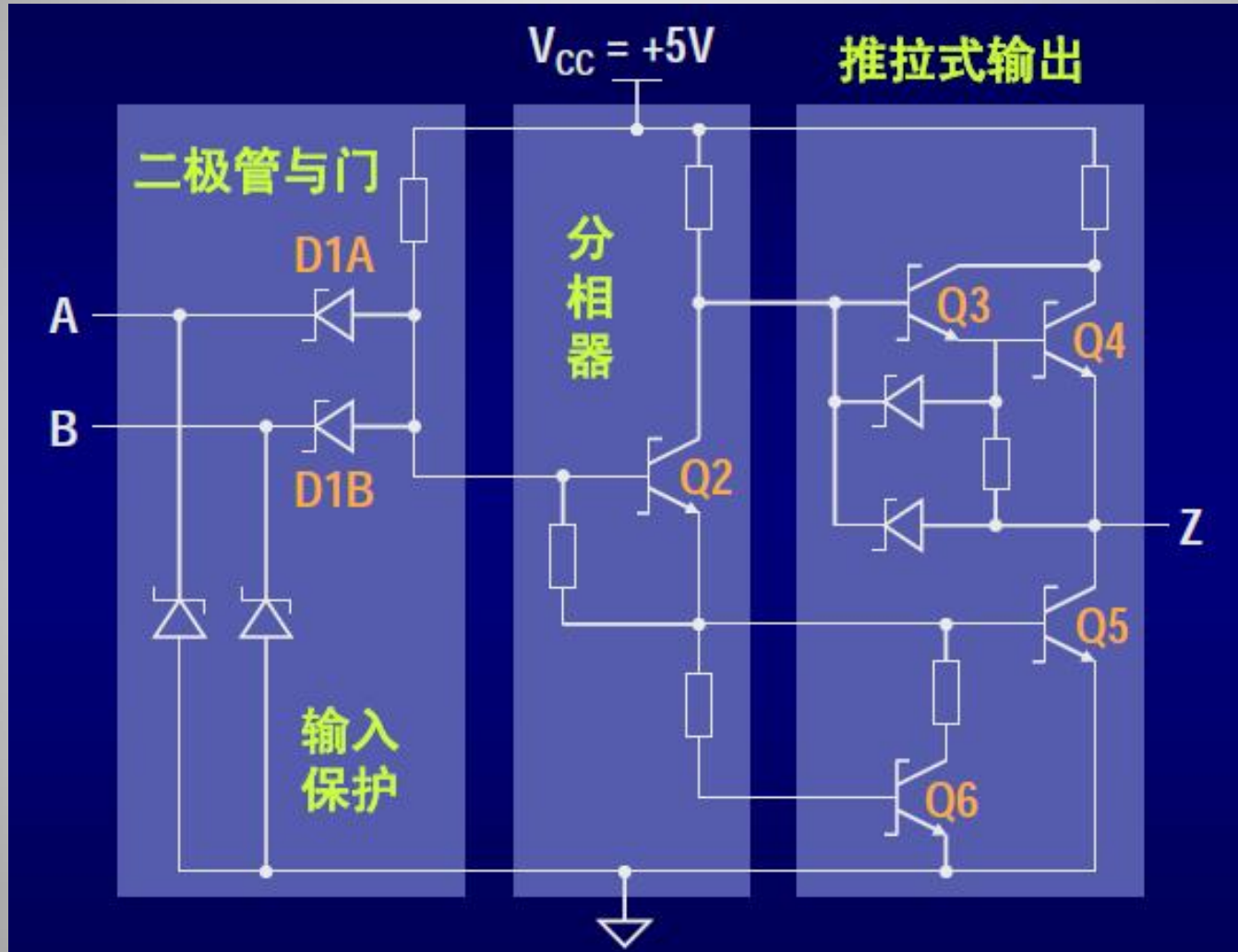
- ***Standard TTL, 74 Series***
- ***Schottky TTL, 74S Series***
- ***Low-Power Schottky TTL, 74LS Series (LS-TTL)***
- ***Advanced Schottky TTL, 74AS Series (AS-TTL)***
- ***Advanced Low-Power Schottky TTL, 74ALS Series***
- ***74F—Fast TTL***
- . . .

Comparison of TTL Series Characteristics

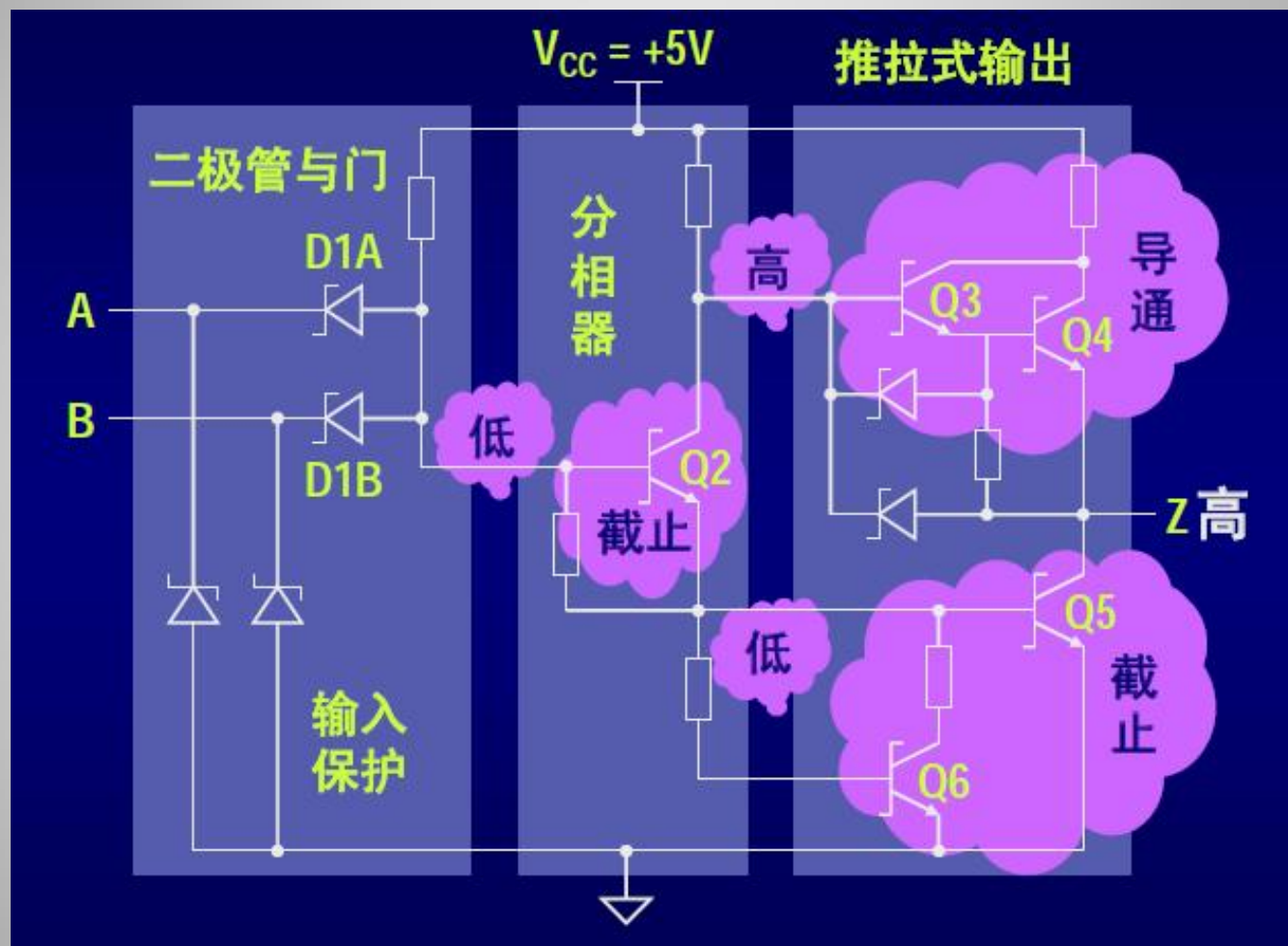
- Typical TTL series characteristics

	74	74S	74LS	74AS	74ALS	74F
Performance ratings						
Propagation delay (ns)	9	3	9.5	1.7	4	3
Power dissipation (mW)	10	20	2	8	1.2	6
Max. clock rate (MHz)	35	125	45	200	70	100
Fan-out (same series)	10	20	20	40	20	33
Voltage parameters						
$V_{OH}(\text{min})$	2.4	2.7	2.7	2.5	2.5	2.5
$V_{OL}(\text{max})$	0.4	0.5	0.5	0.5	0.5	0.5
$V_{IH}(\text{min})$	2.0	2.0	2.0	2.0	2.0	2.0
$V_{IL}(\text{max})$	0.8	0.8	0.8	0.8	0.8	0.8

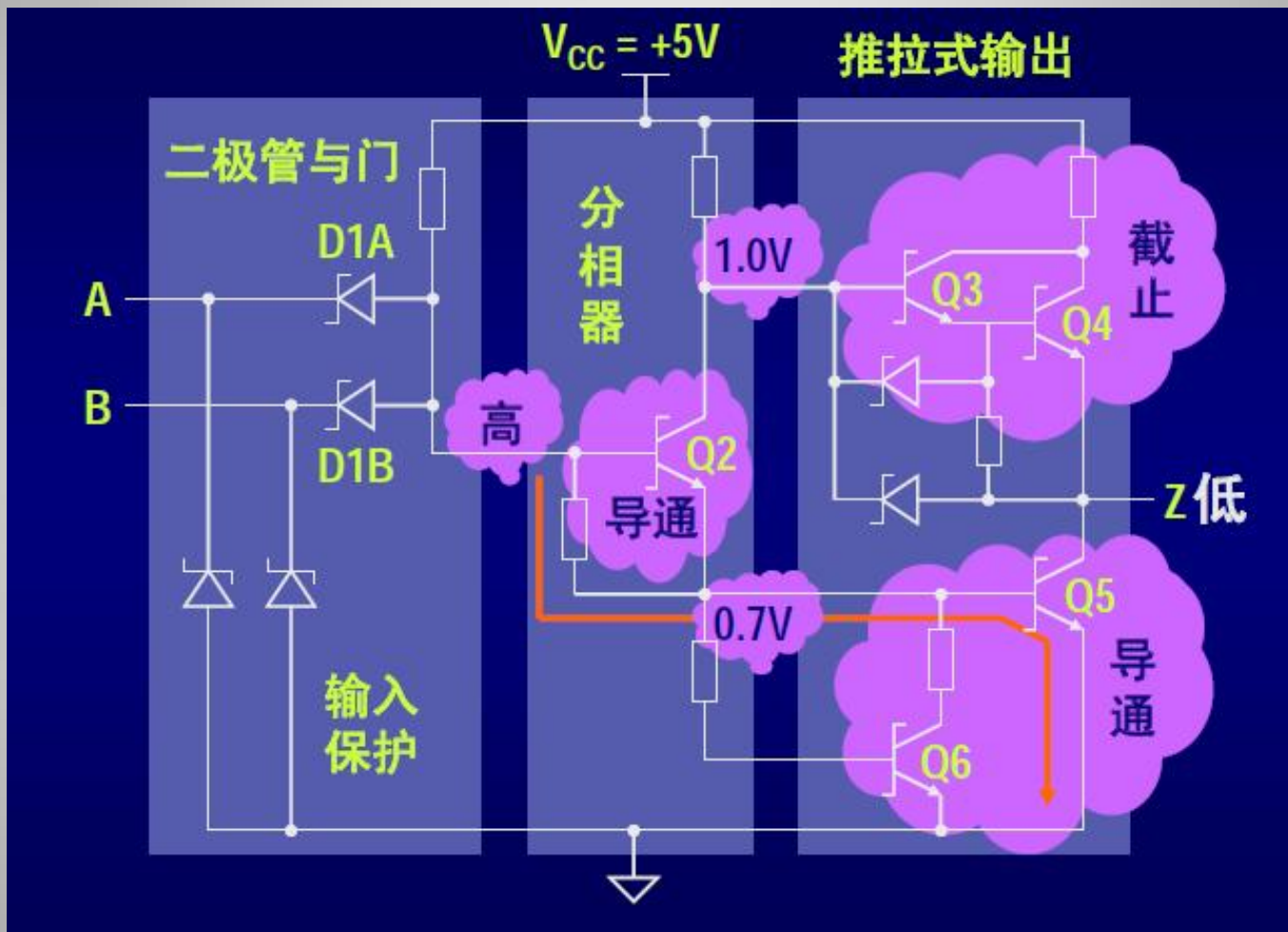
低功耗肖特基系列74LS/54LS (Low-Power Schottky TTL)



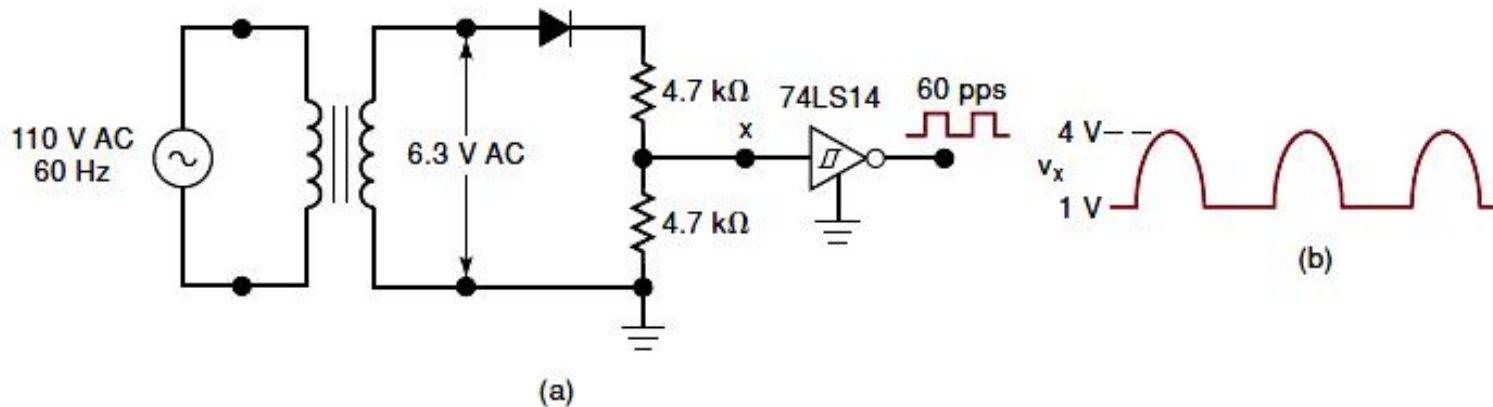
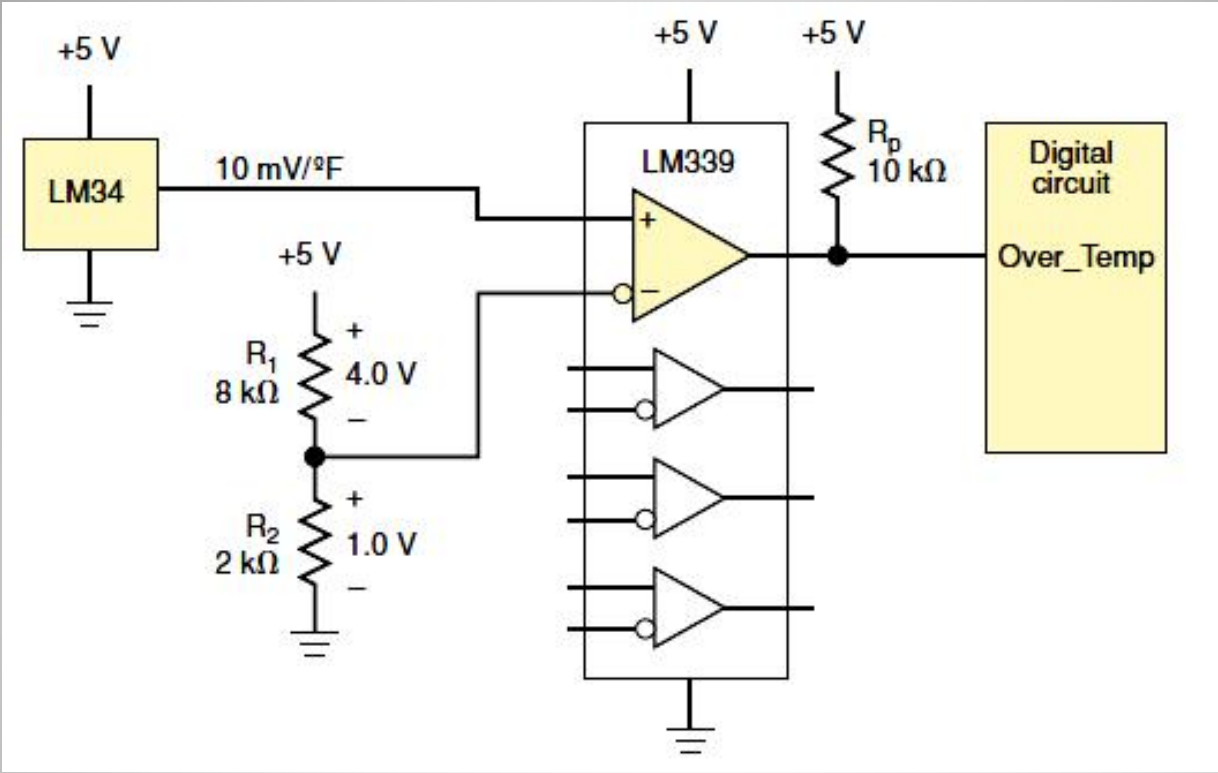
输入A=H, B=L



输入A=H, B=H

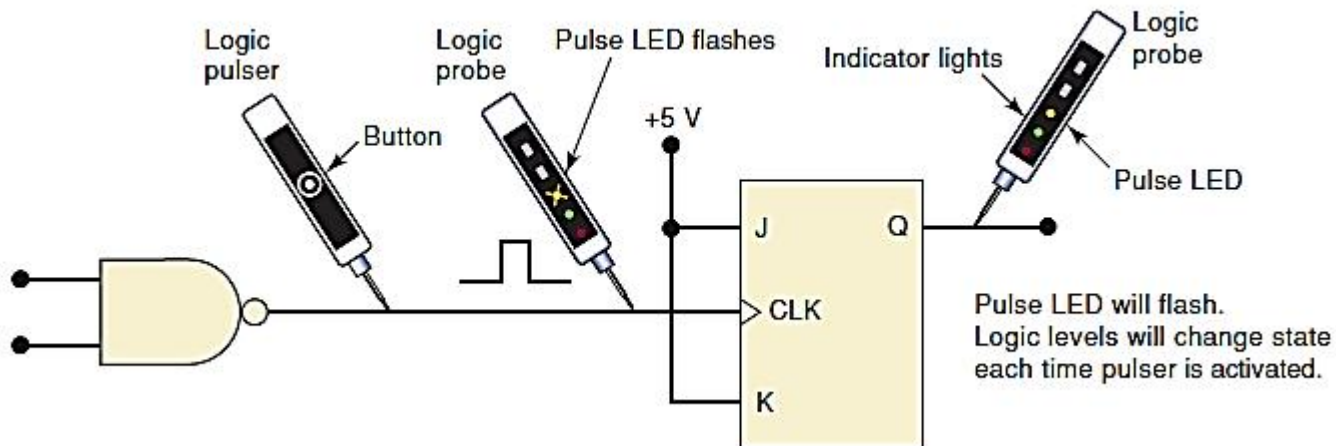
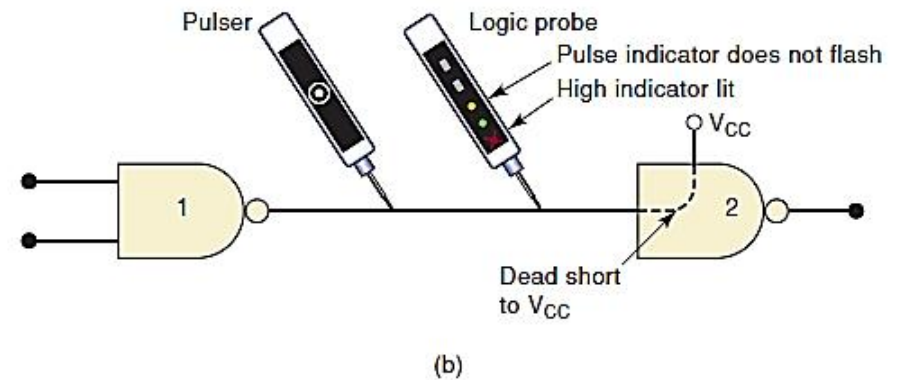
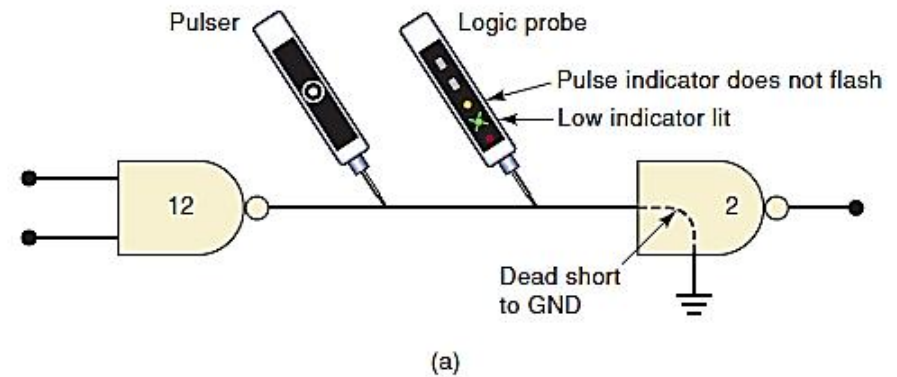


ANALOG VOLTAGE COMPARATORS



TROUBLESHOOTING (故障测试)

- The logic pulser



其他类型的双极型数字集成电路*

DTL: 输入为二极管门电路, 速度低, 已经不用

HTL: 电源电压高, V_{th} 高, 抗干扰性好, 已被CMOS替代

ECL: 非饱和逻辑, 速度快, 用于高速系统

I²L: 属饱和逻辑, 电路简单, 用于LSI内部电路

...

发射极耦合逻辑ECL(emitter-coupled logic)

如何提高速度？

——防止晶体管饱和

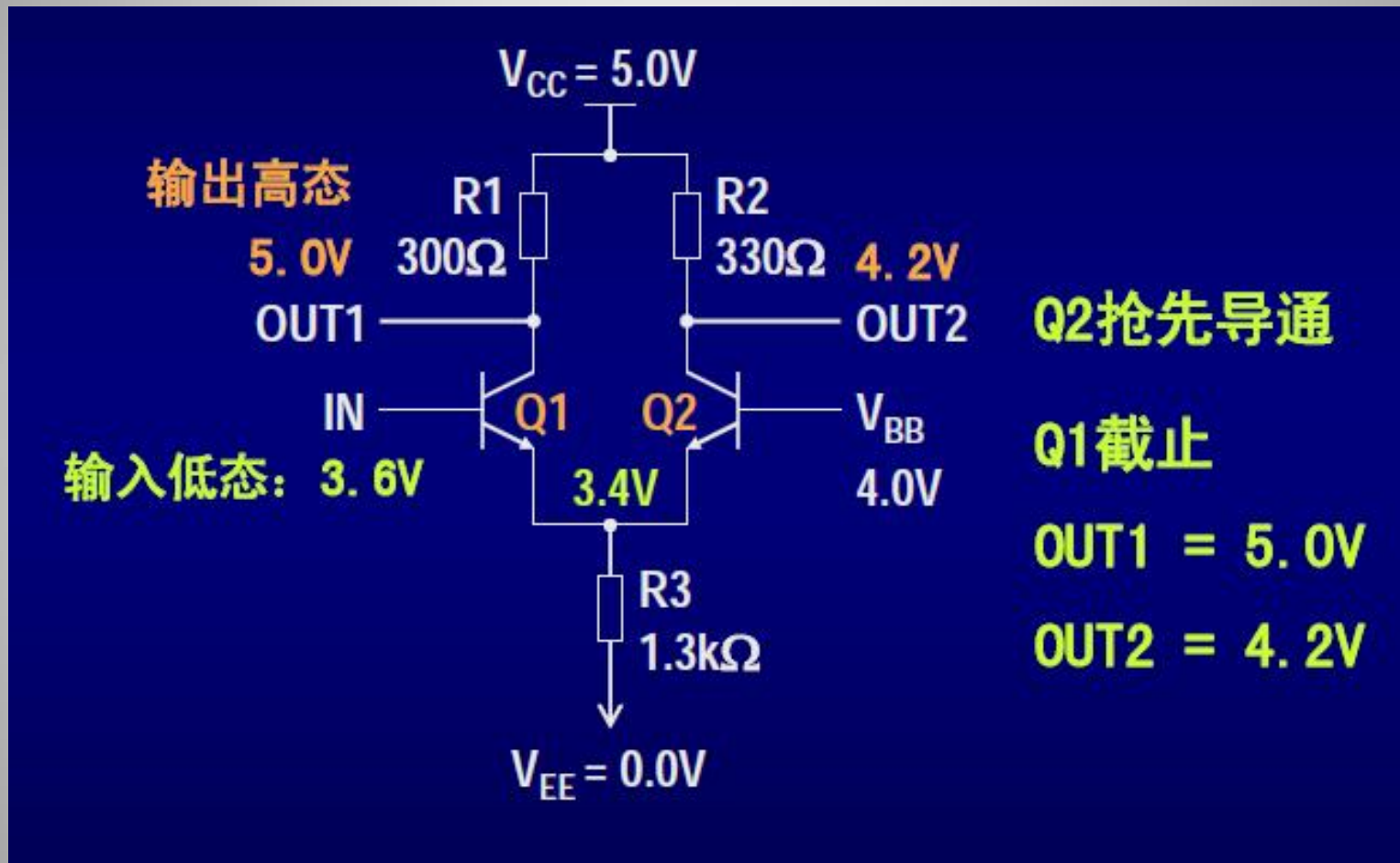
- 电流型逻辑（**CML**）
 - »（**current—mode logic, CML**）
- 也称为：发射极耦合（**ECL**）

TABLE 8-11 High-speed logic comparison.

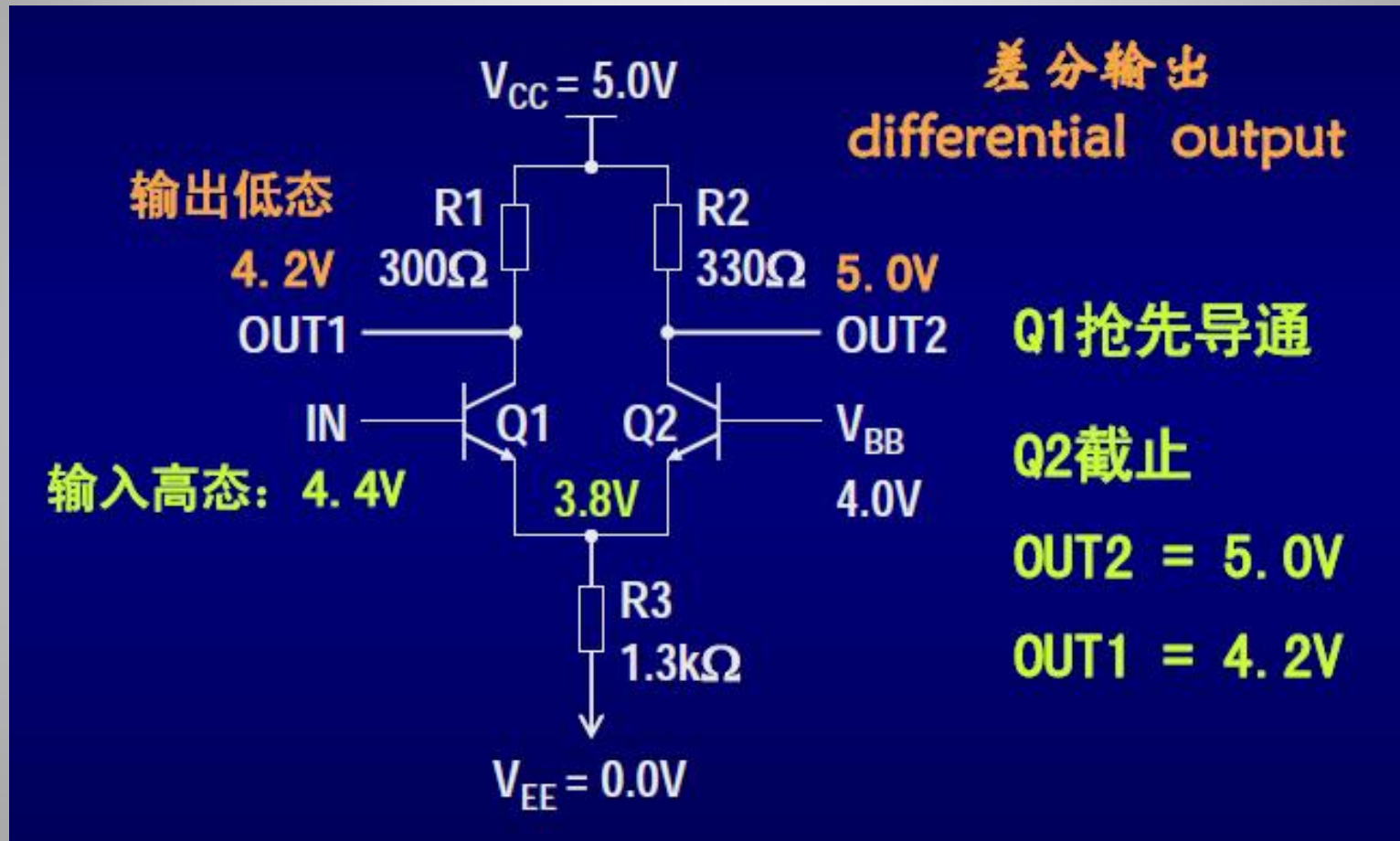
Logic Family	t_{pd} (ns)	P_D (mW) <100 kHz	Worst-Case Noise Margin (mV)	Maximum Clock Rate (MHz)
74AS	1.7	8	300	200
74F	3.8	6	300	100
74AHC	3.7	0.006	550	130
74AVC	2	0.006	250	•
74ALVT	2.4	0.33	400	•
74ALB	2.2	1	400	•
ECL	0.3	25	150	1400

*Flip-flops not available in this series.

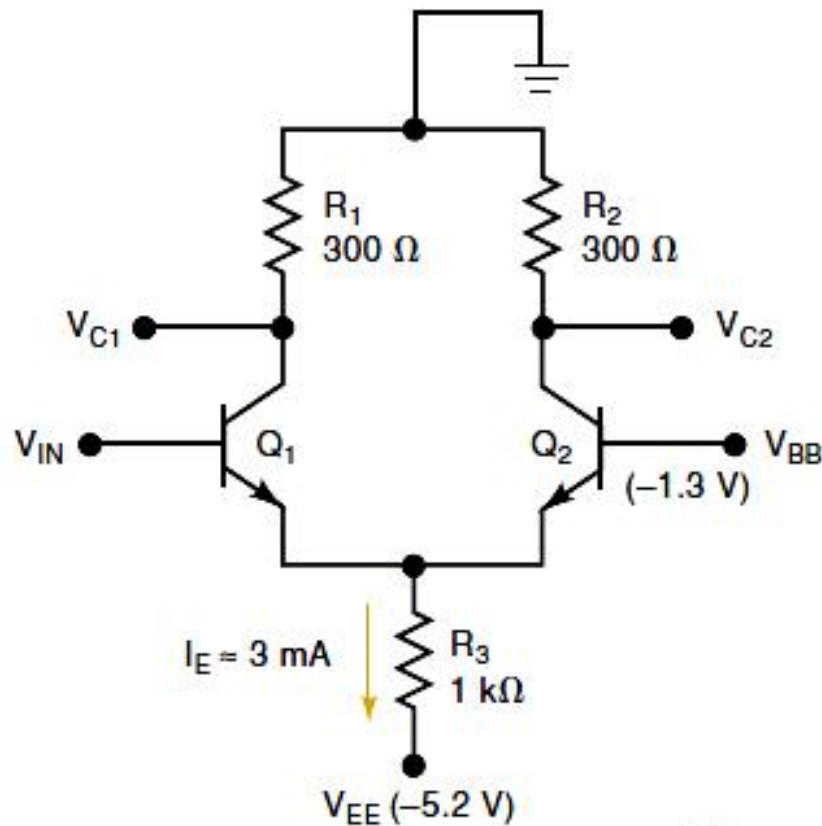
基本CML电路



基本CML电路



Basic ECL circuit



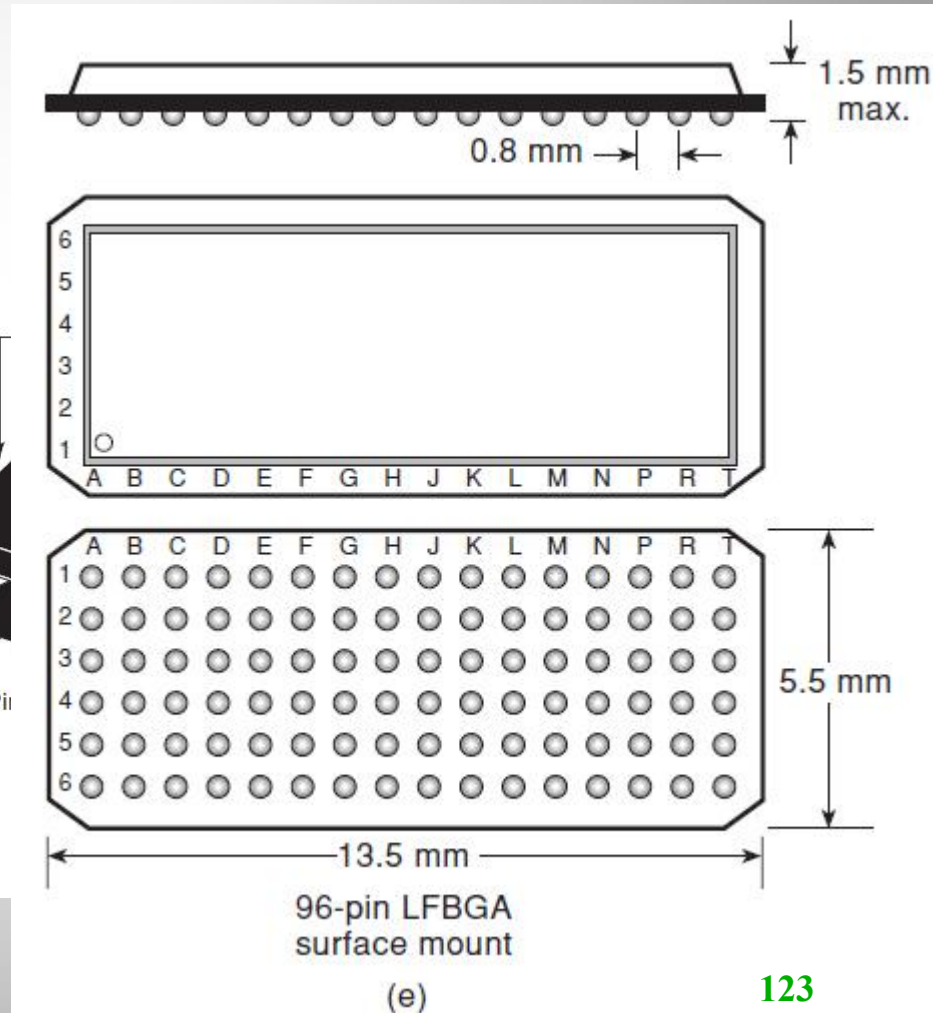
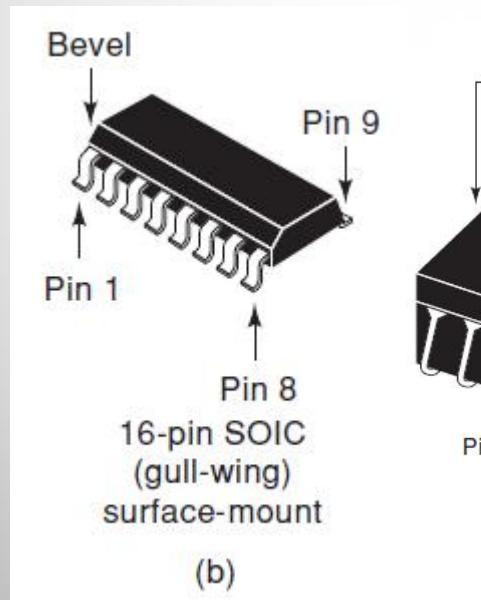
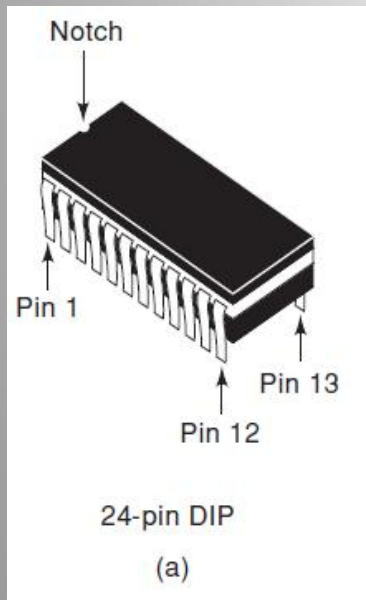
(a)

Operating States

V_{IN}	Outputs
-1.7 V (logic 0)	$V_{C1} = 0$ V $V_{C2} = -0.9$ V } Q_2 conducts
-0.8 V (logic 1)	$V_{C1} = -0.9$ V $V_{C2} = 0$ V } Q_1 conducts

IC Packages

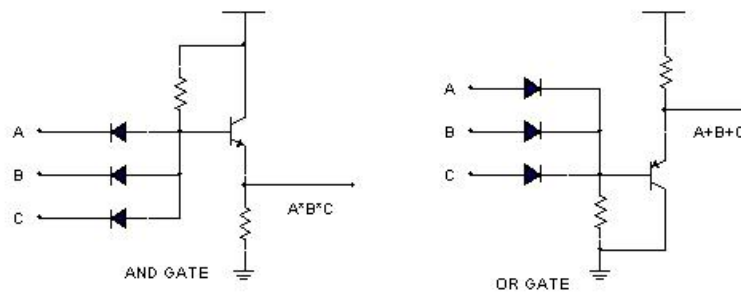
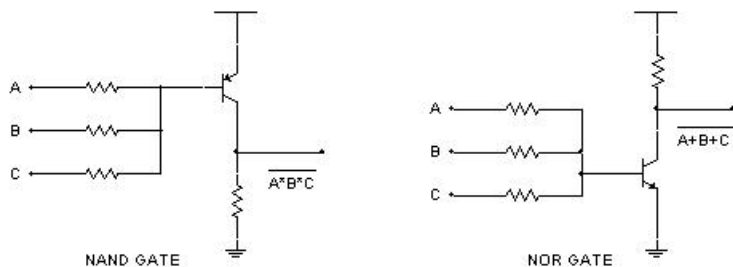
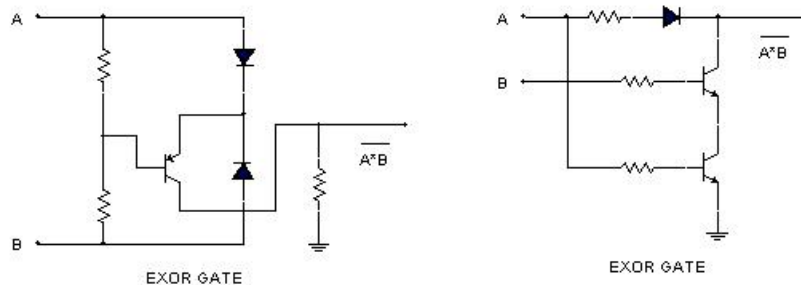
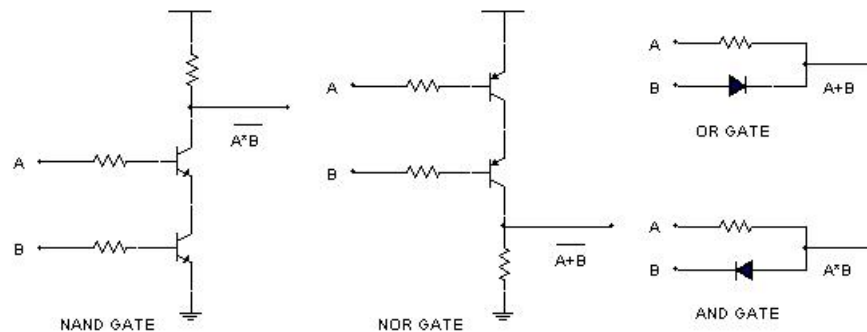
- DIP (dual-in-line package)
- The PLCC has J-shaped leads that curl under the IC
- The ball grid array (BGA)



课后作业

- 1) 调查:
 - 国际上的数字相关集成电路公司有哪些芯片?
 - » TI, Philips, Toshiba, Fairchild, Motorola.....
 - 图书馆资源: 电子器件天地
 - » http://10.15.61.7:8080/poweb/s_search.jsp?searchFiled=5&KWord=tjn&sClass=0&Submit=%BC%EC+%CB%F7
- 2) 作业:
 - 见学在浙大
- 3) 阅读
 - 组合逻辑电路

三极管的简单逻辑 (参考)



什么是电子设计竞赛？什么是SRTP？

- 了解规则
- 了解课题
- 组队
- 联系指导教师
- 申报
- go